AQUAFAB INDUSTRIES

Aquafab Workbench User Guide

Every page, every control, and step-by-step procedures for every experiment the software runs.

For Aquafab Workbench 0.2.1

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Screenshots show simulated machines and modelled example data, not measurements. Aquafab Workbench is not a safety system.

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About this guide

This is the complete user guide for **Aquafab Workbench**, the control, data-acquisition and analysis software for Aquafab Industries flow chambers, respirometry systems and swim tunnels. It covers installing the software, activating a license, every page and every control, and step-by-step procedures for the experiments the software runs.

How to use it. Chapters 1 to 4 are worth reading once, in order: they explain what the software is, how to install it, how licensing works and how the window is laid out. Chapters 5 to 19 are a reference, one chapter per page, describing every control. Chapter 20 is a set of step-by-step procedures ("how do I run a respirometry trial?"). Chapters 21 onward cover remote access, updates, where your data is kept, troubleshooting and the published methods the analysis follows.

Shorter, task-focused guides are published alongside this one: *Getting Started*, *Managing Your Lab's Licenses*, *Running a Respirometry Trial*, *Running a Temperature Program*, *PIV Flow Measurement and Chamber Velocity Calibration*, *Swim Performance (Ucrit) Testing*, *Fish Tracking*, *Analysing and Exporting Results* and *Commissioning a New Machine*. All of them are available from the Resource Library at **aquafabindustries.com**.

Keeping up to date. The software changes, and this guide is rebuilt with it. The edition date is on the cover. The newest edition is always at the same address on the Aquafab website, so a bookmarked link never goes stale.

> **About the screenshots.** Every screenshot in this guide was taken from the real application running *simulated* machines, which is why most of them show the amber "SIMULATED" banner. The trial results shown on the history and analysis pages are *modelled example data* (rainbow-trout-like resting metabolic rates at three temperatures), and the PIV and tracking videos are synthetic. None of the numbers in the screenshots are measurements, and none should be read as a statement of what an Aquafab machine achieves.

> **Safety.** Aquafab Workbench is laboratory control software. It is **not a safety system**: no interlock, limit or emergency stop in it is rated or certified. The on-screen **Stop Everything** button is software stopping software. Every Aquafab machine must also have its own physical means of stopping pumps and heaters, and those, not this software, are what protect people and animals.

Questions, corrections and requests for this guide are welcome at sales@aquafabindustries.com or +1 (513) 781-8010.

1 Introduction

1.1 What Aquafab Workbench is

Aquafab Workbench is one application that runs every machine Aquafab builds. The same program drives a teaching flow chamber, a large research flow chamber with several pumps and thermal loops, and a self-contained respirometry system. It also analyses the data those machines produce, and video filmed anywhere.

It does five broad jobs:

- **Runs the machine.** Pumps, valves, heaters, chillers and flush systems are controlled from one dashboard, with live readings of every temperature probe, flow meter and oxygen probe the machine has.
- **Holds and programs temperature.** A chamber can be held at a set temperature, or made to follow a programmed curve such as a tidal cycle.
- **Runs experiments.** Intermittent-flow and closed respirometry (SMR, RMR, MMR, aerobic scope, P_{crit}, EPOC), critical swimming speed (U_{crit}) and cost-of-transport tests, fish tracking, and particle image velocimetry (PIV) of the flow itself.
- **Calibrates.** A commissioning checklist, a PIV-based chamber velocity calibration, a thermal step test and a two-point dissolved-oxygen probe calibration, each saved with the machine so it survives restarts and updates.
- **Analyses.** Every trial is saved with every cycle and setting it used, and can be compared across treatments and machines, re-analysed with a different selection of cycles, and exported.

1.2 How the software thinks about machines

A **machine** in Aquafab Workbench is described by a **profile**: a file listing what hardware the machine has (relays, pumps, temperature probes, flow meters, oxygen probes, thermal loops) and how its dashboard should present it. The software contains no machine-specific code. When you open a machine, the dashboard is built from its profile, so a teaching chamber with one pump shows one flow slider, and a research chamber with three coupled pumps shows a single blended control and a readout for each pump.

The software ships with profiles for the current Aquafab range:

Machine	What it is
Alpha Flow Chamber	Large-scale research-grade flow chamber: two variable-speed pumps, one thermal loop, flow meter.
Alpha Flow Chamber (Legacy)	The earlier, fully specified Alpha build with a separate flush loop: three pumps, fourteen relays, five temperature probes and two thermal loops.

Machine	What it is
Small Respirometry System	Self-contained respirometry: one chamber, a flush pump, one temperature probe and a dissolved-oxygen probe. No variable-speed pump.
Teaching Flow Chamber	Smaller teaching unit with one pump and one temperature probe.

A new installation lists one machine, **Alpha Flow Chamber (Example)**: the Alpha Flow Chamber, always simulated, to explore. **+ Add Machine** adds the machine you have, either one of the bundled profiles above in one click or your own rig through the setup wizard (chapter 19), and the example goes away once you have added one. A profile can also be exported to a file and imported on another computer.

1.2.1 Simulated and connected machines

Every machine can run in one of two ways:

- **Connected:** on the control computer wired to the machine (a Raspberry Pi inside Aquafab systems), the software drives the real relays and pumps and reads the real probes.
- **Simulated:** anywhere else, the software runs the same machine against a built-in simulator generated from the profile. Readings are synthetic, and controls drive nothing. A simulated respirometry chamber holds a modelled fish: it rests at 110 mg O_2/kg/h, starts well above that after handling and settles over about an hour, with the occasional burst of activity. Those rates are a model, not a measurement of any animal. A simulated machine always shows an amber banner across the top of its dashboard: **"SIMULATED – no real hardware detected, all readings are synthetic"**.

If a profile names real hardware but the hardware cannot be reached, the machine opens in simulation and the software says so in a dialog the first time it is opened. Simulation is how you learn the software, rehearse a protocol, or show it to a colleague, with nothing at risk.

1.3 What you can do with each kind of experiment

Experiment	Where	Gives you	Needs
Respirometry	Experiments → Respirometry	SMR, RMR, MMR, aerobic scope, P_{crit}, EPOC	A dissolved-oxygen probe for oxygen results
Temperature program	Experiments → Respirometry, *Temperature program only*	Temperature and flow following a programmed curve	A thermal loop
Swim performance	Experiments → Swim Performance	U_{crit} in % output, cm/s and body lengths/s; cost of transport with an MO_2	A variable-speed pump; a camera only for tracking

Experiment	Where	Gives you	Needs
Fish tracking	Experiments → Fish Tracking, or Video Analysis	Position over time, swim speed, distance	A camera or video file, an empty-tank background, a scale
Flow field (PIV)	Experiments → Flow Field (PIV), or Video Analysis	A velocity vector field and median chamber velocity	Video of seeded flow, even lighting, a scale

Fish tracking and PIV read video, so they need no machine at all: the **Video Analysis** card on the start screen runs both on footage filmed anywhere.

1.4 The preview and the licensed software

Anyone can download and install Aquafab Workbench. **Without a license it runs as a preview**: the example machine, **Alpha Flow Chamber (Example)**, can be opened, explored and run in simulation, so you can see exactly how the software works before buying it. A license is needed for:

- **adding machines** (**+ Add Machine**, **↥ Import Config…**, **Duplicate**) and seeing machines already added,
- running a trial on a **connected** machine (your own hardware),
- **PIV** flow-field analysis,
- **fish tracking**,
- **trial analysis** (the Analysis screen, a machine's Analysis tab, the single-trial view, and importing readings from a file), and
- **exporting data** (trial CSV and PDF reports, chart exports, track and vector-field files).

On the example machine, thermal and flow control, running respirometry and swim trials, every calibration (including the PIV-based chamber velocity calibration), Saved Experiments, the Trial History list, machine setup and Settings all work without a license. Manual control of a connected machine and the emergency stop are never locked.

A license comes with every Aquafab system, and can be bought for use with your own equipment. It is bought once, never expires, and includes updates and support. Chapter 3 explains codes, seats and activation.

> **A license never stops anything that is running.** It is checked only at the moment a trial is started on connected hardware. A license that lapses during a ten-hour overnight run changes nothing about that run, and no pump, heater, flush or emergency stop ever consults it.

1.5 Words used in this guide

Term	Meaning
Profile	The file that describes a machine's hardware and dashboard.

Term	Meaning
Launcher	The start screen listing your machines and the no-machine tools.
Dashboard	The page set for one open machine: Home, Thermal Control, Experiments, Calibrations, Trial History, Analysis, Settings.
Section rail	The column of section names down the left of a dashboard.
Thermal zone (or loop)	One temperature-controlled body of water, such as the main loop or the flush loop, with its heater, chiller and probe.
Trial	One run of an experiment. Saved trials appear in Trial History.
Experiment (saved)	The *settings* for a trial, kept under a name so it can be run again. Not the results.
Treatment	A free-text grouping label (“control”, “18 °C”, “fed”) that Analysis groups trials by.
Cycle	In intermittent-flow respirometry, one flush, wait and measure sequence. Each cycle gives one MO_2.
MO_2	Rate of oxygen uptake, reported in mg O_2 per kg per hour.
SMR / RMR / MMR	Standard, routine and maximum metabolic rate.
P_{crit}	The oxygen level below which the animal can no longer maintain its oxygen uptake.
U_{crit}	Critical swimming speed, from a stepped-velocity test.
PIV	Particle image velocimetry: measuring water velocity by tracking seeded particles between video frames.
Commissioning	Checking that each relay, pump and probe in the profile is the one physically wired.

2 Installing and opening the software

2.1 What you need

Computer	Installer
Windows 10 or 11, 64-bit	`AquafabWorkbench-Windows.exe`
Mac with Apple silicon (M1 or later)	`AquafabWorkbench-macOS.dmg`
Mac with an Intel processor	`AquafabWorkbench-macOS-Intel.dmg`
Linux, x86-64	`AquafabWorkbench-Linux.AppImage`
Raspberry Pi control computer in an Aquafab system	Installed by Aquafab

Any of the desktop builds can open every machine in simulation, analyse saved trials and process video. Driving real hardware happens on the control computer wired to the machine. In an Aquafab system that is a Raspberry Pi 5 or Compute Module 5 running 64-bit Raspberry Pi OS with desktop, set up by Aquafab before delivery.

PIV and fish tracking are the heaviest work the software does. They run on any of the computers above, but a long clip analyses noticeably faster on a recent laptop or desktop than on the Pi.

2.2 Downloading

Download the installer for your computer from the Aquafab Workbench page:

aquafabindustries.com/pages/aquafab-workbench (the *Downloads* section)

No account or sign-in is needed. The download links always point at the newest release.

2.3 Installing on Windows

1. Run `AquafabWorkbench-Windows.exe`.
2. If Windows shows **“Windows protected your PC”**, click **More info**, then **Run anyway**. (The installer is not yet code-signed, so SmartScreen does not recognise it.)
3. Follow the installer. It installs for your Windows user only, into `%LOCALAPPDATA%\Programs`, and **does not need administrator rights**, so it works on most locked-down lab computers.
4. Open **Aquafab Workbench** from the Start menu.

2.4 Installing on a Mac

1. Open the `.dmg` file and drag **Aquafab Workbench** into **Applications**.

2. Open it from Applications. The first time, macOS may refuse with a warning that it cannot check the app. Click **Done**.
3. Open **System Settings → Privacy & Security**, scroll down to the message about Aquafab Workbench, click **Open Anyway**, and confirm with your password.
4. Open it again. From now on it opens normally.

Use the **Intel** image on a Mac with an Intel processor. To check which you have, choose **Apple menu → About This Mac**: "Chip: Apple M..." is Apple silicon, "Processor: ... Intel" is Intel.

2.5 Installing on Linux

The Linux build is a single AppImage file. Make it executable and run it:

```
chmod +x AquafabWorkbench-Linux.AppImage
./AquafabWorkbench-Linux.AppImage
```

Most file managers can also mark it executable under *Properties → Permissions*.

2.6 The Raspberry Pi in an Aquafab system

The control computer arrives with Aquafab Workbench installed and an **Aquafab Workbench** icon on its desktop (also in the applications menu under *Education* or *Science*). It shows the Aquafab logo while it boots. Unless your system was set up otherwise:

- **Double-click opens it**, as on any computer. (A Pi with only a touchscreen may be set up to open icons with a single tap instead.)
- **It opens by itself at power-on.** Launching the application never opens a machine or starts anything on its own. A site that would rather an operator start it deliberately can ask Aquafab support to turn this off.
- **When it opens by itself, it opens full screen**, filling the touchscreen with no window frame. **Exit full screen** in the top strip returns it to an ordinary window, and the same button, now **Enter full screen**, goes back.

If your Pi ever needs reinstalling, contact Aquafab support.

2.7 Opening it for the first time

The first launch takes a few seconds longer than later ones while the software prepares its fonts and charts. Then:

1. **The license dialog appears** (chapter 3). Enter a license code and press **Activate**, or press **Not now** to use the preview. You are asked once; after that, a license is entered under **Settings → License**.
2. **The Launcher appears**, listing **Alpha Flow Chamber (Example)** and the three no-machine tools (chapter 5). The example is always simulated.

3. **Add your machine** with **+ Add Machine** (needs a license; chapter 5), or open the example to explore.
4. **Open a machine.** On anything other than the machine's own control computer, a machine you added opens in simulation and a dialog titled **"Running in simulation"** explains why: the profile names real hardware, but it could not be reached. That is expected on a laptop or office computer. Press **OK**; the dashboard shows the amber SIMULATED banner while it runs that way. The example machine, and any machine opened in the preview, opens in simulation without this dialog.

> **On a Raspberry Pi**, the "Running in simulation" dialog on a machine that *should* be connected usually means the operating-system user has not yet been given access to the GPIO and serial ports, or has not logged out and back in since being given it. Contact Aquafab support if it persists.

2.8 The window size

The window opens at 1200 × 900 pixels, or maximised on a screen too small for that, such as a 7-inch Pi touchscreen. Every page scrolls, by mouse wheel, scrollbar or by dragging a finger, so nothing is out of reach on a small screen. On a narrow screen, the ☰ button at the top left of a dashboard hides the section rail to give the page more room.

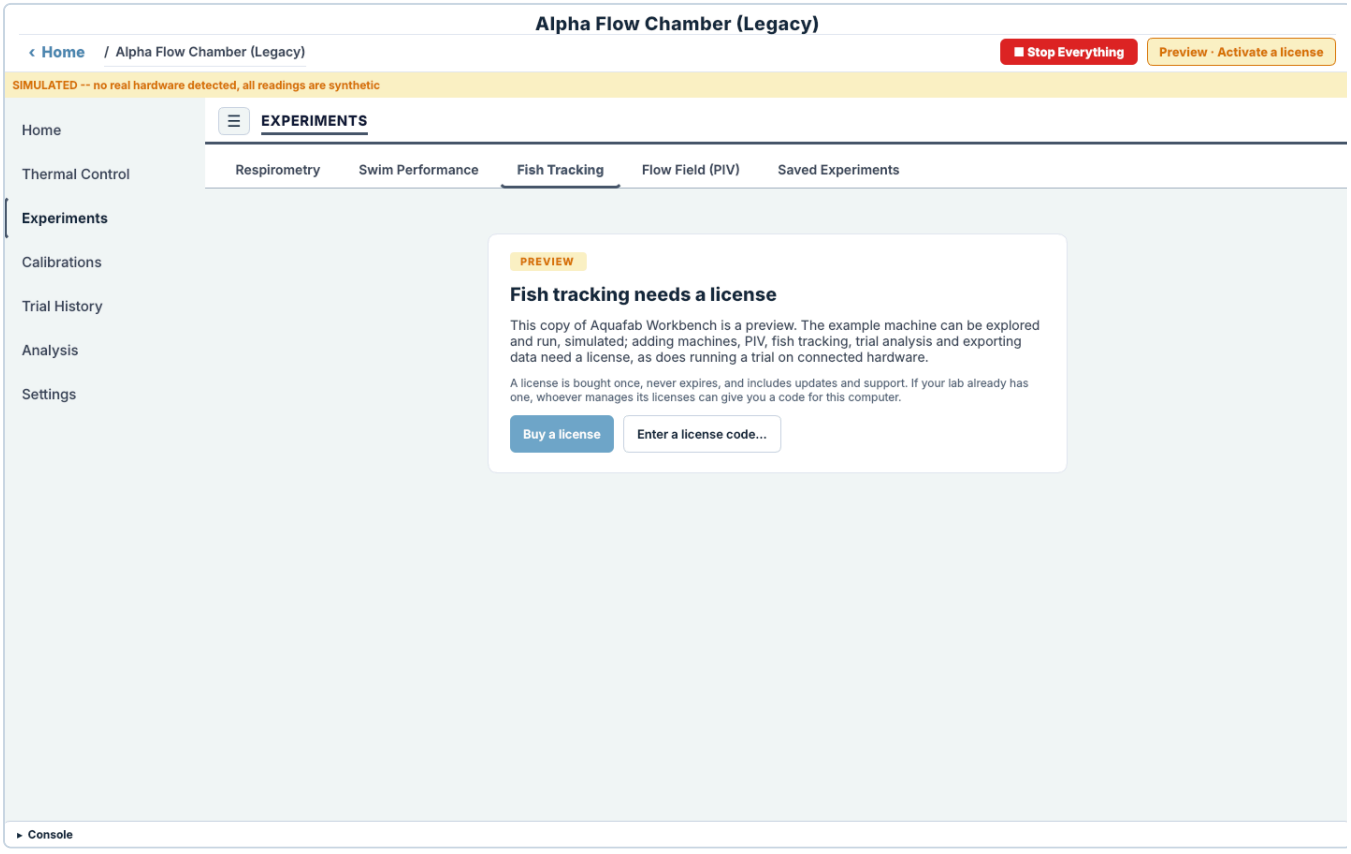
3 Licenses and activation

3.1 What a license does

Without a license, Aquafab Workbench is a **preview** (section 1.4): the example machine, with its setup, calibrations and trial list, all works, but adding machines, running a trial on connected hardware, PIV, fish tracking, trial analysis and exporting data are locked. A license unlocks all of them on the computer it is activated on.

While the software is a preview, the top strip of the window shows **Preview · Activate a license**. A locked page is covered by a card with a **PREVIEW** badge, a title such as *"Fish tracking needs a license"*, a short explanation, and two buttons:

- **Buy a license** opens the Aquafab Workbench license page on the store.
- **Enter a license code…** opens the activation box.



A locked page in the preview: Fish Tracking on a simulated machine.

A locked button, such as an export, shows the same card as a dialog with a **Close** button. Entering a license unlocks every open page immediately; there is no need to restart.

3.2 Codes, seats and keys

Licenses are sold per lab, as a set of **seats**. Each seat is one computer.

- A **teacher seat** is for the people who run the lab. It never expires.
- A **student seat** lasts a year from activation and **renews itself** whenever that computer is online, so a computer in regular use never notices. A seat nobody has renewed for a year can be taken by another computer, so seats pass on as classes graduate without anyone doing anything.

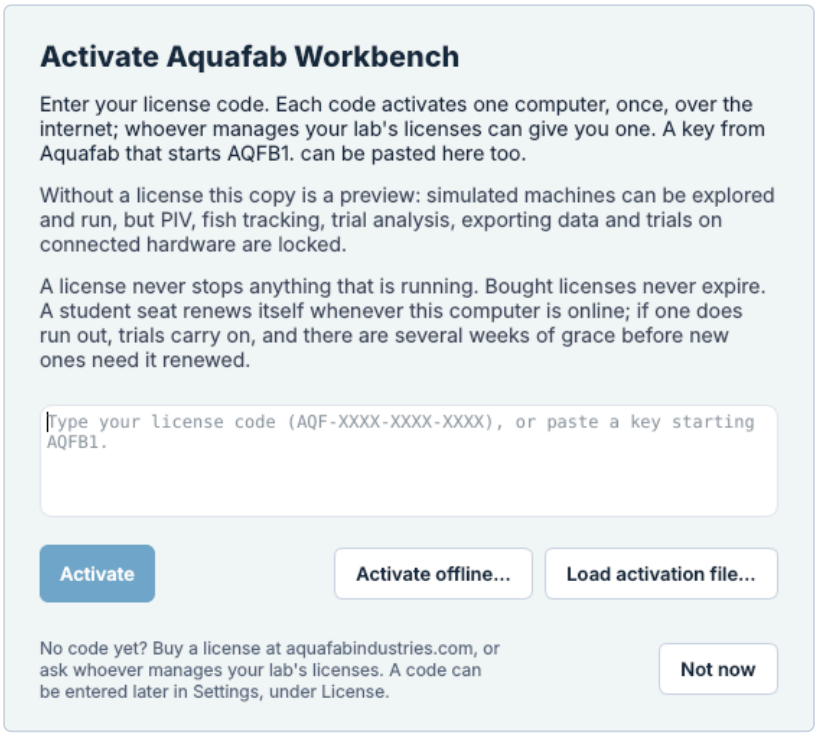
Each seat has its own **license code**, in the form `AQF-XXXX-XXXX-XXXX`. A code activates one computer, once, over the internet. After that the computer holds a certificate signed for it alone and checks it offline at every start; the internet is not needed again except for a student seat's yearly renewal.

Older **license keys**, long text starting `AQFB1.`, are still accepted and can be pasted into the same box. The software tells codes and keys apart itself.

Bought licenses never expire and include updates and support.

3.3 Activating with a code (online)

When the software is first opened it asks once, in the **Activate Aquafab Workbench** dialog, which says what a license unlocks and that it never stops a running trial.



The first-run license dialog.

1. Type or paste the code into the box (`AQF-XXXX-XXXX-XXXX`).
2. Press **Activate**. The computer contacts the license server, which returns a certificate for this computer.

3. The dialog closes and the license panel shows the lab's name and the seat.

Press **Not now** to carry on with the preview. You are not asked again on this computer; enter a code later under **Settings → License** (from the Launcher, or the *Settings* section of any machine). The same box and buttons appear there.

3.4 Activating a computer with no internet

Some lab computers are never online. They are activated with a file instead:

1. On that computer, open **Settings → License**, type the code, and press **Activate offline…**. A dialog shows a **request code**.
2. Press **Copy request code** and give it to whoever manages your lab's licenses (save it to a USB stick, or type it).
3. They sign in to the license dashboard (below), paste it under **Computer with no internet**, and press **Make activation file**. The dashboard shows the activation text; they save it as a file named `aquafab-activation.txt`.
4. Copy that file to the offline computer, press **Load activation file…** and open it.

3.5 The license panel in Settings

Settings → License shows the current state in one line: no key, a licensed lab and seat with its issue date, a warning as an expiry approaches, or a grace period. Under it are:

Control	What it does
Code box	Type a code (`AQF-…`) or paste a key (`AQFB1.…`). Line breaks in a pasted key are ignored.
Activate	Activates the code online, or checks and stores a pasted key.
Activate offline…	Makes a request code for a computer with no internet (above).
Load activation file…	Opens the file made from a request code.
Remove key from this computer	Shown once a license is stored. Removes it from this computer only; the seat stays bound until it is freed in the dashboard.

The license is stored per computer, not per user, so every account on a shared lab computer sees it. A license file copied to another computer is invalid there: each certificate names the computer it was issued for.

3.6 Managing your lab's seats

Whoever bought the license manages its seats from the store: **aquafabindustries.com/apps/licenses**. Sign in with the account the license was bought with. The page lists **Student seats** and **Teacher seats**, each with its **code**, who it is **assigned to**, which

computer holds it, and when it **renews**.

Action	What it does
Name / Email + Save	Records which student a seat belongs to.
Email code	Opens a ready-written email with the code, in your own mail program.
Free seat	Releases the seat from its computer. Its code stops working and a new code is made, so a code once shared cannot be reused.
Free all student seats (end of term)	Frees every student seat at once, each with a new code.
Computer with no internet	Turns a request code into an activation file (above).
Lab name	The name shown in the software's license panel.
Invite another teacher	Lets a colleague manage the same license.
Buy more seats	Opens the store to add seats to this license.

3.7 What happens when a license lapses

Only student seats can lapse, and only on a computer that has not been online for a year. In that case:

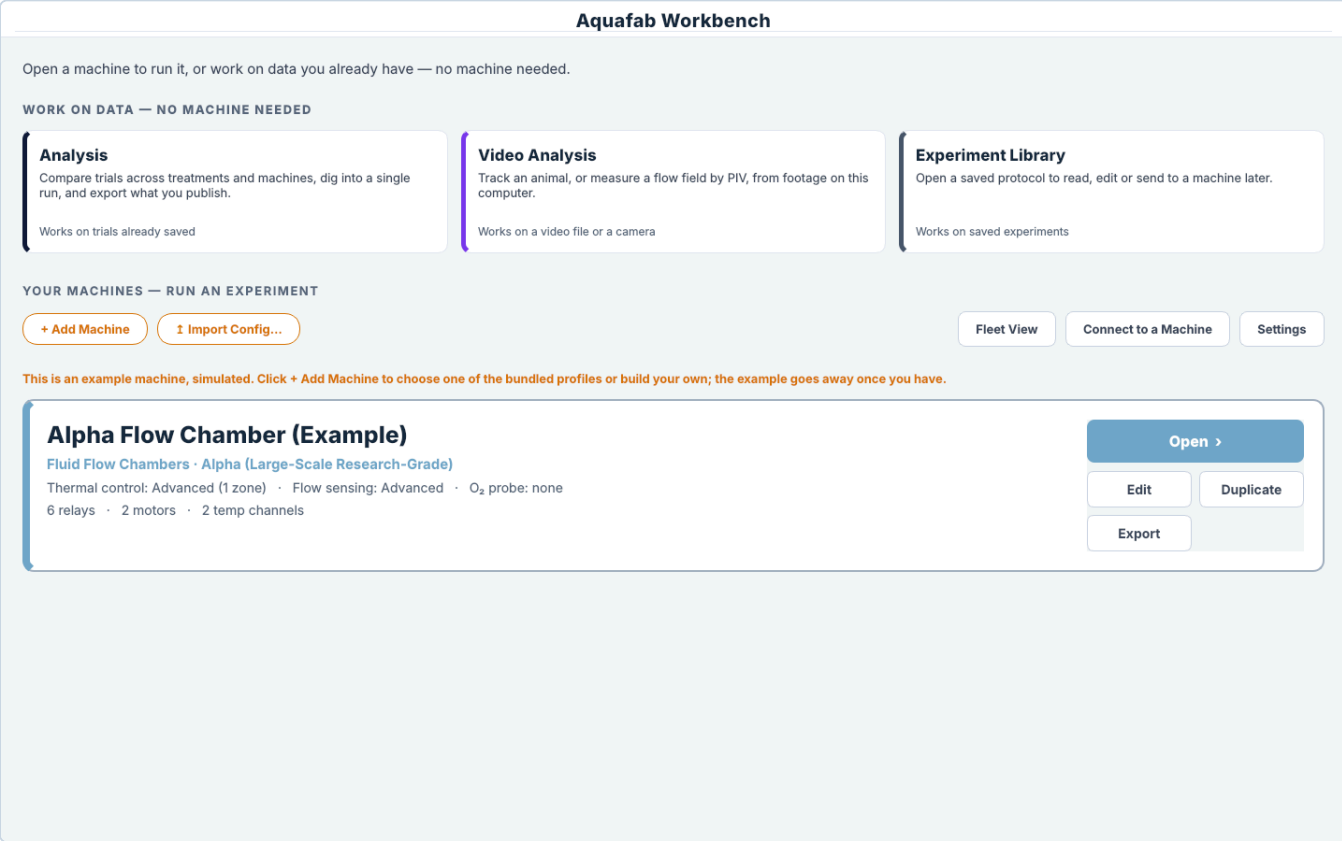
- In the last 30 days before expiry the license panel shows a warning.
- After expiry there are **60 days of grace**, with a notice and everything still working.
- Only after that are **new** trials on connected hardware refused, with the reason shown on the trial page. Simulated machines are never refused.

At no point does a license stop anything already running.

3.8 Moving to a new computer

Free the seat in the dashboard, then activate the new computer with the seat's new code. Reinstalling the operating system counts as a new computer, because the software identifies a computer by its operating system's own machine ID. Reinstalling Aquafab Workbench itself, or updating it, keeps the license.

4 A tour of the window



The Launcher, the first screen after opening the software.

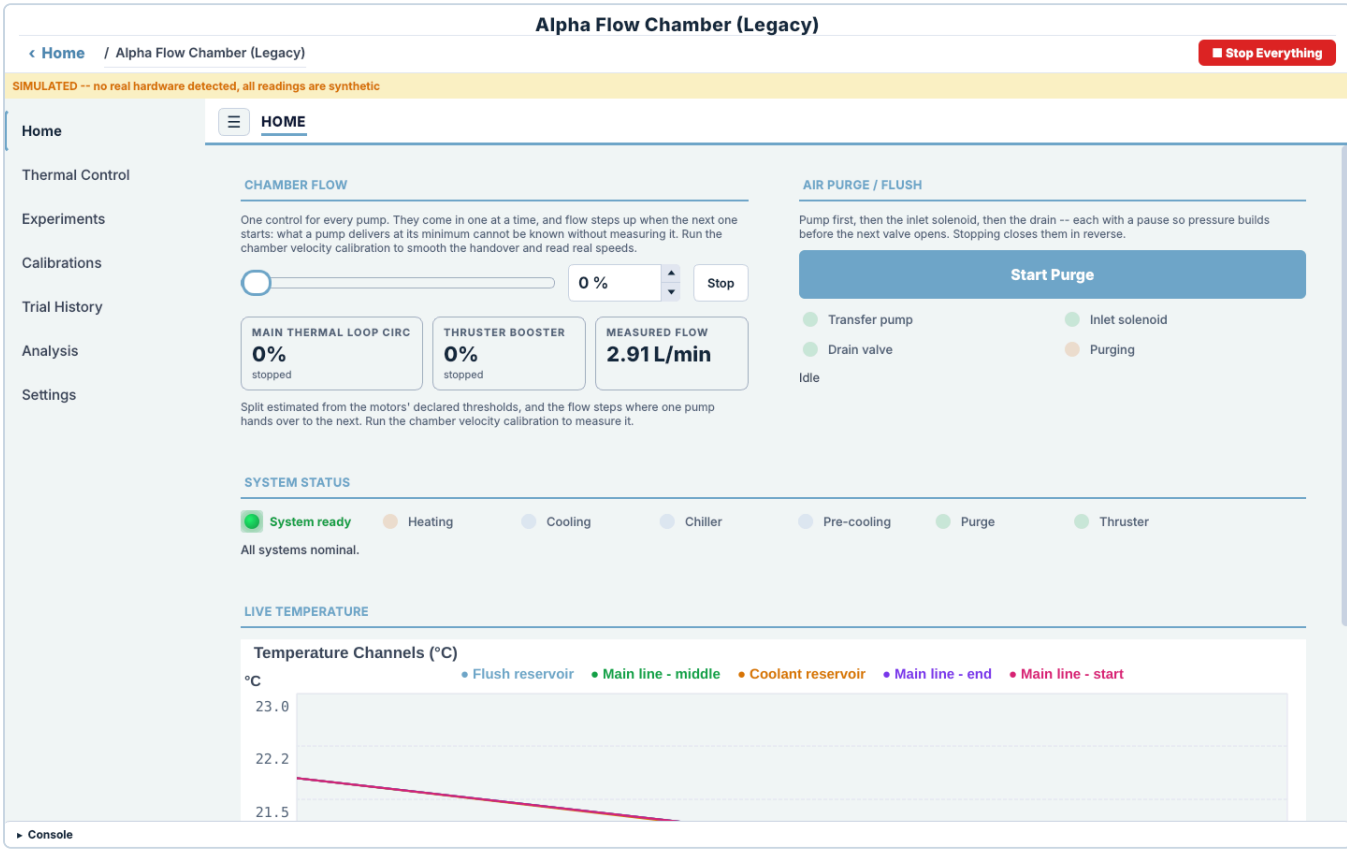
4.1 The top strip

A slim strip runs across the top of every screen.

- **The page title**, centred: the machine's name on its dashboard, or the name of the tool you are in ("Analysis", "Settings").
- **‹ Home** returns to the Launcher. It is hidden on the Launcher itself. Leaving a machine with **‹ Home** does *not* close it: a thermal zone held on one machine keeps running while you look at another, and a running trial carries on.
- **The location**, after the slash, says where you are.
- **■ Stop Everything** (red, shown once a machine is open) stops all activity on every open machine. See section 4.5.
- **Update available** (shown only when a newer version has been found) opens Settings at the update panel. Nothing installs until you press **Download and install** there (chapter 22).
- **Preview · Activate a license** (shown only without a license) opens the activation box.
- **Exit full screen** / **Enter full screen** (Raspberry Pi only) switches the window in and out of full screen, and is named for what it will do. A Pi opens the software full screen at power-on; **Exit full screen** gives back the desktop on a touchscreen with no keyboard, and **Enter full screen**

returns to it. F11 does the same from a keyboard. On a Mac or Windows computer, use the window's own full-screen control; **Exit full screen** appears only if F11 has made the window full screen.

4.2 The machine dashboard



A machine's dashboard: the section rail on the left, the Home section open, and the console drawer at the bottom. This machine is simulated.

Opening a machine shows its **dashboard**:

- **The amber SIMULATED banner** appears across the top when the machine is running in simulation. If you do not see it, the controls drive real hardware.
- **The section rail** down the left lists the machine's sections: **Home, Thermal Control, Experiments, Calibrations, Trial History, Analysis, Settings**. Every section is one click from every other, so you can watch Home while a trial runs on Experiments. Each section keeps its state when you leave it.
- ☰ at the top left of each section hides or shows the rail, to give a page more width on a small screen.
- **The section title** is under the ☰ button.
- ▸ **Console** along the bottom opens the console drawer: the machine's event log as it happens, and a command prompt (section 4.4).

4.3 Controls you will meet everywhere

4.3.1 Tooltips

Almost every control has a tooltip explaining what it does and why. Hover the mouse over a field, label or button for a second to see it. Many pages say so (“Hover any field for what it does”). The tooltip text is the quickest answer to “what does this setting mean?”.

4.3.2 Live charts

Every live chart (temperatures, oxygen, flow, swim speed) has the same toolbar underneath:

Control	What it does
Pause / **Resume**	Freezes the chart so you can read it. Recording continues underneath; nothing is lost.
Window	How much recent history is shown: 15 s to 24 hours. The value axis rescales to what is inside the window, so narrowing it zooms in on both axes.
Auto-scale	On: the value axis fits the data. Off: type fixed **Min** and **Max** values.
Show	Raw samples, or a trailing mean over 10 s, 30 s, 1 min or 5 min. A smoothed chart says so beside its title (for example *SMOOTHED — 30 S MEAN*), and the target line is never averaged.
Export…	Saves what the chart is showing (the visible window, ticked series only, smoothed if smoothing is on) as a CSV file. Needs a license.
Series tick boxes	One per trace. Untick to hide a trace.

Two mouse actions on the plot itself:

- **Left-click** the plot to drop a cursor: every visible series is read at that moment, with how long ago it was. The cursor stays with that sample as the trace scrolls.
- **Right-click** the plot for the same actions as the toolbar (pause, rescale, hide a series, export), next to the trace you are reading.

4.3.3 Stage strips

Pages with several steps (Fish Tracking, Flow Field (PIV), Swim Performance) show the steps as a numbered strip, for example **1 Scale and frame rate, 2 Choose the region, 3 Correlation settings, 4 Run and save**. Click a step to open its settings. A step turns green with a ✓ when it has what it needs, so the strip doubles as a checklist. The steps can be done in any order.

4.3.4 Segmented buttons

Short choices (“Treatment / Machine / Kind of trial / Each trial on its own”, “Intermittent-flow / Closed”) are shown as a row of buttons; the dark-filled one is selected. An option that cannot be used on this machine is greyed out, with the reason in its tooltip.

4.3.5 Experiment headers

Each experiment page opens with a short card: what the experiment measures, **Gives you:** (its results) and **Needs:** (the hardware or inputs it requires).

4.4 The console

Open the console with ▸ **Console** at the bottom of a dashboard, or **Ctrl+`**. The upper part is the machine's event log, written as things happen: a zone engaged, a trial started, a relay switched. The prompt below takes typed commands. Type `help` for the list:

Command	What it does
`status`	Everything this machine is doing right now.
`temps`	Current temperature readings.
`flow`	The flow-sensor reading, if the machine has one.
`o2`	Dissolved-oxygen readings.
`relay <name> on|off`	Switches a relay.*
`motor <name> <percent>`	Sets a motor's output.*
`zone <name> <target_c>` or `zone <name> off`	Engages or disengages a thermal zone.*
`trial` or `trial stop`	Reports on the running trial, or stops it.*
`help [command]`	Lists commands, or explains one.

* Changes the machine's state, and is written to the machine's audit log exactly as a button press would be. Relay and motor names are the ones in the profile (shown on the Commissioning checklist and in the profile editor).

The **Console** tab under **Settings** holds the same prompt.

4.5 Stop Everything

■ **Stop Everything**, at the top right whenever a machine is open (keyboard **Ctrl+.**), stops all activity on **every open machine**:

1. It asks first: *"Stop all activity on N open machine(s)?"* Press **Yes** to stop, **Cancel** to carry on.
2. Running trials stop, every motor goes to zero, thermal control disengages and relays turn off. A red banner under the top strip says what was stopped.
3. The button becomes ↻ **Undo Stop**. Pressing it (or Ctrl+. again) puts back the motor speeds, relay states and thermal setpoints from just before the stop. **A stopped trial is not resumed**; start it again.

A machine can be set to keep its flush pump running through an emergency stop, because cutting the flush can seal an animal into a chamber whose oxygen is falling. That is set per machine under **Settings → Emergency stop** (chapter 16).

> Stop Everything is a convenience, **not a substitute for a physical emergency stop**. It is software stopping software.

4.6 Keyboard shortcuts

Keys	Where	What
Ctrl+.	Anywhere	Stop Everything, or Undo Stop. Asks first.
Ctrl+`	A machine dashboard	Open or close the console.
Ctrl+Return	Experiments → Respirometry	Start, or stop, the trial.
F	Experiments → Swim Performance, during a test	Fish Fatigued.
Arrow keys	A video with a scale marker selected	Move the marker one pixel; with **Shift**, ten.

On a Mac, press **Command (⌘)** wherever this table says Ctrl.

5 The Launcher

The Launcher is the first screen, and **‹ Home** always returns to it. It says at the top: *"Open a machine to run it, or work on data you already have — no machine needed."* It has two parts.

5.1 Work on data — no machine needed

Three cards open tools that need no machine at all:

Card	What it opens
Analysis	*"Compare trials across treatments and machines, dig into a single run, and export what you publish."* Every saved trial from every machine on this installation (chapter 15). Works on trials already saved. Needs a license.
Video Analysis	*"Track an animal, or measure a flow field by PIV, from footage on this computer."* The Fish Tracking and Flow Field (PIV) pages without a machine attached (chapter 18). Works on a video file or a camera. Needs a license.
Experiment Library	*"Open a saved protocol to read, edit or send to a machine later."* Every saved experiment (chapter 18).

These are what someone at an office computer uses to analyse data recorded elsewhere.

5.2 Your machines — run an experiment

Above the machine list are five buttons:

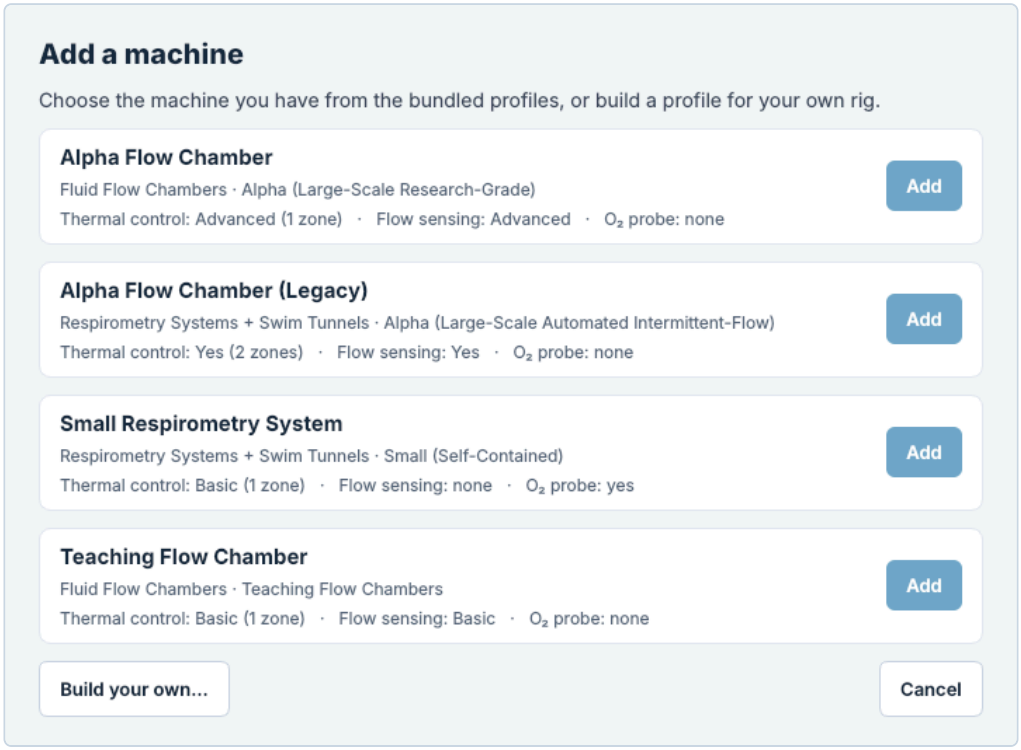
Button	What it does
+ Add Machine	Opens **Add a machine** (below): one of the bundled machines, or **Build your own…**. Needs a license.
↥ Import Config…	Adds a machine from a configuration file (`.aquafab.yaml`) exported from another computer or downloaded from the Aquafab website. It checks the file first and asks *"Import this machine?"*. An import never overwrites a machine that already exists. Needs a license.
Fleet View	Live status of every machine side by side (chapter 18).
Connect to a Machine	Reach a machine somewhere else, such as the one in the lab from a laptop at home, through Aquafab remote access (chapter 21).
Settings	App-wide settings: appearance, units, license, updates, deleted trials (chapter 17).

5.3 The example machine

A new installation lists one machine, **Alpha Flow Chamber (Example)**, with the note *"This is an example machine, simulated. Click + Add Machine to choose one of the bundled profiles or build your own; the example goes away once you have."* It is the Alpha Flow Chamber, always run by the simulator, so it never looks for hardware and never shows the *"Running in simulation"* dialog. It has no **Delete** button: it disappears by itself when the first real machine is added, and comes back if every machine is removed.

In the preview (no license) the example is the only machine listed, with the note *"Preview: the example machine, simulated. Adding your own machines, or one of the bundled profiles, needs a license."*

5.4 Adding a machine



Add a machine: the bundled profiles, or build your own.

+ Add Machine opens **Add a machine**: *"Choose the machine you have from the bundled profiles, or build a profile for your own rig."* Each bundled machine (Alpha Flow Chamber, Alpha Flow Chamber (Legacy), Small Respirometry System, Teaching Flow Chamber) has an **Add** button that puts it on the Launcher, ready to open with **Open ›**; one already listed shows **Added** instead. A licensed installation starts with no machines of its own, even on a computer that has used one before: add the one you have. Its saved calibrations and trials are found again when you do. **Build your own...** opens the setup wizard (chapter 19). For a second copy of a machine you already have, use **Duplicate**.

5.5 Machine rows

Each machine is one row, showing:

- its **name**, for example *Alpha Flow Chamber (Legacy)*;
- its **product line**, for example *Respirometry Systems + Swim Tunnels · Alpha (Large-Scale Automated Intermittent-Flow)*;
- its **capabilities**: thermal control tier and number of zones, flow sensing tier, and whether it has an O_2 probe;
- its **hardware count**: relays, motors and temperature channels.

And five buttons:

Button	What it does
Open ›	Opens the machine's dashboard. A machine already open is brought back exactly as it was left.
Edit	Opens the machine's profile in the profile editor (chapter 19).
Duplicate	Creates another machine just like this one. Asks for a **Machine ID** for the copy (lowercase with underscores) and a **display name**. Useful for a second, identical rig.
Export	Saves the machine's whole configuration to one `.aquafab.yaml` file, to install on another computer with Import Config.
Delete	For a bundled machine, takes it off the Launcher after asking (*"Remove … from this computer's machines? Its saved trials and calibrations are kept, and + Add Machine can add it back."*). For a machine of your own, deletes it after asking *"Delete …? This cannot be undone."* Saved trials are never deleted. The example machine has no Delete.

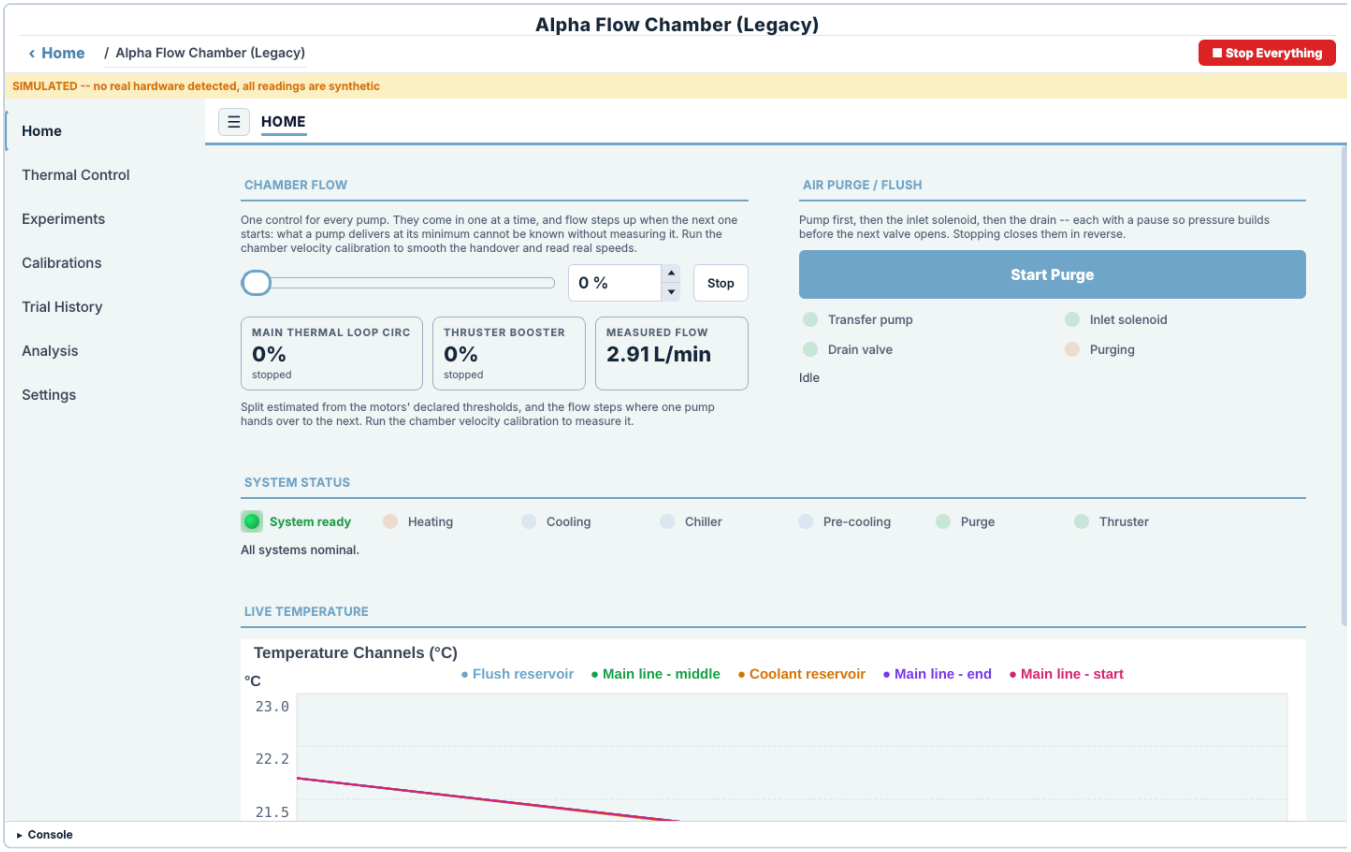
On a touchscreen nothing on the Launcher needs a double-tap: every row has an **Open ›** button.

5.6 Machine configuration downloads

Every shipped machine profile is also published on the Aquafab website as a download (**Resource Library → Machine profile**). A fresh install already includes them all, offered by **+ Add Machine**, so you need a download only to set up another computer by hand.

6 Home: live readings and manual control

Home is the first section of every machine: what the machine is doing now, and direct control of its pumps, flush and indicators. What appears depends on the machine's profile and dashboard layout (chapter 19), so a machine without pumps has no flow card and a machine without a purge group has no purge card.



Home on the Alpha Flow Chamber (Legacy), simulated: chamber flow, air purge, system status and live temperature.

6.1 Chamber flow

One control sets the water speed through the chamber.

Control	What it does
Slider	Drag to set the flow demand, 0 to 100 %.
Number box	Type an exact value. It moves the slider, and the slider moves it.
Stop	Sets the demand to zero.

Under the control is a readout for each pump (its name from the profile, its output in %, and *running* or *stopped*) and, on a machine with a flow meter, **MEASURED FLOW**.

What the slider means depends on the machine:

- **One pump, not yet calibrated:** the slider sets the pump across its usable range. Below a pump's starting threshold the motor does not turn, so the slider begins there rather than at zero. *"Slider position is motor output across its usable band, not a measured speed."*
- **Several coupled pumps, not yet calibrated:** *"One control for every pump."* The pumps come in one at a time, and flow steps up when the next one starts, because what a pump delivers at its minimum cannot be known without measuring it. The text under the readouts says the split is estimated from the motors' declared thresholds.
- **After a chamber velocity calibration** (section 13.3): the pumps are coordinated against the measured velocity surface, so the water speed follows the slider, and the card shows *"Measured chamber velocity at this setting: … cm/s."*

> Until the chamber velocity calibration has been run on your machine, the slider is a pump command, not a water speed. Run the calibration before reporting a speed.

On machines whose dashboard layout lists individual motors, a **MOTORS** card gives each one its own slider and % box instead.

6.2 Air purge / flush

Machines with a flush transfer pump, inlet solenoid and drain valve have a grouped purge control. *"Pump first, then the inlet solenoid, then the drain – each with a pause so pressure builds before the next valve opens. Stopping closes them in reverse."*

- **Start Purge** runs the sequence; the button becomes **Stop Purge**.
- Lamps show **Transfer pump**, **Inlet solenoid**, **Drain valve** and **Purging**, lit while each is active.
- The line under the lamps says which step the sequence is in, or *Idle*.

On a machine without the full valve group, the purge (if the layout shows it) is a single **Air Purge** toggle in the **MANUAL CONTROLS** card, alongside any relays the layout lists for manual switching. Indicator lamps are never offered as manual switches: each lamp has exactly one owner, the status logic below.

6.3 Flow rate

On machines whose layout lists flow sensors, a **FLOW RATE** card shows each meter's reading with a **Measurement confirmed** lamp. A slow-pulsing meter at low flow can only extrapolate from a few pulses, and below its rated minimum the figure is not a measurement at all. The lamp is lit only when the reading is a real measurement.

The unit (L/min, L/s, m^3/h, GPM, gal/h, or chamber velocity) is chosen under **Settings → Units** (chapter 17).

6.4 System status

A row of indicators mirrors the machine's panel lamps, computed in one place so that what is lit on the bench is what is lit on the screen:

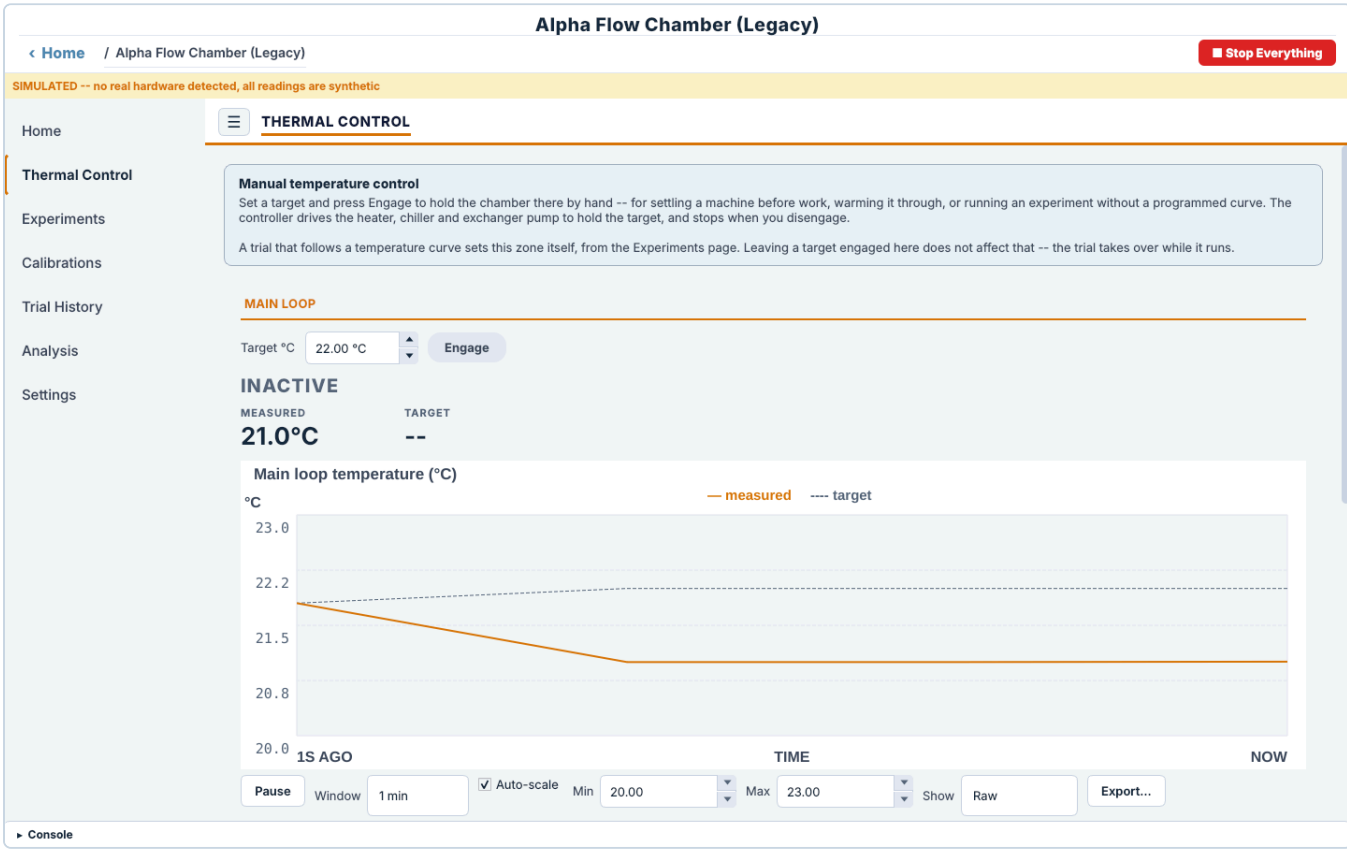
Indicator	Lit when
System ready	The machine has started and no sustained fault is present.
Heating	A thermal zone is heating.
Cooling	A thermal zone is cooling through the heat exchanger.
Chiller	The chiller compressor is running.
Pre-cooling	The coolant reservoir is being chilled ahead of a demand, so cooling capacity is ready when a curve needs it.
Purge	A purge or flush is running.
Thruster	The thruster pump is running.

Only the indicators this machine has are shown. Under the row, a line reads **All systems nominal.** or lists the current faults, for example a probe that has stopped reporting. A brief glitch does not change it; only a sustained fault does.

6.5 Live temperature

A live chart of every temperature channel the layout lists (for example *Flush reservoir, Main line - start, Main line - middle, Main line - end, Coolant reservoir*). It has the standard chart toolbar (section 4.3): pause, time window, auto-scale, smoothing, export, and a tick box per channel.

7 Thermal Control: holding a temperature by hand



Thermal Control on the Alpha Flow Chamber (Legacy), simulated, with its two zones.

Thermal Control holds a chamber at a temperature you set, for as long as you want. It is for settling a machine before work, warming it through, or running an experiment without a programmed curve. A trial that follows a temperature curve sets its zone itself from the Experiments section (chapter 8); leaving a target engaged here does not get in its way, because the trial takes over the zone while it runs.

There is one card per **thermal zone**. A small chamber has one (for example *CHAMBER LOOP*); the Alpha Flow Chamber (Legacy) has two, **MAIN LOOP** and **FLUSH LOOP**, so the flush water can be held at the chamber's temperature and a flush is not a thermal shock to the animal.

7.1 The zone card

Control	What it does
Target °C	The temperature the zone holds while engaged, in the dashboard's units (°C or °F). It is limited to the range set for this machine under **Settings → This machine's limits** (2 to 30 °C unless changed). The tooltip says the range.

Control	What it does
Engage / **Disengage**	Engage hands the zone to closed-loop control: the controller drives its heater, chiller and pumps to hold the target. Disengage turns them all off. Changing the target while engaged takes effect immediately.
Status	**INACTIVE** (disengaged), **HEATING**, **COOLING** or **HOLDING** (engaged, at target).
MEASURED	The zone's control probe, now.
TARGET	The setpoint being held, or `--` while disengaged.
Chart	Measured temperature with the target overlaid, with the standard chart toolbar.

7.2 How a zone is controlled

The strategy is set per zone in the machine's profile:

- **Simple on/off** (`simple_bangbang`): one heater or chiller relay switched against one probe. Teaching chambers and the Small Respirometry System.
- **Heat/cool with a coolant gate** (`main_loop_heat_cool`): heater and chiller, never both at once.
- **Split-range PID** (`pid_heat_cool`): one controller whose output is signed. Positive output drives heater duty; negative output drives the heat-exchanger pump, pushing pre-chilled reservoir water through the exchanger. Never both at once.

On machines with a coolant reservoir, the reservoir is supervised separately: it is pre-cooled ahead of demand (the **Pre-cooling** lamp on Home), the exchanger pump keeps a small idle trickle so its downstream probe reads moving water, and the chiller never runs against the heater.

How fast the water actually follows a change is a property of the machine. The **Thermal Step Test** (section 13.4) measures it.

7.3 Good practice

- **Engage the flush loop at the chamber's temperature** before a respirometry trial, so the first flush does not shock the animal. The Respirometry page's **Prepare system** (chapter 8) does this for you.
- **A large reservoir is slow.** Pulling a reservoir of tens of litres down by 10 °C can take over an hour. Start early, or hold a standby temperature between trials.
- **The controller holds what its probe reads.** If the control probe sits right after the heater it will see the heater, not the chamber. The Commissioning checklist (section 13.2) confirms which probe is which.

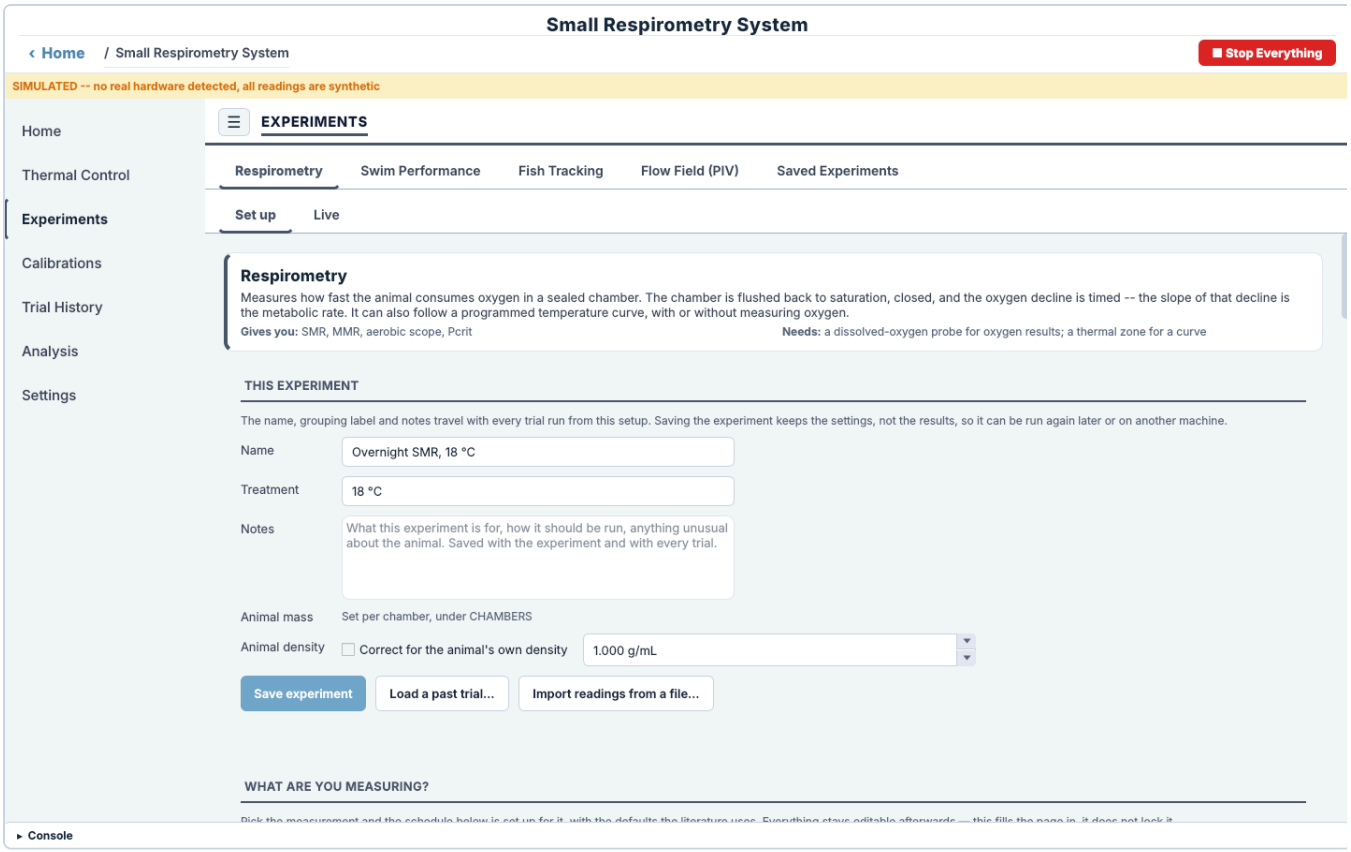
8 Experiments: Respirometry and temperature programs

The **Experiments** section of a machine holds every kind of experiment it can run, as tabs across the top: **Respirometry**, **Swim Performance**, **Fish Tracking**, **Flow Field (PIV)** and **Saved Experiments**. This chapter covers the Respirometry tab, which also runs temperature programs. The next chapters cover the others.

> *"Measures how fast the animal consumes oxygen in a sealed chamber. The chamber is flushed back to saturation, closed, and the oxygen decline is timed – the slope of that decline is the metabolic rate. It can also follow a programmed temperature curve, with or without measuring oxygen."*
>
> **Gives you:** SMR, MMR, aerobic scope, P_{crit}. **Needs:** a dissolved-oxygen probe for oxygen results; a thermal zone for a curve.

The tab has two views, switched by the strip under the tabs:

- **Set up**: everything that defines the trial, top to bottom.
- **Live**: preparing the machine, starting and watching the trial, and its results. Starting a trial switches to Live by itself.

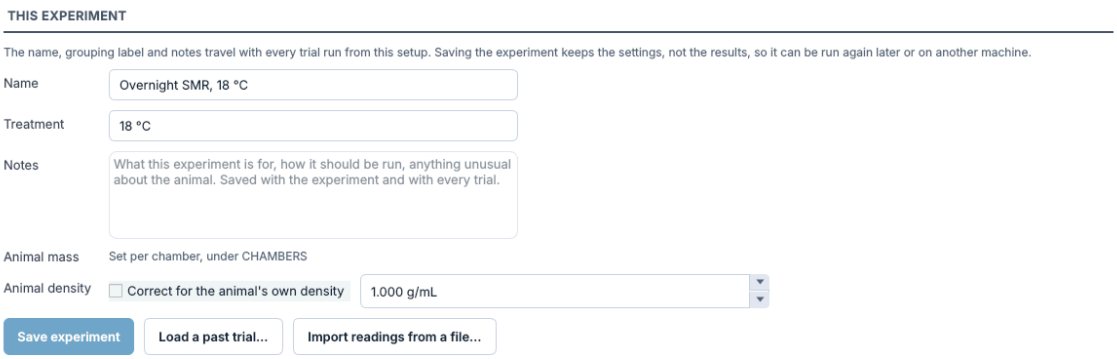


The top of the Respirometry Set up view on the Small Respirometry System, with the SMR measurement chosen. Simulated machine.

8.1 Set up

The Set up view is a column of cards. Work down it in order; every card has a one-line explanation at its top and tooltips on every field.

8.1.1 This experiment



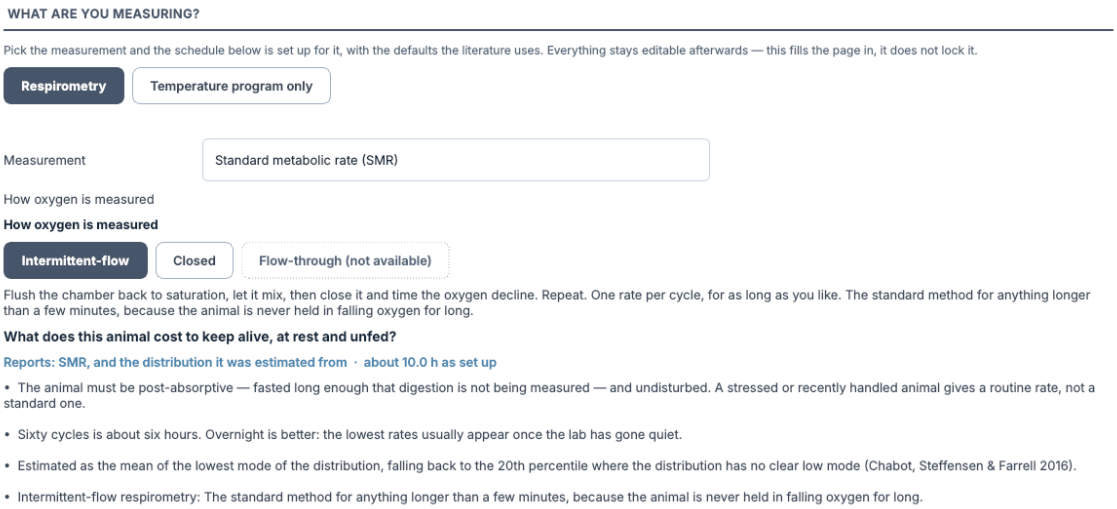
"The name, grouping label and notes travel with every trial run from this setup. Saving the experiment keeps the settings, not the results, so it can be run again later or on another machine."

Field	What it is for
Name	Names the saved experiment and every trial run from it, for example *Tidal cycle 12–24 °C, six hours*.
Treatment	A grouping label, for example *control*, *18C*, *fed*. Trials that share one are compared together on the Analysis page. Type it exactly the same way each time.
Notes	What the experiment is for, how it should be run, anything unusual about the animal. Saved with the experiment and with every trial.
Animal mass	Set per chamber, under **Chambers** below.
Animal density	Tick **Correct for the animal's own density** to enter a measured density (g/mL), used to work out how much water the animal displaces. Off, the animal is taken as 1.0 g/mL, which is fresh water and within a few per cent for most fish.

Three buttons:

- **Save experiment** keeps everything configured on this page under the name, in **Saved Experiments** (chapter 12). Results are saved separately when a run finishes.
- **Load a past trial…** picks a saved run and puts its settings back on this page, ready to run again. Only runs whose settings this page can load are listed.
- **Import readings from a file…** reads a CSV of readings over time (from another logger or system) and keeps it as an *imported* trial, marked as imported wherever it appears (section 20.11). Needs a license.

8.1.2 What are you measuring?



"Pick the measurement and the schedule below is set up for it, with the defaults the literature uses. Everything stays editable afterwards — this fills the page in, it does not lock it."

Respirometry / Temperature program only chooses which half of the page applies:

- **Respirometry**: oxygen decline measured in a closed chamber, giving MO_2 and P_{crit}. Needs a dissolved-oxygen probe.
- **Temperature program only**: follows a programmed temperature curve and records temperature and flow. No oxygen measurement, so no MO_2 or P_{crit}. This is the default, because it is the run that is ready without a probe, an animal mass and a schedule.

Switching keeps everything you set on the other half, so nothing is lost by switching back.

Measurement fills in the page for a standard measurement. Each choice explains the question it answers, what it reports, and how long it takes as set up:

Measurement	Question	Defaults it sets
Standard metabolic rate (SMR)	What does this animal cost to keep alive, at rest and unfed?	Intermittent-flow; 1 h acclimation; 60 cycles of 180 s flush, 60 s wait, 300 s measure; min R^2 0.95; SMR by MLND with a 0.2 quantile fallback. About 10 h.
Routine metabolic rate (RMR)	What is this animal using, moving about as it pleases?	30 min acclimation; 24 cycles; min R^2 0.9; reports the mean of accepted cycles.
Maximum metabolic rate (MMR), after a chase	What is the most oxygen this animal can take up?	No acclimation; 12 short cycles (30 s flush, 15 s wait, 120 s measure). Start immediately after the chase.
Aerobic scope	How much capacity does this animal have above maintenance?	30 min acclimation; 48 cycles; reports SMR, MMR, absolute and factorial scope.
Critical oxygen level (P_{crit})	How low can oxygen go before this animal cannot keep up?	**Closed** respirometry: one long 2 h closed phase. Set the oxygen floor before starting.
Recovery after exercise (EPOC)	How much oxygen does this animal owe after being pushed?	No acclimation; 48 short cycles to follow the recovery curve.
Temperature program (no oxygen needed)	What happens as the water warms, cools or cycles?	Switches to Temperature program only.
Critical swimming speed (U_{crit})	How fast can this animal swim before it fatigues?	A stepped water-speed staircase (30 % start, 10 % steps, 20 min each). Needs a variable-speed pump.
Cost of transport, and optimal speed (U_{opt})	What is the cheapest speed for this animal to travel at?	A staircase of 8 steps of 18 min with oxygen measured at each. Needs a pump and a probe.

Choices that need hardware this machine does not have say so in the list (for example *"(needs a dissolved-oxygen probe)"*). **Leave everything as it is** changes nothing.

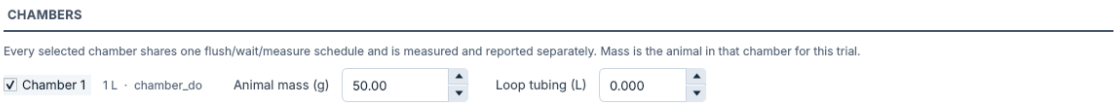
Under the measurement, a panel lists the preset's practical notes. For SMR, for example: the animal must be post-absorptive and undisturbed; overnight is better than six hours; SMR is estimated as the mean of the lowest normal distribution (MLND), falling back to the 20th percentile (Chabot, Steffensen & Farrell 2016).

How oxygen is measured (respirometry only):

- **Intermittent-flow**: *"Flush the chamber back to saturation, let it mix, then close it and time the oxygen decline. Repeat. One rate per cycle, for as long as you like."* The standard method for anything longer than a few minutes.
- **Closed**: the chamber stays closed and oxygen falls through the whole trial. Used for P_{crit}.
- **Flow-through**: shown greyed, *not available*.

If the machine has no oxygen probe, a notice says so: trials still run (the schedule and thermal program work normally), but nothing oxygen-derived is recorded, so there is no MO_2, no P_{crit} and no oxygen-triggered flushing. **Add one in Settings** opens the profile editor.

8.1.3 Chambers



"Every selected chamber shares one flush/wait/measure schedule and is measured and reported separately. Mass is the animal in that chamber for this trial."

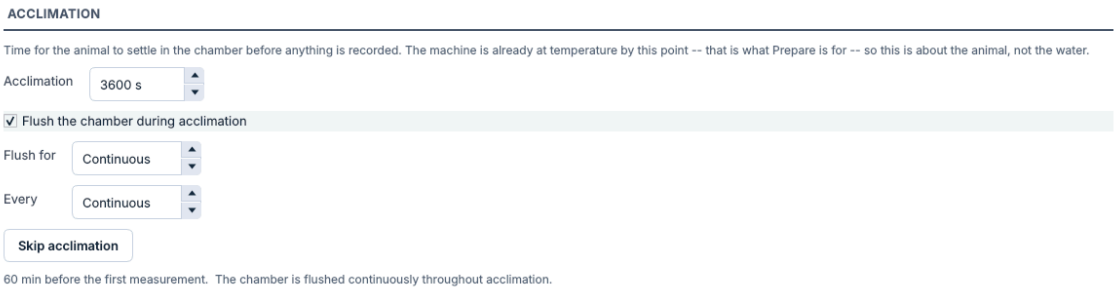
One row per physical chamber on the machine:

Control	What it does
Tick box with the chamber's name	Include this chamber in the trial. The line under it shows the chamber's volume and its oxygen probe, for example *1 L · chamber_do*.
Animal mass (g)	The animal in this chamber for this trial. Oxygen uptake is scaled by it, so it must be this animal's mass, not a nominal one.
Loop tubing (L)	Water in this chamber's recirculation loop. It is part of the closed volume the animal breathes from, so it adds to the volume every rate is multiplied by. Zero in a swim tunnel.

Under the fields a line states the **respirometric volume** used for every rate, for example *"0.953 L of water: 1.000 L chamber − 47 mL displaced by the animal (4.7% of it). Water-to-animal ratio 20x."* Check it before every trial. The chamber volume itself is set under **Settings → This machine's limits → Working section volume**.

On a machine with several chambers, **Stagger their schedules instead of running them in step** offsets each chamber by an even share of one cycle, so their flush pumps and measurement windows don't all coincide.

8.1.4 Acclimation

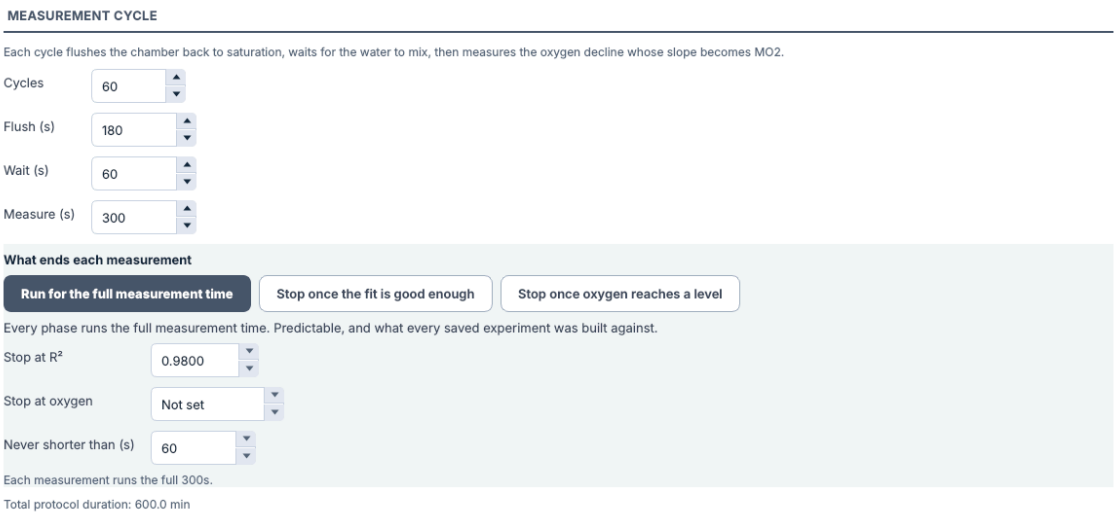


"Time for the animal to settle in the chamber before anything is recorded. The machine is already at temperature by this point – that is what Prepare is for – so this is about the animal, not the water."

Control	What it does
Acclimation (s)	How long the animal settles before the first measurement. 0 skips it.
Flush the chamber during acclimation	Keeps the chamber oxygenated while the animal settles. Untick for a closed acclimation, or where the flush would disturb the animal.
Flush for (s)	How long each acclimation flush runs. 0 means continuous.
Every (s)	How often an acclimation flush starts. Acclimation always begins with one.
Skip acclimation	Sets the time to zero, so the trial begins measuring immediately. For a rerun on a settled animal, or for testing.

The note under the fields explains the choice: a stressed animal's first cycles read high, and how long it needs is a question about the species.

8.1.5 Measurement cycle



"Each cycle flushes the chamber back to saturation, waits for the water to mix, then measures the oxygen decline whose slope becomes MO_2."

Field	What it is
Cycles	How many flush–wait–measure cycles to run.
Flush (s)	How long the flush pump runs to bring oxygen back to saturation.
Wait (s)	Time after the flush closes for the water to mix before measuring.
Measure (s)	The closed measurement phase whose oxygen slope becomes one MO_2.

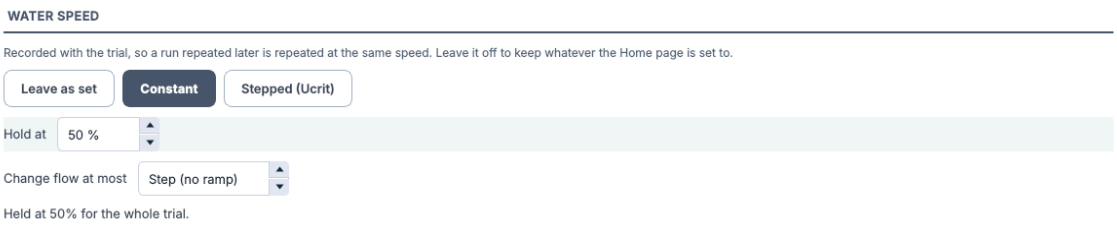
What ends each measurement chooses how a measurement phase finishes:

- **Run for the full measurement time**: every phase runs the full Measure time. Predictable, and what every saved experiment was built against.
- **Stop once the fit is good enough**: the phase stops as soon as its oxygen decline fits a straight line with **Stop at R^2** (for example 0.98).
- **Stop once oxygen reaches a level**: the phase stops when oxygen reaches **Stop at oxygen** (mg/L), a planned endpoint that completes the cycle.

With either early stop, **Never shorter than (s)** sets a minimum, so a phase cannot stop at the first good-looking fit seconds into a smooth trace. Early stops need an oxygen probe.

Total protocol duration under the card adds everything up, including acclimation.

8.1.6 Water speed

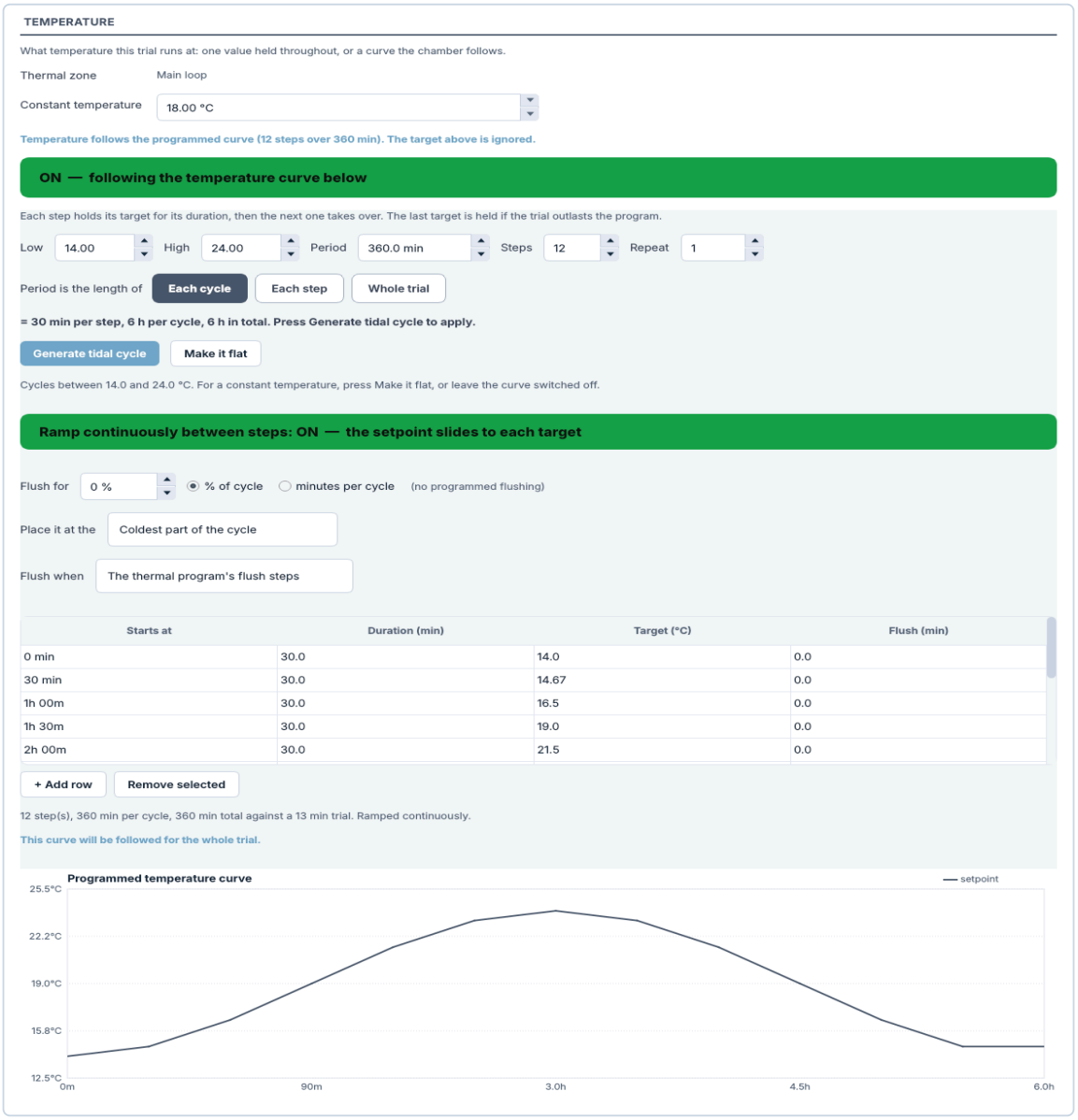


"Recorded with the trial, so a run repeated later is repeated at the same speed. Leave it off to keep whatever the Home page is set to."

- **Leave as set**: the trial does not touch the flow. Whatever Home is set to carries on.
- **Constant**: **Hold at** a set demand (%, or cm/s once the chamber velocity calibration has been run) for the whole trial.
- **Stepped (U_{crit})**: a staircase. **Start**, **Step by**, **Hold each** (minutes) and **Steps**. Used for swimming respirometry and cost of transport.

Change flow at most (%/s) limits how fast commanded flow may change. A step startles the animal, and the oxygen uptake measured just after includes the cost of that startle; on a staircase every step would carry it. 0 commands each change as a step. A line under the card says what the trial will do, for example *"Held at 50% for the whole trial."*

8.1.7 Temperature



The Temperature card with a tidal cycle generated and the curve switched on. Simulated machine.

"What temperature this trial runs at: one value held throughout, or a curve the chamber follows."

Control	What it does
Thermal zone	The zone the trial drives. With one test chamber it is fixed (for example *Main loop*); a machine with several chambers offers a choice.
Constant temperature	Held for the whole trial, unless the curve below is switched on.

Control	What it does
Curve button (**ON** / **OFF**)	A large button under **Constant temperature**. **ON — following the temperature curve below** (filled green): the zone setpoint follows the program table for the length of the trial. **OFF — holding one temperature** (outlined): the trial holds the constant temperature and ignores the table. Press it to switch.

Generating a curve. The generator fills the table with a smooth rise and fall. Changes to these fields take effect when you press **Generate tidal cycle**:

Field	What it is
Low, **High**	The two extremes of the cycle.
Period (min)	A length of time, to a tenth of a minute. What it measures is set by **Period is the length of**, below.
Steps	How many holds the smooth curve is cut into. More gives a smoother approximation and a longer table.
Repeat	How many times the whole program runs back to back.
Period is the length of	**Each cycle** (the default): one full rise and fall. **Each step**: one hold or ramp, so a cycle is the period times **Steps**. **Whole trial**: every cycle together, so each cycle is the period divided by **Repeat**. A line underneath works out the rest, for example *"= 25 s per step, 5 min per cycle, 60 min for all 12 cycles."* Switching between them keeps the same cycle and changes the number shown.
Generate tidal cycle	Fills the table with a rise and fall over one period, sampled from a sine so the temperature changes most slowly at the extremes, the way a real pool behaves.
Make it flat	Sets High to Low, for a long hold at one temperature. (Simpler: leave the curve off and use the constant temperature.)
Ramp continuously between steps	A large button. **ON** (filled green): the setpoint slides from each target to the next. **OFF** (outlined): each target is held, then jumps. Press it to switch.

Programmed flushing (for long thermal cycles on a closed chamber):

Field	What it is
Flush for	How much of each generated cycle runs the flush pump, as **% of cycle** or **minutes per cycle**. The equivalent in the other unit is shown beside it. A long thermal cycle on a closed chamber drives the animal hypoxic; a real pool is refilled, not just re-tempered.
Place it at the	Where the flush windows sit: *Coldest part of the cycle* (the tide returning), *Warmest part*, *Start of each cycle*, *End of each cycle*, or *Spread evenly through the cycle*.
Flush when	*The thermal program's flush steps*, *When oxygen drops below a setpoint* (with a mg/L value; needs a probe), or *Either, whichever asks first*.

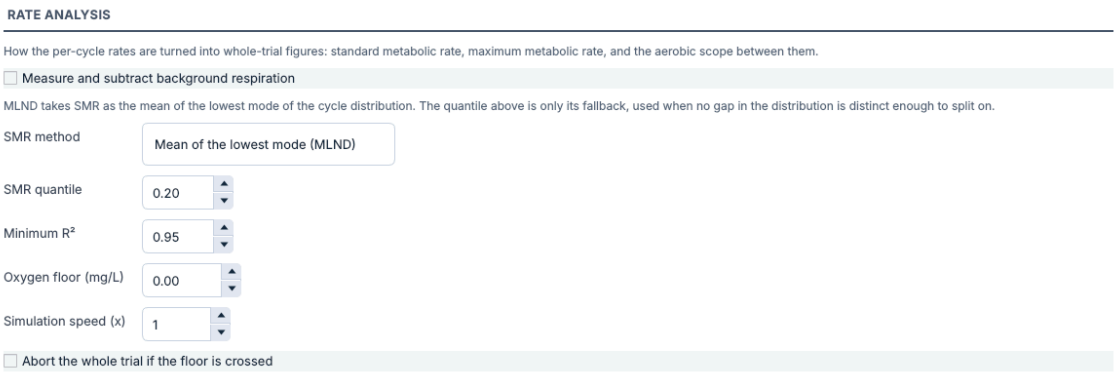
Flushing is suspended automatically during any respirometry measurement window.

The program table lists every step: **Starts at**, **Duration (min)**, **Target (°C)** and **Flush (min)**. Edit any cell; **+ Add row** and **Remove selected** change the steps. Each step holds its target for its duration, then the next takes over; the last target is held if the trial outlasts the program. A preview chart draws the curve.

Under the table, the page checks the curve against the machine:

- **A rate warning**, if a step is steeper than the machine's measured heating or cooling rate (from the Thermal Step Test): *"This curve is steeper than measured: … The water will lag behind it."*
- **A reservoir note**, on machines with a chilled reservoir: how cold the reservoir must be before starting for the curve's hardest cooling step, for example *"Chill the reservoir to about … °C before starting."*

8.1.8 Rate analysis

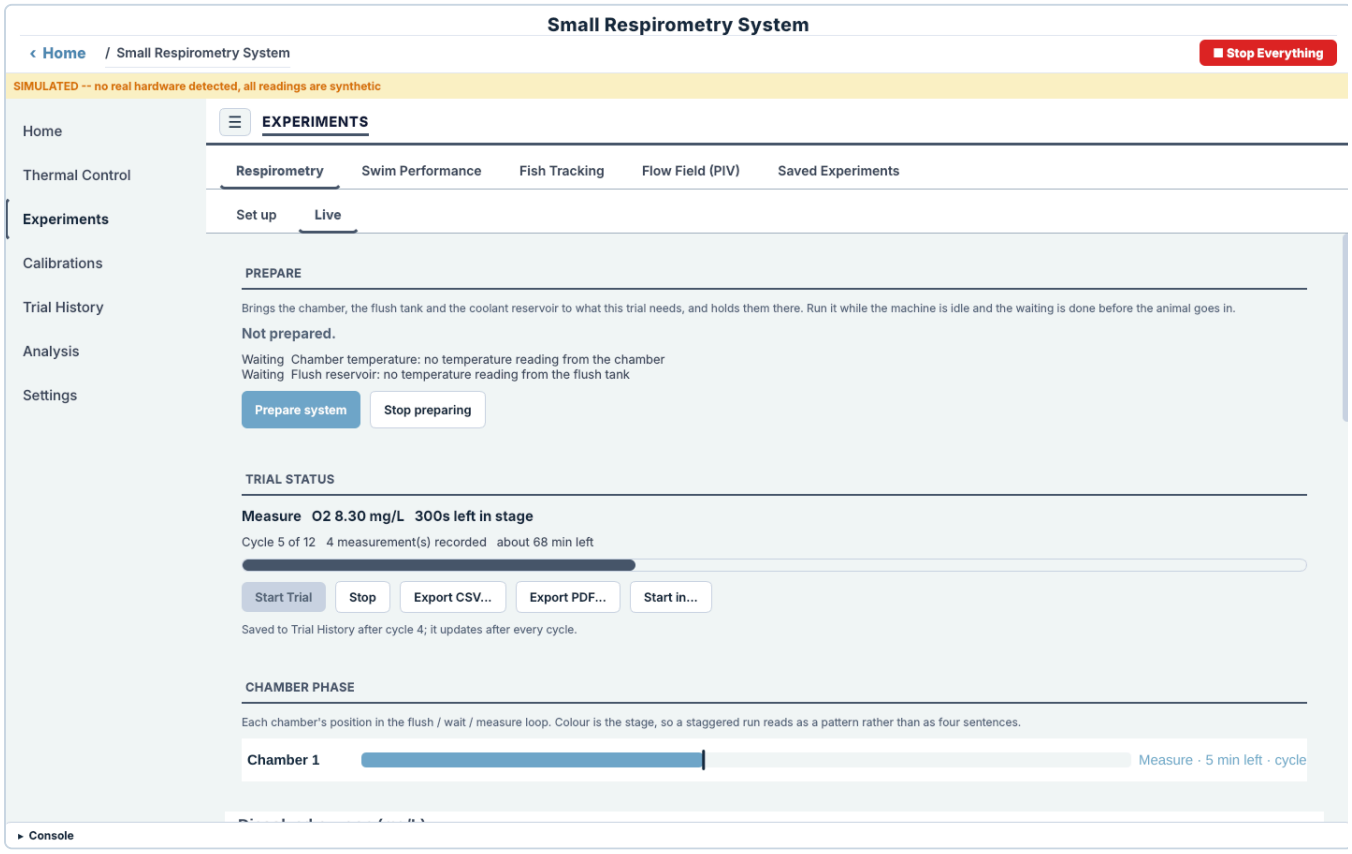


"How the per-cycle rates are turned into whole-trial figures: standard metabolic rate, maximum metabolic rate, and the aerobic scope between them."

Field	What it does
Measure and subtract background respiration	Runs one extra blank cycle first and subtracts its rate from every measurement (Svendsen et al. 2016).
SMR method	**Low quantile of all cycles**, or **Mean of the lowest mode (MLND)**. Chabot, Steffensen & Farrell (2016) accept either; the quantile is simpler and reproducible, MLND is better when the animal was intermittently active.
SMR quantile	The quantile used (0.2 or 0.25 are recommended). With MLND it is the fallback when the distribution has no distinct low mode.
Minimum R^2	Cycles whose oxygen decline fits a line worse than this are recorded but excluded from SMR and MMR. A poor fit usually means the animal moved or the chamber had not mixed. 0 accepts everything.
Oxygen floor (mg/L)	Safety floor: if oxygen falls below this during a closed phase, the chamber is flushed immediately instead of being driven further into hypoxia. 0 disables it. **Always set it for a closed or P_{crit} run.**

Field	What it does
Abort the whole trial if the floor is crossed	Off: the chamber is flushed and the trial continues. On: the run ends there.
Simulation speed (x)	Simulated machines only: runs the trial clock up to 300× faster so a protocol can be previewed without waiting.

8.2 Live



The Live view during a simulated SMR trial: trial status, chamber phase and the oxygen trace.

8.2.1 Prepare

"Brings the chamber, the flush tank and the coolant reservoir to what this trial needs, and holds them there. Run it while the machine is idle and the waiting is done before the animal goes in."

The card lists what is being waited for, each line marked *Waiting* or ready, for example *"Chamber temperature: at 21.0 °C, needs 18.0 °C"* and *"Flush reservoir: at 21.0 °C, needs about 18.0 °C – flushing with untempered water is a thermal shock to the animal..."*.

Button	What it does
Prepare system	Drives the chamber, flush tank and reservoir to this trial's conditions and holds them. Safe with nothing in the chamber.

Button	What it does
Stop preparing	Asks first, then lets the heaters and chiller go back to what they held before. Nothing is damaged, but a reservoir part-way down has to be pulled down again.
Start anyway	Starts the trial without waiting. The first cycles will be measured on a chamber that has not arrived, so treat them accordingly.

Start Trial refuses to start an unprepared machine. It says what is being waited for and begins preparing; press **Start anyway** to override.

8.2.2 Trial status

The status line shows the stage (*Acclimation*, *Flush*, *Wait*, *Measure*, *Complete*), the latest oxygen reading and the time left in the stage. Under it: the cycle number, measurements recorded, time remaining, and a progress bar.

Button	What it does
Start Trial	Starts the trial (**Ctrl+Return**). On connected hardware, needs a license.
Stop	Stops the trial (**Ctrl+Return** while running). Completed cycles are kept, and the trial is saved as it stands.
Export CSV...	Writes the per-cycle results plus the full oxygen trace. Needs a license.
Export PDF...	A printable report: settings, derived statistics, the cycle table and the traces. Needs a license.
Start in...	Arms the trial to start by itself after a delay you type in minutes. The button becomes **Cancel (...)** with the time left; press it to disarm. Completion and any oxygen-floor abort are written to the machine's activity log.

A trial saves itself. After the first cycle it is written to **Trial History**, with every cycle, the settings and the notes, and it is written again after every cycle after that. The line under the buttons says how far it has got: *"Saved to Trial History after cycle 4; it updates after every cycle."*, then *"Trial saved to Trial History (12 cycle(s))."* when the run ends or you press **Stop**. While it runs, its Trial History card says *Still recording*. If the computer stops before the run ends (a power cut, a crash), the cycles measured up to then are kept, and the next time the software starts the card says *Interrupted* instead (chapter 14).

8.2.3 Chamber phase

"Each chamber's position in the flush / wait / measure loop. Colour is the stage, so a staggered run reads as a pattern rather than as four sentences." One bar per chamber, with its stage and time left.

8.2.4 Charts

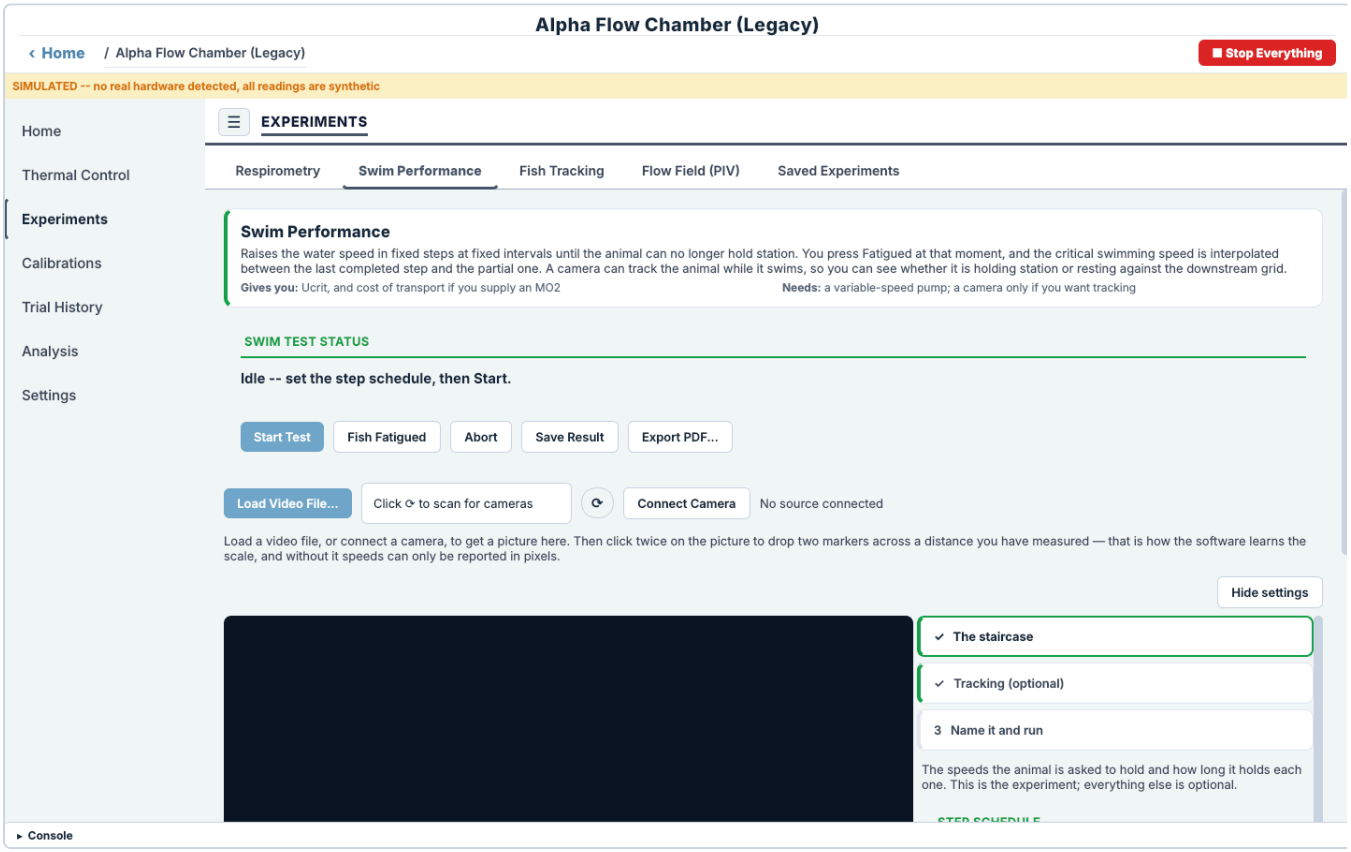
- **Dissolved oxygen (mg/L)**: the live oxygen trace, falling through each measurement and recovering with each flush.

- **MO_2 (mg O_2/kg/h)**: one point per completed cycle.
- **Chamber temperature**: the measured temperature.
- **Programmed curve** (when a curve is in use): the curve the trial is following with the measured temperature drawn over it, and beneath it how hard the heater, exchanger pump and chiller are working to hold it.
- **Oxygen uptake vs water speed** (when the speed changes during the trial): one point per measured cycle, rate against the speed it was measured at. No curve is fitted, because the shape is the question being asked. A hollow point is a cycle whose fit fell below the R^2 threshold. Until the chamber velocity calibration has been run the speeds are pump output, so no cost of transport can be computed.

8.2.5 Derived statistics

Updated as cycles complete: **SMR** (with the method), **MMR**, **aerobic scope** (absolute and factorial), **mean regression R^2** over the cycles, and **P_{crit}** where the oxygen range allows a breakpoint fit (with the fit's R^2 and the number of points).

9 Experiments: Swim Performance (U_{crit})



Swim Performance on the Alpha Flow Chamber (Legacy), simulated, before a test.

> *"Raises the water speed in fixed steps at fixed intervals until the animal can no longer hold station. You press Fatigued at that moment, and the critical swimming speed is interpolated between the last completed step and the partial one. A camera can track the animal while it swims, so you can see whether it is holding station or resting against the downstream grid."*
>
> **Gives you:** U_{crit}, and cost of transport if you supply an MO_2. **Needs:** a variable-speed pump; a camera only if you want tracking.

The tab appears only on machines with a variable-speed pump. The Small Respirometry System, which has only a flush pump, does not show it.

9.1 How U_{crit} is calculated

The stepped-velocity test of Brett (1964). Speed is held for a fixed interval T_i, then raised by a fixed step U_i. When the animal fatigues:

> $U_{crit} = U + (T / T_i) \times U_i$

where U is the highest speed held for a full interval and T is how long the animal lasted at the final, partial step. Step size and interval are settings, because published protocols tailor both to the species and the question.

The motor is commanded in per cent of its output, not in cm/s. To get U_{crit} in cm/s, the software needs to know how many cm/s one per cent produces. That is the **cm/s per % output** setting below, and it is measured, not guessed (section 13.3).

9.2 The page

Under the **status** card is the video source (section 10.1), then two columns. On the left are the **video** (for optional tracking), the **water speed** chart and the **Result** card. On the right is a stage strip with three stages. The whole page scrolls as one, so scroll down to see more of the picture and the charts below it. When the stage strip is longer than the window, it also scrolls on its own: point at it and scroll.

Hide settings, above the right-hand column, hides the stage strip so the picture and the charts take the full width. **Show settings** brings it back. The page remembers which you chose.

9.2.1 Swim test status and buttons

Control	What it does
Status line	*"Idle – set the step schedule, then Start."* While running: the stage, the current speed (cm/s if calibrated, otherwise % output) and the seconds left in the step, then the number of completed steps and the elapsed time.
Start Test	Starts the staircase at the start output. On a connected machine this needs a license (chapter 3).
Fish Fatigued	Press the moment the animal can no longer hold station. U_{crit} is interpolated from how far into the current step this happens, and the motor stops. Keyboard: **F**.
Abort	Stops the test and the motor without a result: *"Test aborted – motor stopped."*
Save Result	Saves the finished test to Trial History. A finished test is not saved until you press this, so that an aborted or mistaken run is not kept.
Export PDF…	A one-page report of the protocol, the result and the speed chart. Needs a license.

If the speed reaches **Maximum output** before the animal fatigues, the test holds that speed and keeps timing; press **Fish Fatigued** when it stops holding station, or **Abort**.

9.2.2 Stage 1: The staircase

"The speeds the animal is asked to hold and how long it holds each one. This is the experiment; everything else is optional."

Field	What it does
MO_2 at speed (mg/kg/h)	Oxygen uptake at the speeds being swum, if you are measuring it alongside. Used to report cost of transport at each completed step and the speed that minimises it (U_{opt}). 0 skips both.
Motor	Which variable-speed pump the staircase drives.
Start output (%)	The first speed. Default 20 %.
Step size (%)	How much each step adds. Default 10 %.
Interval T_i (s)	How long each speed is held. Default 1200 s (20 minutes). Published protocols use anywhere from a couple of minutes to an hour.
Maximum output (%)	The staircase never commands more than this.
cm/s per % output	How many cm/s of water speed one per cent of motor output produces. Measure it with the Chamber Velocity calibration (section 13.3). Left at 0, results stay in % output.
Fish length (cm)	Used to report U_{crit} in body lengths per second. 0 skips it.
Simulation speed (x)	Runs the test clock up to 300 times faster than real time. Simulated machines only; it lets you rehearse a two-hour staircase in a minute.

9.2.3 Stage 2: Tracking (optional)

"Connect a camera above to follow the animal while it swims. The track is recorded with the trial, and the measured speed is charted under the commanded water speed. A swim test runs perfectly well without this."

1. Connect a camera with the controls over the video (**Connect Camera**; ⟳ rescans for cameras). A video file can be loaded instead to try the settings.
2. **Capture background** with the tank empty, before the animal goes in. The detector finds whatever is different from this frame.
3. Tick **Track the animal during the test**. It needs a camera and a captured background.
4. Optionally set the **Detection** and **Scale & frame rate** cards. They are the same controls as on the Fish Tracking tab and are explained there (sections 10.3 and 10.2). Without a scale the speed trace is in pixels per second rather than cm/s.

Why it is worth doing: U_{crit} rests on a judgement about when the animal stopped holding station. A saved test with tracking records how many detections were made, so the evidence for that call is kept with the result.

9.2.4 Stage 3: Name it and run

The **This experiment** card (name, treatment, notes, **Save experiment**), as described in section 8.1.1. Saving keeps the staircase settings for reuse; it does not save a result.

9.3 The Result card

After **Fish Fatigued**, the Result card shows:

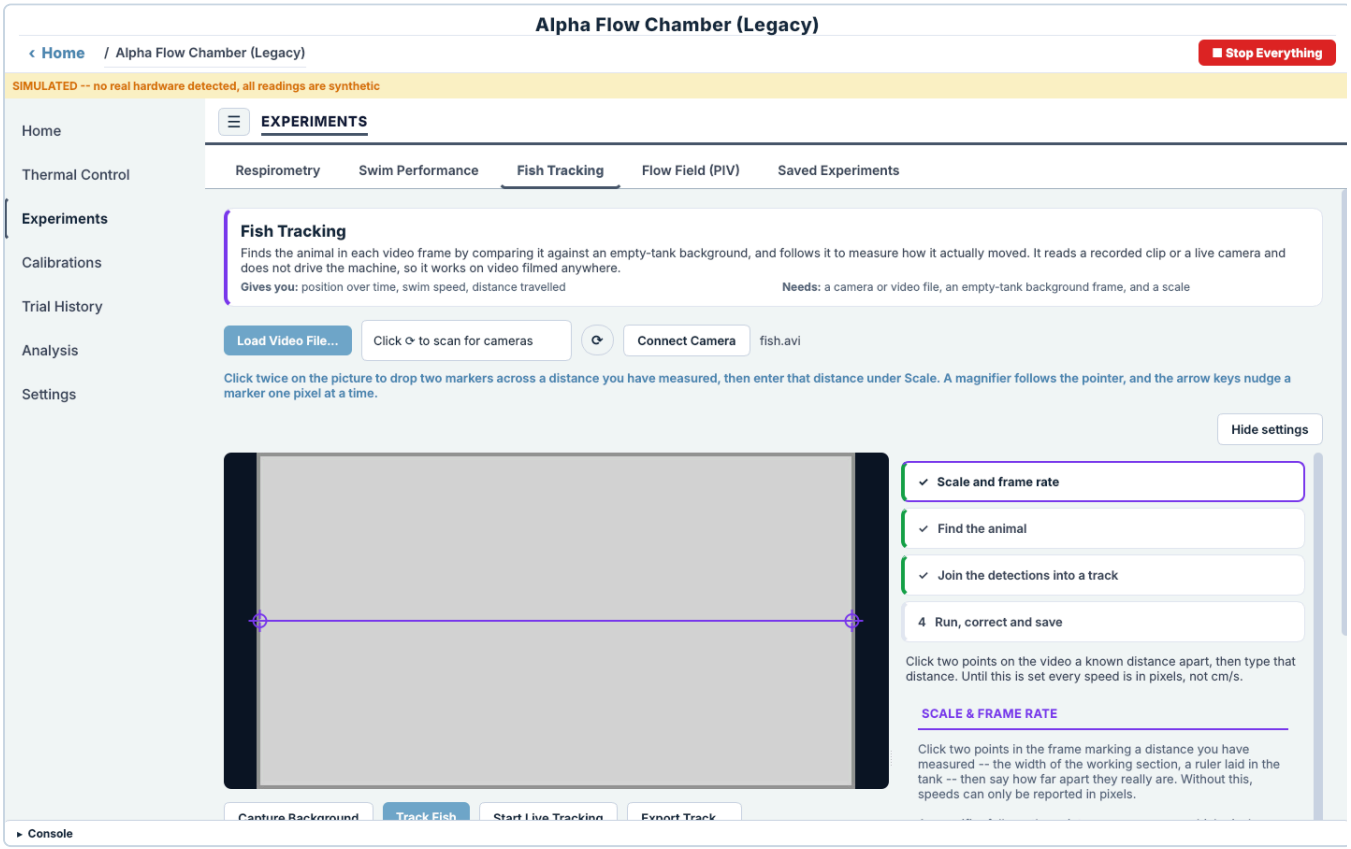
- $\mathbf{U_{crit}}$ in cm/s (if calibrated), in body lengths per second (if a fish length was entered) and in % output;
- the number of completed steps and the seconds spent at the final speed;
- *"No velocity calibration set – measure cm/s per % on the Flow Calibration page."* if cm/s per % was left at 0;
- if an MO_2 was given, the cost of transport (mg O_2 per kg per metre) at each completed step, and $\mathbf{U_{opt}}$, the speed with the lowest cost of transport.

Press **Save Result** to keep it. The saved trial's summary reads, for example, *"Ucrit 42.50 cm/s (55.0% output), 3.40 BL/s, tracked (812 detections)"*.

9.4 Good practice

- **Calibrate the velocity first.** A U_{crit} in % output cannot be compared between machines, or even between two pumps on the same machine.
- **Acclimate the animal in the tunnel** at a low speed before starting. The staircase starts at the start output immediately.
- **Watch the animal, not the screen.** Fatigue is usually defined as the animal resting against the downstream grid and not resuming after gentle encouragement. Decide your criterion before the trial and apply it the same way every time.
- **Blocking.** A large animal in a small working section swims in faster water than the nominal speed. The Fish Tracking tab has a solid-blocking correction (section 10.5).

10 Experiments: Fish Tracking



Fish Tracking with a clip loaded and the scale stage open. The clip is a synthetic test video made for this guide.

> *"Finds the animal in each video frame by comparing it against an empty-tank background, and follows it to measure how it actually moved. It reads a recorded clip or a live camera and does not drive the machine, so it works on video filmed anywhere."*
>
> **Gives you:** position over time, swim speed, distance travelled. **Needs:** a camera or video file, an empty-tank background frame, and a scale.

Fish Tracking needs a license. Without one, the tab shows the preview cover (section 3.1).

The same page, without a machine, is on the Launcher's **Video Analysis** card (chapter 18). Use that to analyse footage on an office computer; its runs are saved under *video_analysis* in the Launcher's **Analysis** card rather than in a machine's Trial History.

10.1 The video source

The controls above the picture are shared by every page that reads video (Swim Performance, Fish Tracking, Flow Field, Chamber Velocity calibration):

Control	What it does
Load Video File…	Opens a clip: `.avi`, `.mp4`, `.mov`, `.mkv` or `.m4v`. The dialog opens in the folder you last used.
Camera list	The cameras this computer can see.
⟳	Rescans for connected cameras, after plugging one in.
Connect Camera	Starts the live picture from the chosen camera.
Pause	Freezes the preview on the current frame. The camera stays connected, so resuming is instant.
Disconnect	Releases the camera so another page, or another program, can use it.

Until something is connected the picture says *"No source connected"* and explains what to do. Only one page can hold a camera at a time; **Disconnect** on one page before connecting on another.

The page then has four stages in a strip on the right. A stage shows a tick when it has what it needs. The whole page scrolls as one, the picture and the stages together, and a stage strip longer than the window also scrolls on its own.

Hide settings, above the stage strip, hides it so the picture and the charts take the full width; **Show settings** brings it back. Swim Performance and Flow Field have the same button, and each page remembers its own choice.

10.2 Stage 1: Scale and frame rate

"Click two points on the video a known distance apart, then type that distance. Until this is set every speed is in pixels, not cm/s."

1. Click once on the picture at one end of a distance you have measured: the width of the working section, or a ruler laid in the tank. Click again at the other end. Two markers appear with a line between them.
2. Type the real distance in **Distance between markers** (cm).
3. Press **Apply scale from markers**. The status line changes from *"Not calibrated – speeds are reported in pixels, not cm/s."* to the scale in pixels per centimetre.

Control	What it does
Distance between markers	The real-world distance between the two markers, measured on the machine.
Frame rate	Leave at 0 to use the frame rate the camera or file reports. Set it only when that is wrong: some USB cameras report a nominal rate rather than the one they deliver, and every speed scales directly with it.
Magnifier	Shows an enlarged view of the pixels under the pointer, in the corner away from it, so you can see exactly which pixel you are on. Turn it off once the markers are placed.

Control	What it does
Zoom	How much the magnifier enlarges: 4× to 16×.
Clear markers	Removes both markers so they can be placed again.

To fine-tune a marker, drag it, or click it and use the **arrow keys** to move it one pixel at a time (**Shift** + arrow for ten).

> **Accuracy here scales every speed the software reports.** Use the longest distance you can measure. A 2 cm reference with a one-pixel error is many times worse than a 30 cm reference with the same error.

If you set the scale after tracking, the result is marked *"Scale applied after this result"*: track again to get it in cm/s.

10.3 Stage 2: Find the animal

"Capture an empty-tank background first, then raise the threshold until only the animal is found."

Press **Capture Background** (under the picture) while the frame shows the tank **without** the animal: before it goes in, or a frame of the clip where it is out of view. The detector finds whatever is different from this frame, so the background must match the lighting and camera position of the run.

The **Detection** card:

Field	What it does
Threshold	How different from the background a pixel must be to count as animal (1–255, default 25). Lower picks up more, including noise; higher risks losing the animal.
Min blob area (px²)	Ignore detected shapes smaller than this. The main defence against specks of noise or debris being tracked as a fish.
Limit max blob area / **Max blob area (px²)**	Ignore shapes larger than this, such as a shadow or a lighting change that covers a large part of the frame.
Blur kernel size	Smooths the image before comparing it with the background, so fine texture and sensor noise do not register as movement. 1 turns it off.
Morphology kernel size	Cleans up the detected shape: removes stray specks and closes small holes, so the animal reads as one solid blob. 1 turns it off.
Frame stride	Analyse every Nth frame. Faster on a long clip, at the cost of time resolution. Speeds are corrected for it.

10.4 Stage 3: Join the detections into a track

"Joins the per-frame detections into a continuous track, which is what speed is measured from."

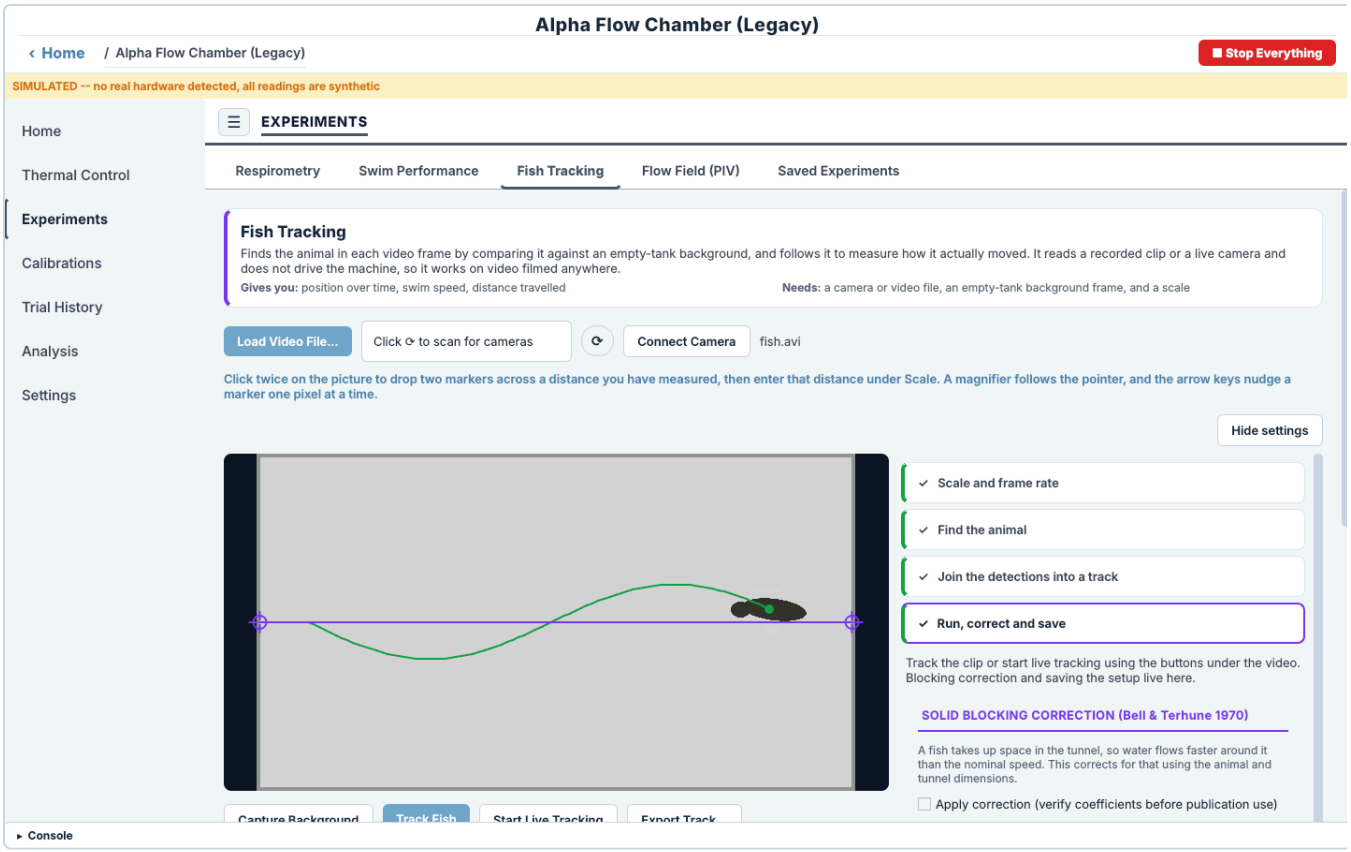
Field	What it does
Multiple fish (track every animal in view)	Off: the largest detection in each frame is taken as the animal. On: every detection is followed, and the search range decides which belongs to which track.
Search range (px, multi-fish)	The furthest an animal is assumed to move between frames. Too large and tracks swap between animals; too small and a fast animal is lost.

10.5 Stage 4: Run, correct and save

The buttons under the picture:

Button	What it does
Track Fish	Tracks the loaded clip from start to end, with a progress bar and **Cancel**. The track is drawn on the picture, the speed is charted, the **Swim track** box gives a summary, and the run is **saved to Trial History automatically**.
Start Live Tracking / **Stop Live Tracking**	Tracks and plots swim speed continuously from the running camera or clip.
Export Track…	Saves the track as CSV. Needs a license.

The **Swim track** summary reads, for one animal, *"Tracked 400 frames. Mean speed: 6.12 cm/s Max speed: 14.80 cm/s"*, and for several, one line per animal. With **Multiple fish** on, the speed chart is hidden and the summary lists each animal.



A tracked clip: the track drawn on the frame, and the swim speed chart. Synthetic test video.

The exported CSV has one row per frame: `frame`, `time_s`, `x_px`, `y_px`, `speed_cm_s`, and `corrected_speed_cm_s` when the blocking correction is on. With several animals the columns are `animal_id`, `frame`, `time_s`, `x_px`, `y_px`.

10.5.1 Solid blocking correction (Bell & Terhune 1970)

"A fish takes up space in the tunnel, so water flows faster around it than the nominal speed. This corrects for that using the animal and tunnel dimensions."

Field	What it is
Apply correction	Adds corrected speeds alongside the raw ones. The label says *verify coefficients before publication use*, and that is meant.
Fish length, **Fish thickness** (cm)	Measured on the animal.
Fish cross-section (cm^2)	The animal's largest cross-sectional area.
Tunnel cross-section (cm^2)	The working section. Taken from the machine's profile when it has a measured value; otherwise a placeholder of 50 cm^2, and the tooltip tells you to measure the working section and enter it in **Settings** (section 16.2).

Field	What it is
Blockage exponent	1.0 (linear) or 1.5 (3/2-power). Both appear in secondary sources; check Bell & Terhune (1970) directly before publishing.

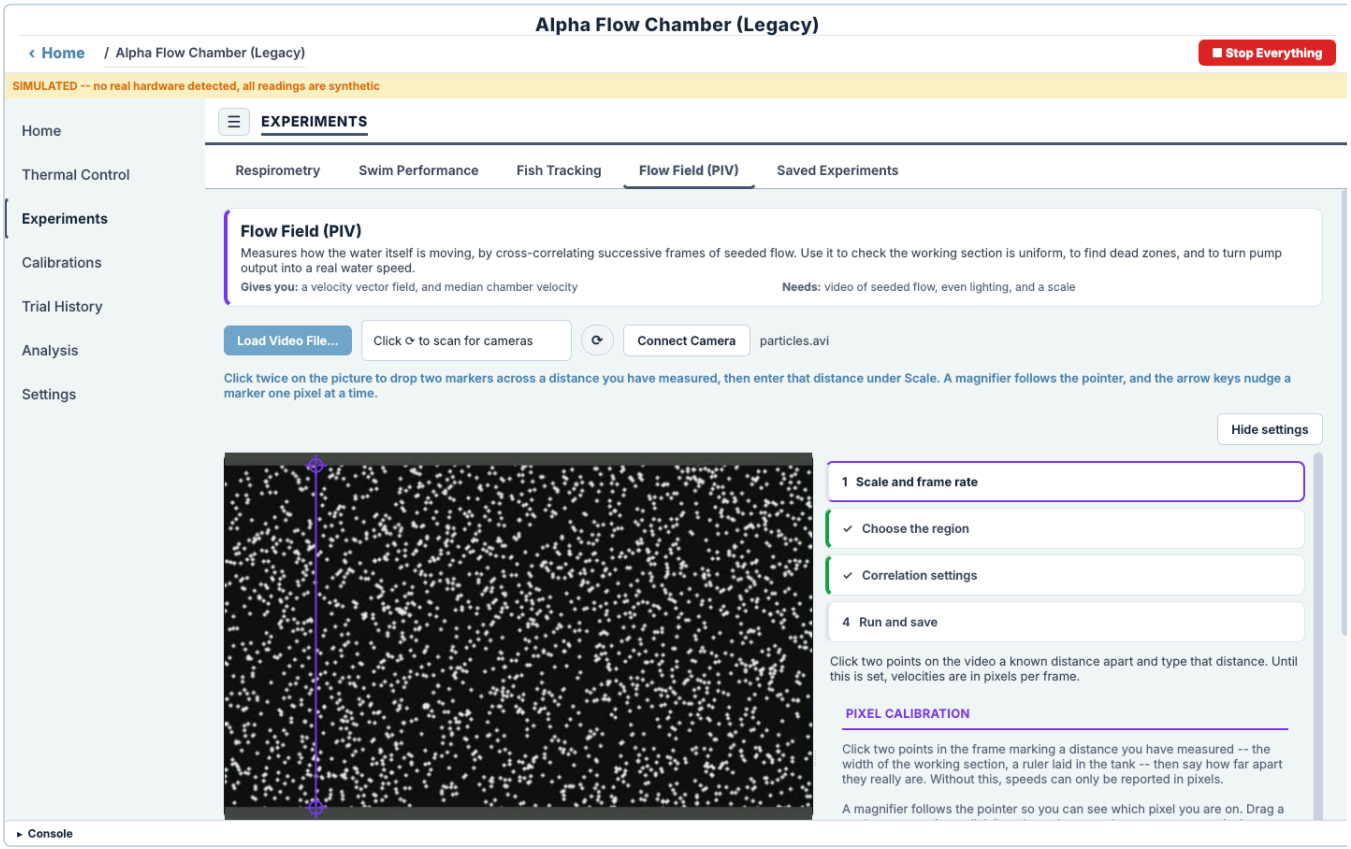
10.5.2 This experiment

Name, treatment, notes and **Save experiment** (section 8.1.1). A tracking setup keeps the detection, linking and blocking settings, so a setup that took a few videos to settle can be reused. These are settings for reading video, so they load on any machine.

10.6 Tips for good tracking

- **Light evenly, from below or behind if you can.** A backlit tank gives a dark animal on a bright, even field, which is the easiest case.
- **Fix the camera.** If it moves after the background is captured, everything that moved becomes “animal”.
- **Capture a fresh background** whenever the lighting changes.
- **Raise the threshold before reaching for the other settings.** Most noise goes away with threshold and minimum area alone.
- **Look at the drawn track.** A track that jumps to the corner of the tank and back is a detection of something that is not the animal.

11 Experiments: Flow Field (PIV)



Flow Field (PIV) with a seeded-flow clip loaded and the scale markers placed. The clip is a synthetic test video made for this guide.

> *"Measures how the water itself is moving, by cross-correlating successive frames of seeded flow. Use it to check the working section is uniform, to find dead zones, and to turn pump output into a real water speed."*
>
> **Gives you:** a velocity vector field, and median chamber velocity. **Needs:** video of seeded flow, even lighting, and a scale.

Particle image velocimetry (PIV) splits each video frame into small square windows, finds where the pattern of tracer particles in each window moved to in the next frame, and turns that shift into a velocity vector. The software uses OpenPIV, and every setting OpenPIV exposes is a control on this page.

PIV needs a license. Without one, the tab shows the preview cover (section 3.1). The PIV-based **Chamber Velocity** calibration (section 13.3) is *not* locked: it is what sets a machine up.

The same page, without a machine, is on the Launcher's **Video Analysis** card (chapter 18). Its runs are saved under *video_analysis* in the Launcher's **Analysis** card rather than in a machine's Trial History.

11.1 Filming for PIV

Good PIV starts at the camera, not in the settings:

- **Seed the water** with neutrally buoyant tracer particles, densely enough that every window (32 × 32 pixels by default) holds several.
- **Light a thin sheet** of the flow, or at least light it evenly. Glare and reflections correlate as "particles" that do not move.
- **Fix the camera**, square to the working section, so the scale is the same across the frame.
- **Match the frame rate to the flow.** Particles should move a few pixels, roughly a quarter of the window size or less, between frames. Faster flow needs a higher frame rate, or several **passes** (below).
- **Film a ruler or a known width** in the plane of the flow, for the scale.

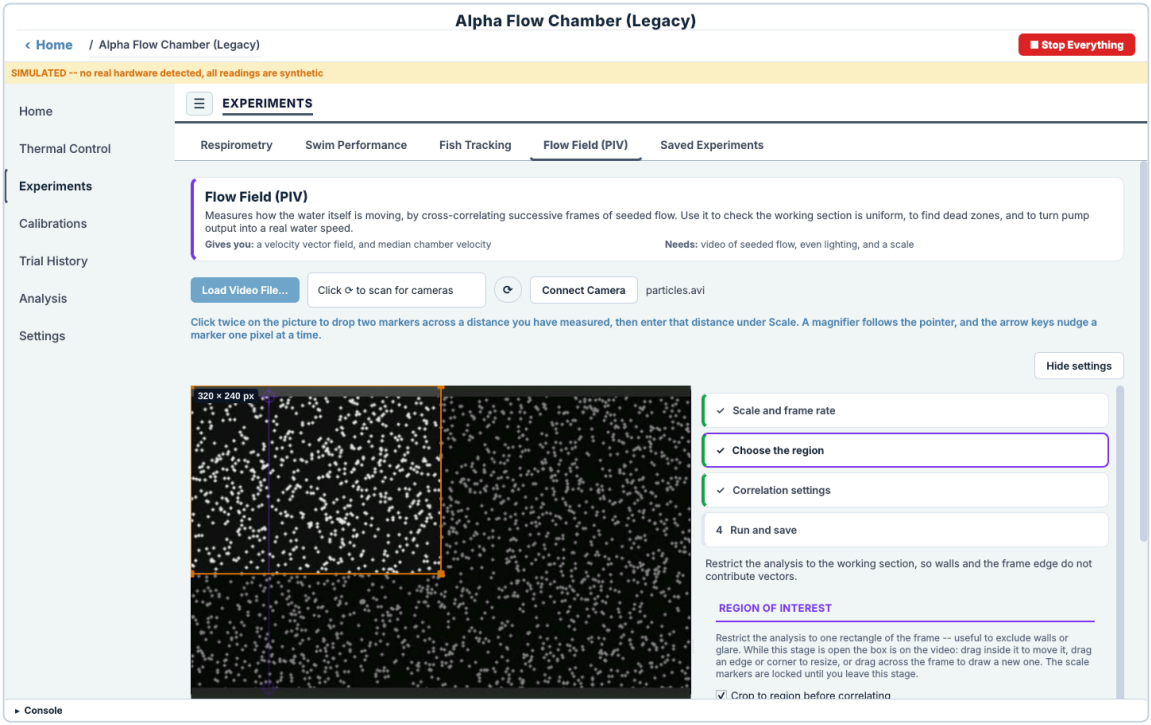
11.2 The video source

The controls over the picture (**Load Video File…**, camera list, ⟳, **Connect Camera**, **Pause**, **Disconnect**) are the same on every video page; see section 10.1. As on Fish Tracking, the page scrolls as one, and **Hide settings** gives the picture the full width.

11.3 Stage 1: Scale and frame rate

The **Pixel calibration** card. *"Click two points on the video a known distance apart and type that distance. Until this is set, velocities are in pixels per frame."* It works exactly like the Fish Tracking scale (section 10.2): two clicks, type **Distance between markers**, press **Apply scale from markers**. **Frame rate** overrides what the file reports, and every velocity scales directly with it.

11.4 Stage 2: Choose the region



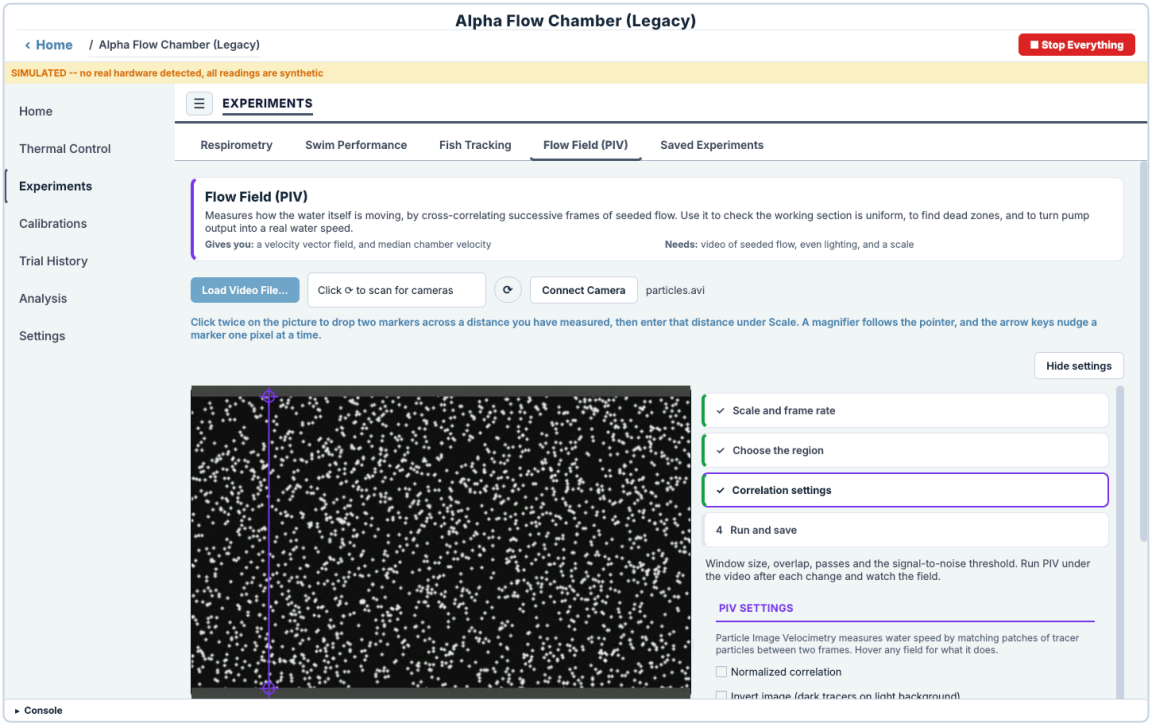
The region of interest box on the video.

"Restrict the analysis to one rectangle of the frame – useful to exclude walls or glare."

While this stage is open the box is drawn on the video. Drag inside it to move it, drag an edge or corner to resize it, or drag across the frame to draw a new one. The scale markers are locked while this stage is open, so you cannot move them by accident.

Control	What it does
Crop to region before correlating	Analyse only the box. Off, the whole frame is used.
X, **Y**, **Width**, **Height**	The box in pixels, if you prefer typing to dragging.

11.5 Stage 3: Correlation settings



The PIV settings card.

The defaults suit most well-seeded video. Change one thing at a time, and look at the vectors after each change.

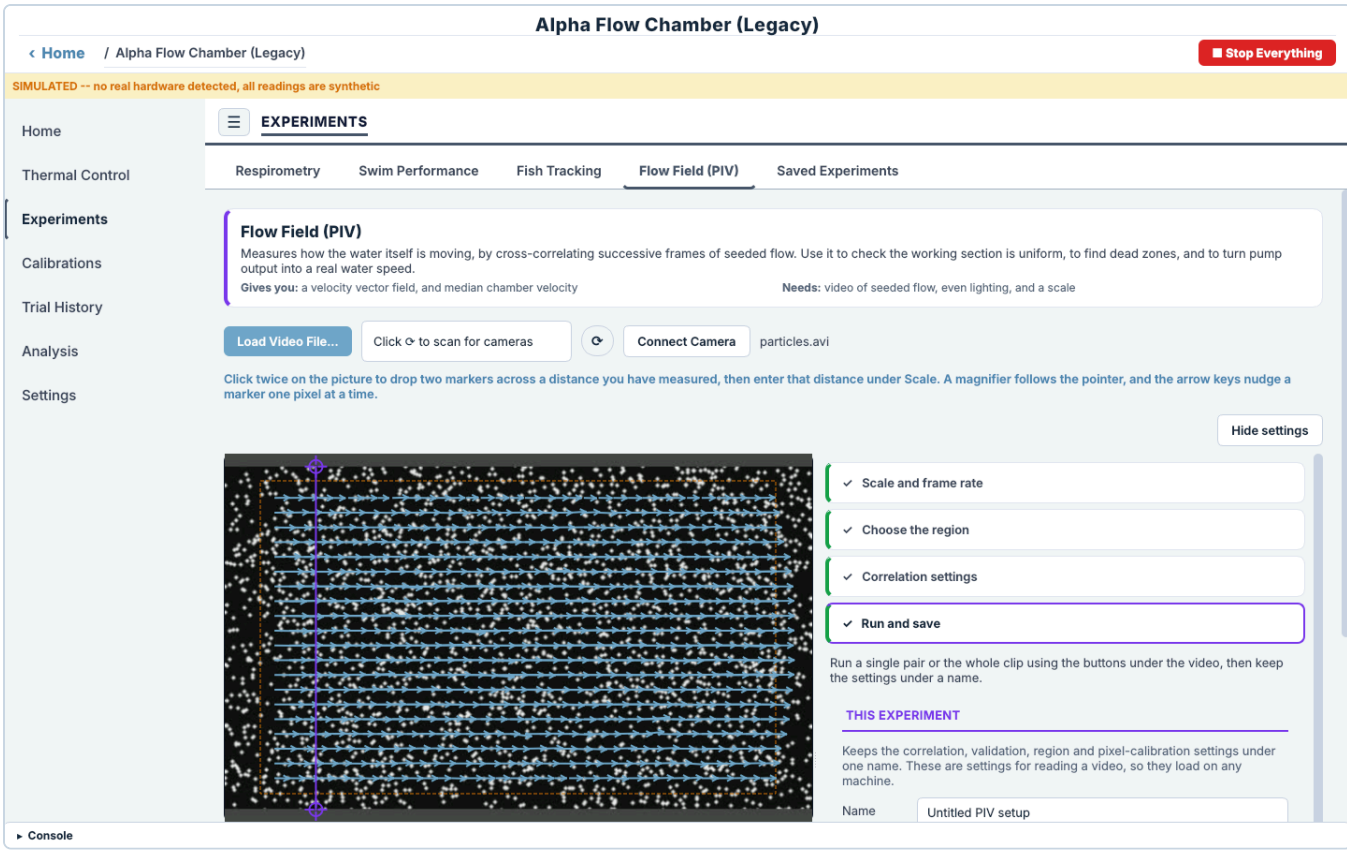
Setting	What it does
Window size (px)	Size of each square patch the image is divided into. Each patch gives one vector. Bigger is more reliable but coarser; it must comfortably contain several particles. Default 32.
Overlap (px)	How much neighbouring windows share. More overlap gives a denser vector field without shrinking the window. Typically half the window size (default 16).
Passes (coarse → fine)	Runs the analysis repeatedly, each pass using the previous result to shift the search window. Handles faster flow than a single pass can resolve. 1 to 4.
Correlation method	*circular* (FFT, fast, the default) or *linear* (avoids wrap-around artefacts at window edges).
Subpixel method	How the correlation peak is located below one pixel: *gaussian* (the usual choice, for round particle images), *centroid* or *parabolic*.
Normalized correlation	Normalises each window's brightness before matching. Helps when lighting is uneven across the frame.
Invert image	For dark tracers on a light background.
Validate & clean outliers	Scores each vector and replaces the unreliable ones. On by default.

Setting	What it does
Sig2noise method	How confidence is scored: the correlation peak against the mean (*peak2mean*) or against the second-highest peak (*peak2peak*).
Sig2noise threshold	Vectors scoring below this are replaced. Raise it to be stricter. Default 1.0.
Sig2noise width (peak2peak)	Pixels around the main peak excluded when looking for the second peak. Only used by *peak2peak*.
Outlier replacement method	How a rejected vector is filled in from its neighbours: *localmean*, *disk* or *distance* (distance-weighted).
Replacement kernel size	How many neighbouring vectors are averaged to fill a rejected one.
Replacement max iterations	How many times the fill-in is repeated, for clusters of rejected vectors with no valid neighbours on the first try.

11.6 Stage 4: Run and save

The buttons under the picture:

Button	What it does
Next Frame Pair	Steps the video forward one frame and clears the vectors.
Run PIV	Analyses the current pair of frames and draws the vector field on the picture. Quick, for trying settings. Not saved.
Analyze Whole Video	Analyses successive frame pairs through the clip (up to 200 frames), with a progress bar and **Cancel**. Charts the median velocity of each pair over time (the **u** and **v** components, which can be hidden with their tick boxes), draws the last field, and **saves the run to Trial History**. A cancelled run is not saved.
Export Vector Field…	Saves the current vector field as CSV: `x_px`, `y_px`, `u_px_per_frame`, `v_px_per_frame`, and `was_outlier` when validation is on. Needs a license.



A vector field after Analyze Whole Video, with the median velocity chart. Synthetic test video.

The **Result** card reads, for example:

> Analyzed 60 frame pairs. Median flow: u=4.21 px/frame, v=0.03 px/frame 12/600 vectors flagged & cleaned Calibrated: u=12.63 cm/s, v=0.09 cm/s (at 30 fps)

u is velocity along the frame's horizontal axis and *v* along its vertical axis. The last line appears only once a scale is set.

11.6.1 This experiment

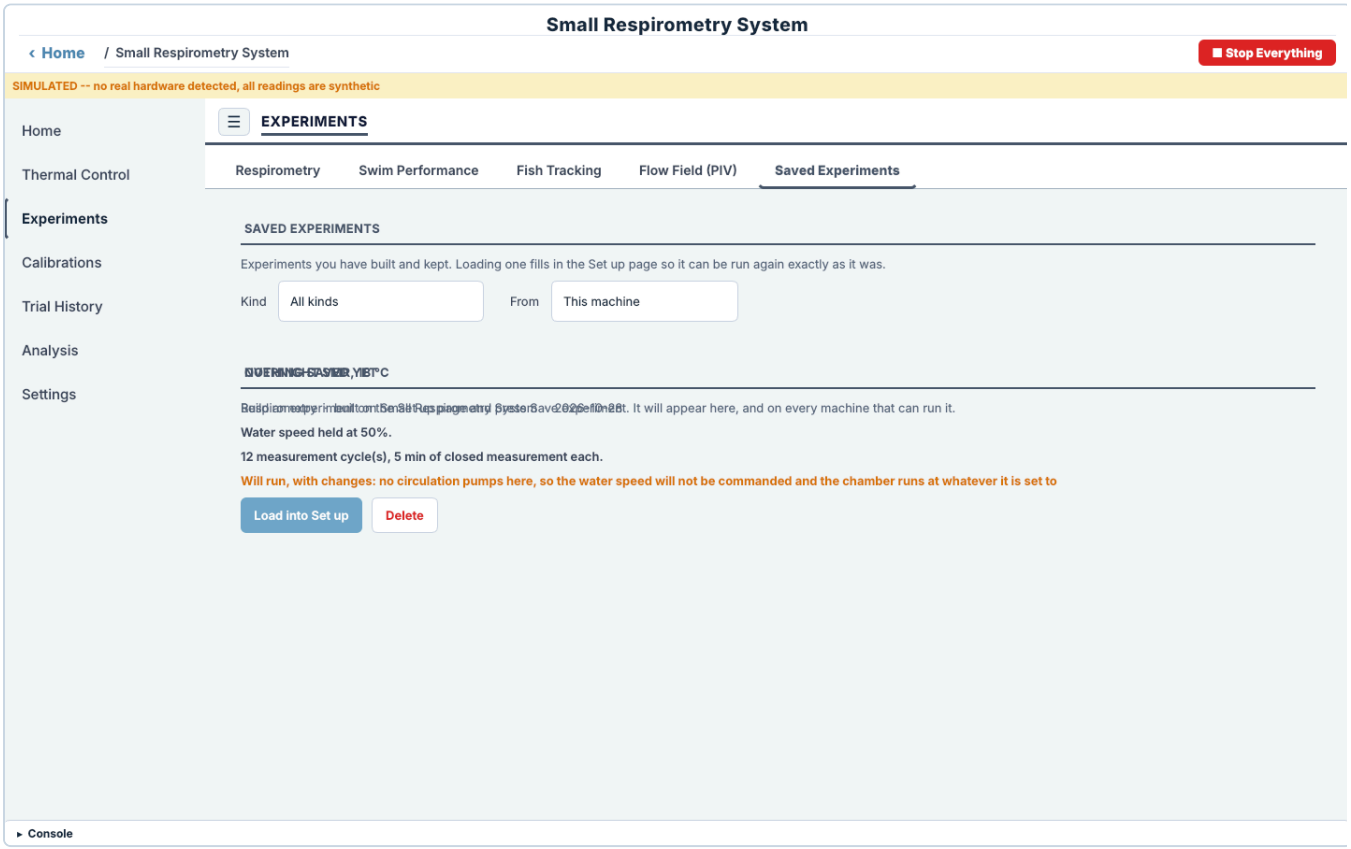
Name, treatment, notes and **Save experiment** (section 8.1.1). A PIV setup keeps the correlation, validation, region and scale settings. These are settings for reading video, so they load on any machine.

11.7 Reading the result

- **Many flagged vectors** mean the settings do not suit the video: too few particles per window, particles moving too far between frames, or glare. Try a larger window, more passes, or a smaller region.
- **Vectors along a wall pointing the wrong way** are usually reflections. Crop them out with the region.

- **The median** is robust to a few bad vectors, which is why the software reports it rather than the mean.
- To map the working section, film several positions and compare medians. To turn pump settings into chamber velocity across the machine’s whole range, use the Chamber Velocity calibration (section 13.3), which drives the pumps and runs PIV at each setting for you.

12 Experiments: Saved Experiments



Saved Experiments on the Small Respirometry System, with one saved temperature program.

"Experiments you have built and kept. Loading one fills in the Set up page so it can be run again exactly as it was."

A **saved experiment** is a recipe: every setting from a Set up page, kept under a name. It holds no results. Results are trials, in Trial History (chapter 14). Keeping the two apart means one well-tested protocol can be run every week, by anyone, on any machine that can run it.

12.1 Saving an experiment

On any experiment page (Respirometry, Swim Performance, Fish Tracking, Flow Field), fill in the **This experiment** card and press **Save experiment**. Saving again under the same name updates the saved copy. Saved experiments are shared by every machine on this computer.

12.2 The list

Control	What it does
Kind	Show only one kind: *Respirometry*, *Temperature program*, *Swim performance*, *Fish tracking* or *Flow field (PIV)*. *All kinds* shows everything.
From	*This machine* shows experiments built on the machine you have open; *Every machine* shows them all.

Each saved experiment is a card showing:

- its **name**;
- its **kind**, the machine it was **built on**, and the date;
- its **notes**, if it has any;
- a **small chart** of the temperature curve, flow program or swim staircase, where there is one, so the six-hour version is easy to tell from the two-hour one;
- a **verdict** on whether *this* machine can run it (below);
- **Load into Set up**, which fills in the matching Set up page and switches to it, and **Delete**, which asks *"Delete '…'? This removes the recipe. Trials already run from it are not affected."*

12.3 Will it run here?

Machines differ. A teaching chamber has no chiller; a swim tunnel has no oxygen probe; a respirometry box has no circulation pumps. Every saved experiment is checked against the machine you have open, and the card says one of three things:

- **"This machine can run it as written."**
- **"Will run, with changes: …"**, followed by what will not happen here. For example, a program with a water-speed curve loaded on the Small Respirometry System says *"no circulation pumps here, so the water speed will not be commanded and the chamber runs at whatever it is set to"*. It still runs; the rest of the experiment is still worth doing. Loading it shows the same message in a **Loaded with changes** dialog, so nobody finds out from a flat trace afterwards.
- **"Cannot run here: …"** when nothing meaningful would be left, such as a swim staircase on a machine with no variable-speed pump.

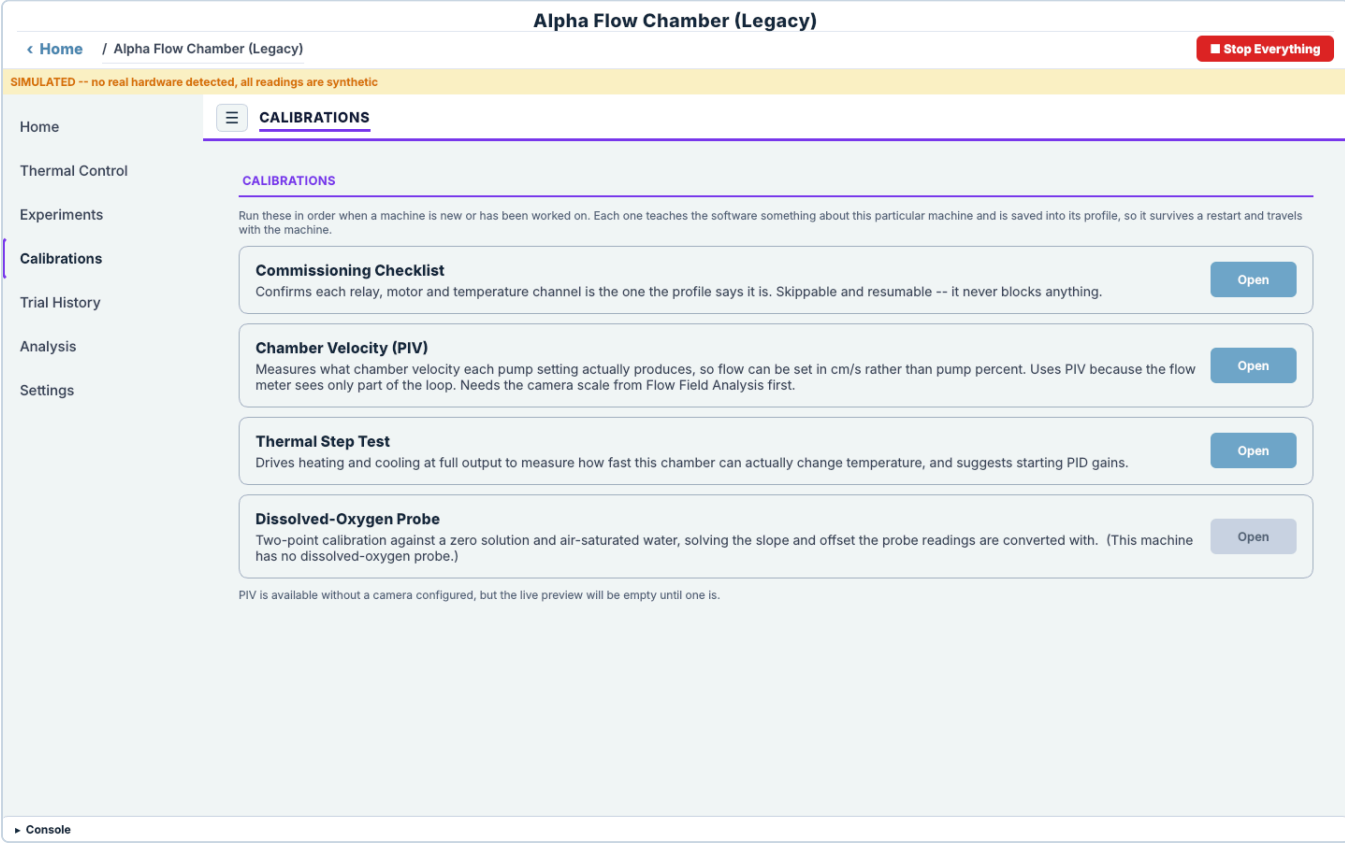
12.4 The Experiment Library

The same list, for every machine, is on the Launcher's **Experiment Library** card (chapter 18). Opened there it has no machine to load into, so **Load into Set up** explains that you should open it from inside a machine.

12.5 Where saved experiments live

One file per experiment in `~/.aquafab/protocols/` (chapter 23). To give a protocol to a colleague on another computer, copy the file into the same folder there.

13 Calibrations



The Calibrations menu on the Alpha Flow Chamber (Legacy).

"Run these in order when a machine is new or has been worked on. Each one teaches the software something about this particular machine and is saved into its profile, so it survives a restart and travels with the machine."

13.1 The Calibrations menu

The section opens on a list of four calibrations, each with a description and an **Open** button. A calibration that does not apply to this machine is greyed out and says why, for example *"(This machine has no dissolved-oxygen probe.)"*. Inside a calibration, **‹ All Calibrations** returns to the list.

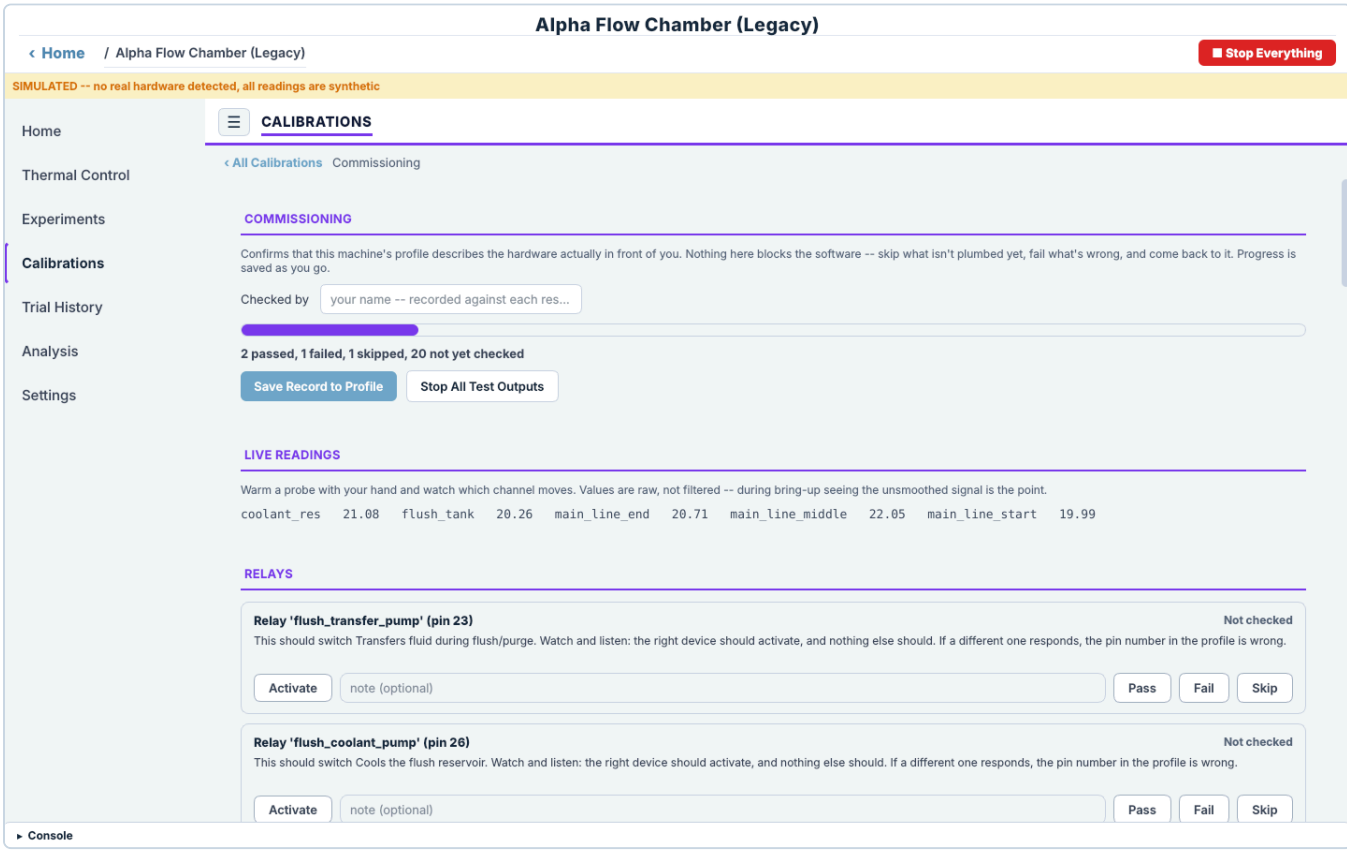
Calibration	What it teaches the software	Run it
Commissioning Checklist	That each relay, motor and temperature channel is the one the profile says it is.	First, on any new or rewired machine.
Chamber Velocity (PIV)	What chamber velocity each pump setting really produces.	After commissioning, on machines with variable-speed pumps.
Thermal Step Test	How fast the chamber can change temperature, and starting PID gains.	After commissioning, with the chamber empty.

Calibration	What it teaches the software	Run it
Dissolved-Oxygen Probe	The slope and offset that convert the probe's reading to mg/L.	Whenever the probe is new, serviced or drifting.

All four run on a simulated machine too, so they can be rehearsed. On a simulated machine the results describe the simulator, not your hardware, and the software does not pretend otherwise.

None of the calibrations needs a license.

13.2 Commissioning Checklist



The Commissioning checklist, with results recorded for some items. Simulated machine.

"Confirms that this machine's profile describes the hardware actually in front of you. Nothing here blocks the software – skip what isn't plumbed yet, fail what's wrong, and come back to it. Progress is saved as you go."

A profile is a description, typed by a person. A relay wired to pin 23 that the profile says is on pin 24 will switch the wrong device every time, and nothing in software can tell. The only way to know is to drive each output and watch.

13.2.1 At the top

Control	What it does
Checked by	Your name. Recorded against each result.
Tally	For example *"2 passed, 1 failed, 1 skipped, 20 not yet checked"*.
Save Record to Profile	Writes the results into the machine's profile, so the commissioning record travels with the machine.
Stop All Test Outputs	Turns off anything this page has switched on.
Live readings	Every temperature channel, raw and unfiltered, so you can see which one moves when you warm a probe.

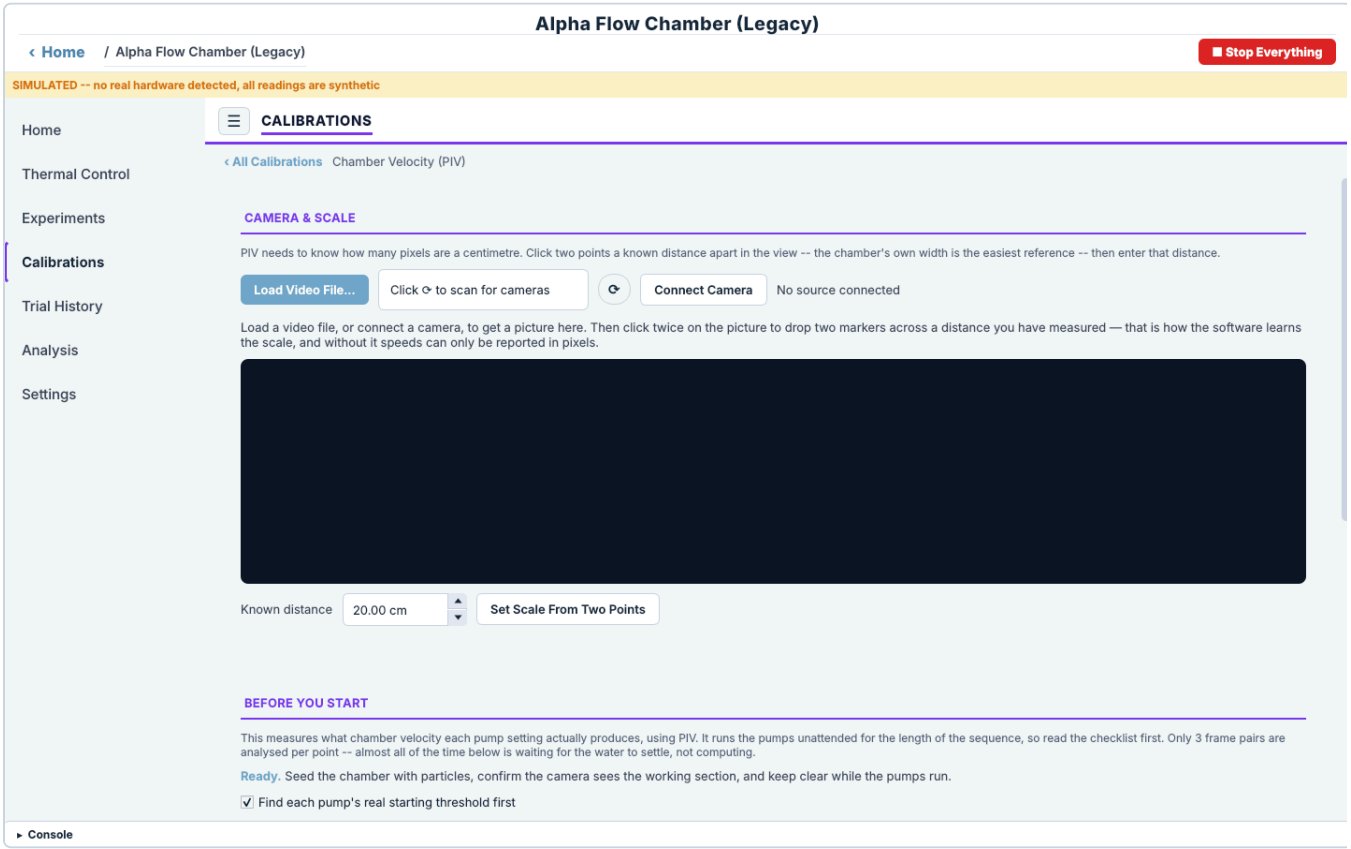
13.2.2 The items

The checklist has a group for each kind of hardware in the profile. Each item says what it should do, and has **Pass**, **Fail** and **Skip** buttons and a **note** field. Items marked · **critical** are ones the machine cannot be trusted without.

Group	How to check it
Relays	Press **Activate**. The device named should switch: watch and listen. If a different one responds, the pin number in the profile is wrong.
Motors	Press **Activate** to run the pump briefly at its declared starting threshold (for example 37 %). Confirm the right pump turns, in the right direction, and that it actually starts at that setting. If it does not, its minimum start is wrong, and every flow calibration built on it will be too.
Temperature probes	Warm the probe you believe is the named channel; a hand around it is enough. The reading should rise within a few seconds. If a different channel moves, the wires are swapped, and no later measurement can be trusted until they are not.
Flow	With the loop running, press **Run Check**. The reading should be stable and plausible. A value that jumps between zero and something enormous means the meter's settings are wrong, not that the flow is unsteady.
Configuration	*Plant specifications*: checks the declared chiller power, reservoir volume and working-section area for sanity. Note the values you confirmed, for example *"chiller 1800 W, reservoir 189 L, working section 500 cm2"*.

Record what you saw in the note, for example *"clicked, heater lit"* or *"purge LED lit instead"*. A failed item is a fix for the profile (chapter 19) or the wiring; re-check it after the fix.

13.3 Chamber Velocity (PIV)



The Chamber Velocity calibration. Simulated machine.

"Measures what chamber velocity each pump setting actually produces, so flow can be set in cm/s rather than pump percent. Uses PIV because the flow meter sees only part of the loop."

Until this is run, a flow slider is a motor output, not a speed. After it is saved, the Home page's **Chamber flow** slider reads as chamber velocity, with *"Measured chamber velocity at this setting: … cm/s"* under it. On a machine with several coupled pumps, the slider is solved against the measured velocity surface, so the pumps are coordinated to make the water speed follow it.

> **Why a surface, not a curve per pump.** Where two pumps feed one chamber through connected legs, they are hydraulically coupled: one pump running harder can push water backwards through the other's leg and *reduce* chamber velocity. Measuring each pump alone and adding them gives a calibration that looks reasonable and is wrong where it matters. So the calibration measures a grid of combinations.

13.3.1 Camera & scale

"PIV needs to know how many pixels are a centimetre."

1. Load a camera (or a clip, to rehearse) with the usual video controls (section 10.1).

2. Click two points a known distance apart in the view. The chamber's own width is the easiest reference.
3. Enter **Known distance** (cm) and press **Set Scale From Two Points**. The scale appears in px/cm.

If a scale has already been set on the Flow Field (PIV) tab for the same camera, that one is used.

13.3.2 Before you start

"It runs the pumps unattended for the length of the sequence, so read the checklist first."

The checklist lists everything that would make the result untrustworthy, and **Start Calibration** stays disabled until it is clear:

- no video source;
- no scale (*"without it PIV reports pixels per frame, and the whole calibration would be in the wrong units while looking entirely plausible"*);
- the machine declares no pumps that drive chamber velocity;
- the working section's dimensions are not in the profile (needed to cross-check against the flow meter; set them in **Settings**, section 16.2);
- the vision components are not installed.

When it is clear it says **Ready**: *"Seed the chamber with particles, confirm the camera sees the working section, and keep clear while the pumps run."*

Find each pump's real starting threshold first (on by default) ramps each pump up from below its declared minimum until the chamber actually moves. The declared figure is a number somebody typed, and every sweep starts from it, so it is worth measuring. It adds a few minutes.

Under it, the plan, for example *"18 measurements, about 19 minutes – 8 single-pump – 9 combined"*. Only three frame pairs are analysed per point; almost all the time is waiting for the water to settle.

13.3.3 Calibration run

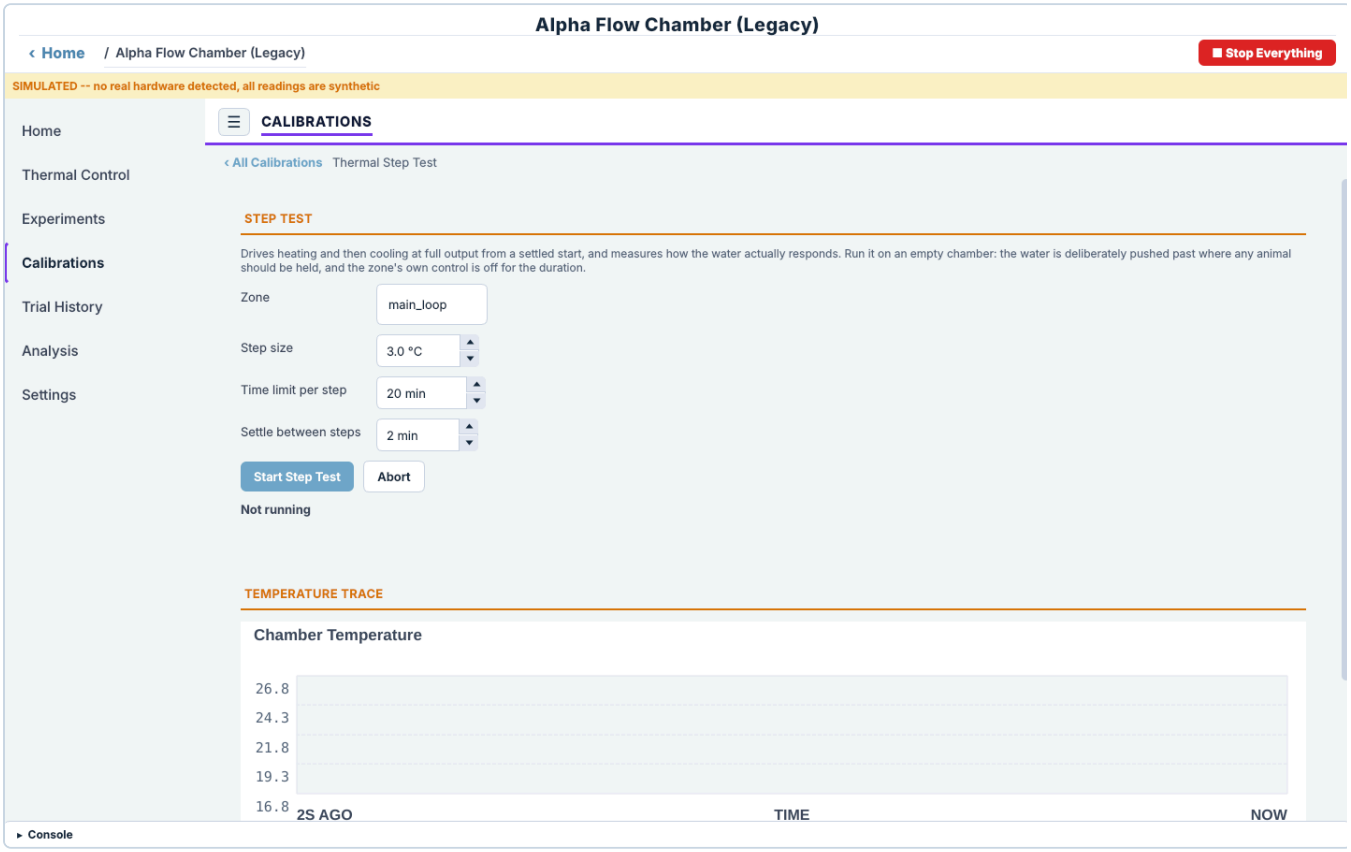
Control	What it does
Start Calibration	Starts the sequence. Each point shows *"Point 3 of 18 — …"*, then *"Settling — 40s"* while the water reaches steady state, then *"Measuring with PIV…"*.
Abort & Stop Pumps	Stops every pump at once. *"Aborted — pumps stopped. N point(s) measured. Nothing was saved."*
Progress bar	How far through the plan.

13.3.4 Measured points and saving

The **Measured points** table has one row per point: **Pump settings**, **Velocity** (cm/s by PIV), **Implied flow** (L/min, velocity × working-section area) and **Meter** (what the flow meter read). A large difference between implied flow and the meter is expected where the meter sits in only one leg of the loop; a meter reading of zero while PIV sees flow is worth investigating.

When it finishes: *"Calibration complete. Review the points below, then save."* Press **Save to Profile**. The calibration is stored in the machine's profile, and every flow slider and program on the machine uses it from then on.

13.4 Thermal Step Test



The Thermal Step Test. Simulated machine.

"Drives heating and then cooling at full output from a settled start, and measures how the water actually responds. Run it on an empty chamber: the water is deliberately pushed past where any animal should be held, and the zone's own control is off for the duration."

> **Never run this with an animal in the chamber.**

13.4.1 Settings

Field	What it does
Zone	Which thermal zone to test, for example *main_loop* or *flush_loop*.
Step size	How far to move the water in each direction (default 3 °C). Bigger is easier to fit accurately, but takes longer and pushes the chamber further from room temperature.
Time limit per step	Give up on a direction after this long (default 20 min) and fit whatever it managed.

Field	What it does
Settle between steps	Pause between the heating and cooling steps (default 2 min).

Start Step Test begins. The phase reads *Heating at full output*, *Settling between steps*, *Cooling at full output*, then *Finished*. **Abort** stops it. Either way, every heater, pump and relay the test touched is put back exactly as it was.

13.4.2 Measured response

When the test finishes, the **Measured response** card gives:

- **Heating** and **Cooling** rates in °C/min: what a programmed ramp can realistically ask for. The Temperature card's feasibility check (section 8.1.7) uses these.
- **Dead time**, **time constant** and **process gain** of the heating curve, from a first-order-plus-dead-time fit.
- **Suggested PID** gains: Kp, Ki and Kd.
- **Cooling coefficient**, on zones with a coolant reservoir probe: *"the chamber loses that fraction of its lead over the reservoir each minute"*. The Temperature card uses it to work out how cold the reservoir must be for a programmed curve.
- Loop details where the profile has the probes for them: heater power and the rise across it, and heat-exchanger effectiveness.

> **Warning: the suggested PID gains are not valid for these chambers as they stand.** They come from the Ziegler–Nichols open-loop rules, which were validated only for plants whose dead time is roughly 0.1 to 1 times their time constant. Aquafab chambers are **lag-dominant**: a large body of water responds slowly, with a dead time that is a small fraction of its time constant. On such a plant the rules prescribe a very high Kp, a proportional band under a tenth of a degree, which will almost certainly oscillate. When that is the case, the page says so in amber under the gains. **Start well below the suggested Kp and raise it only if the chamber is too slow to reach its setpoint.** Treat every suggested gain as a starting point, never as an answer.

The page also says: *"Suggested gains come from the Ziegler-Nichols open-loop rules, which target roughly quarter-amplitude damping and are known to be aggressive. Treat them as a starting point and reduce the proportional gain if the chamber overshoots."*

If no gains are suggested, the heating curve showed no measurable dead time. That means the test was too short or too coarse, not that the chamber responds instantly: try a larger step.

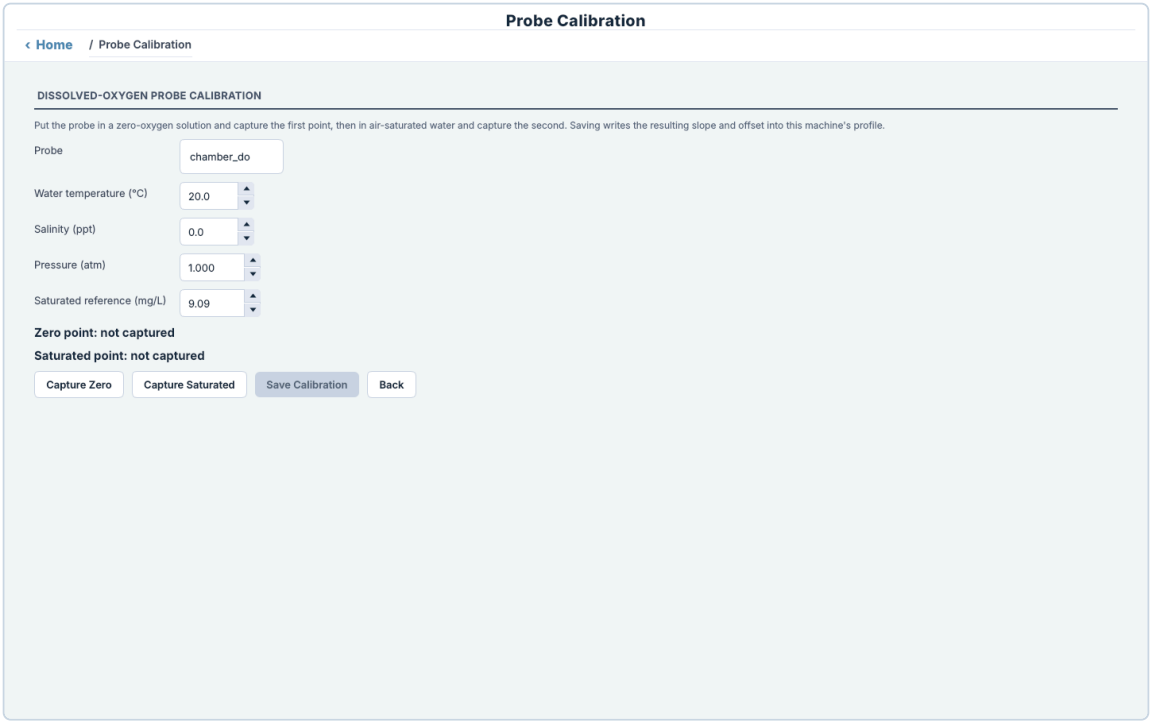
13.4.3 Saving

Button	What it does
Save to Profile	Records the measured response on the machine's profile. **Does not change how the zone is controlled.**

Button	What it does
Apply Gains to Zone	Writes the suggested Kp/Ki/Kd into this zone's control settings. Deliberately a separate step, for the reason in the warning above. Only meaningful for a zone using split-range PID (section 7.2).

Use a probe that represents the chamber's bulk water. A probe right after the heater sees the heater and makes a slow chamber look fast.

13.5 Dissolved-Oxygen Probe



The dissolved-oxygen probe calibration. Simulated machine.

"Put the probe in a zero-oxygen solution and capture the first point, then in air-saturated water and capture the second. Saving writes the resulting slope and offset into this machine's profile."

The software converts the probe's raw reading as *mg/L = slope × raw + offset*. Two points fix the line.

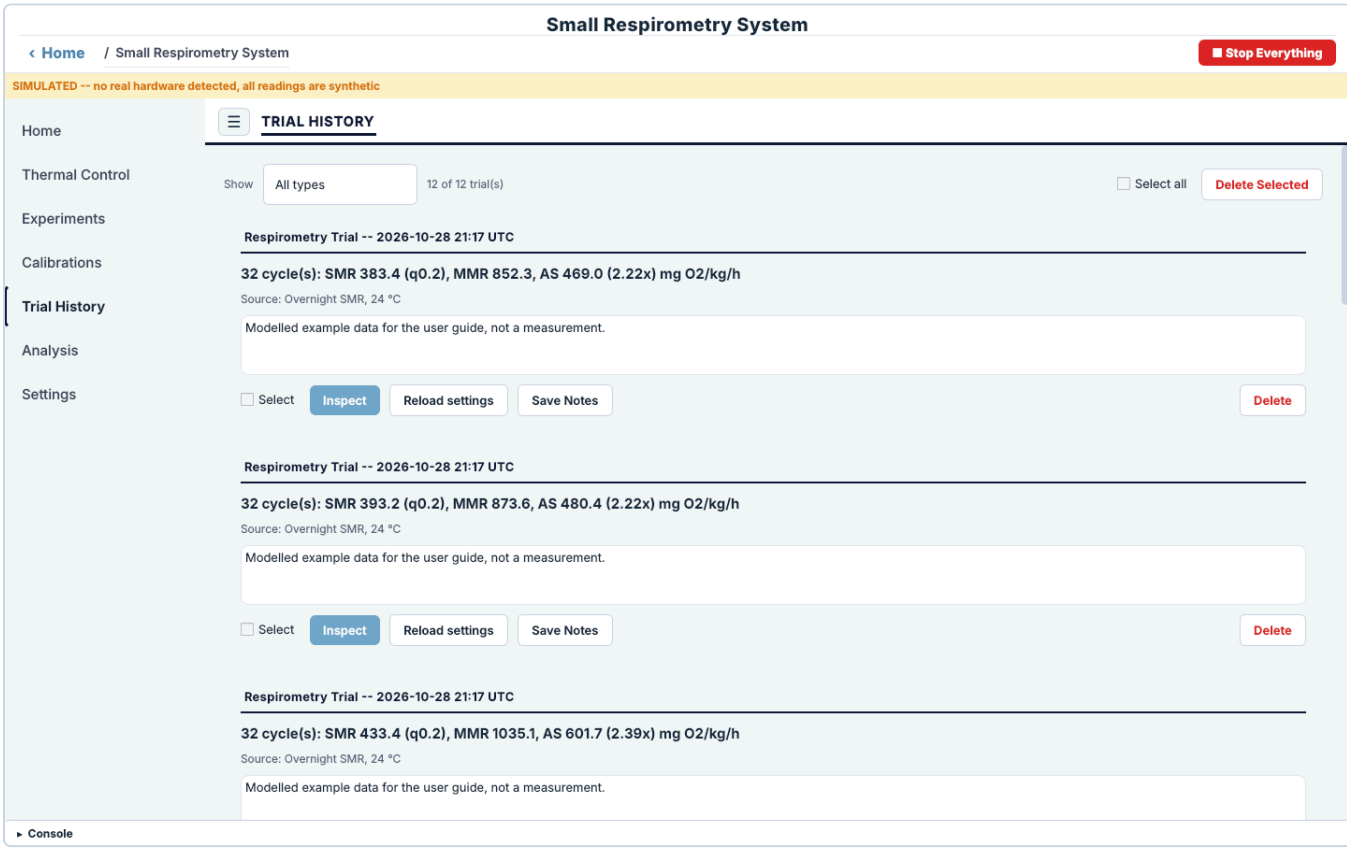
Field	What it does
Probe	Which oxygen probe, on a machine with more than one.
Water temperature (°C)	Of the saturated water.
Salinity (ppt)	0 for fresh water; full-strength seawater is about 35.
Pressure (atm)	Barometric pressure; 1.0 at sea level.

Field	What it does
Saturated reference (mg/L)	What fully air-saturated water holds at that temperature, salinity and pressure. Computed for you from García & Gordon (1992); type over it if you prefer your own table value.
Capture Zero	Takes the zero-oxygen point.
Capture Saturated	Takes the air-saturated point.
Save Calibration	Writes slope and offset into the profile. Enabled once both points are captured.
Back	Returns to the menu.

Procedure (section 20.12 has the full version):

1. Put the probe in a **zero-oxygen solution** (for example fresh sodium sulfite solution). Wait for the reading to settle, then **Capture Zero**.
2. Rinse the probe and put it in **air-saturated water** at the temperature you will measure at (bubble air through it for several minutes, or shake a half-full bottle). Set temperature, salinity and pressure, wait for the reading to settle, then **Capture Saturated**.
3. **Save Calibration**.

14 Trial History



Trial History on the Small Respirometry System. The trials shown are modelled example data made for this guide, not measurements.

Every trial a machine saves appears here, newest first: respirometry and temperature-program runs, swim tests, fish tracking runs and PIV analyses. Trial History is never locked, so a lab without a license can still see what it has recorded.

A trial is saved:

- **Respirometry**: by itself, after the first cycle, and again after every cycle until the run ends. Nothing to press.
- **Swim Performance**: when you press **Save Result**.
- **Fish Tracking**: automatically after **Track Fish**.
- **Flow Field (PIV)**: automatically after **Analyze Whole Video** completes.

14.1 The list

Control	What it does
Show	Narrows the list to one kind of trial (*Respirometry Trial*, *Swim Performance (Ucrit)*, *Fish Tracking*, *Flow Calibration*), or shows **Deleted (trash)**.

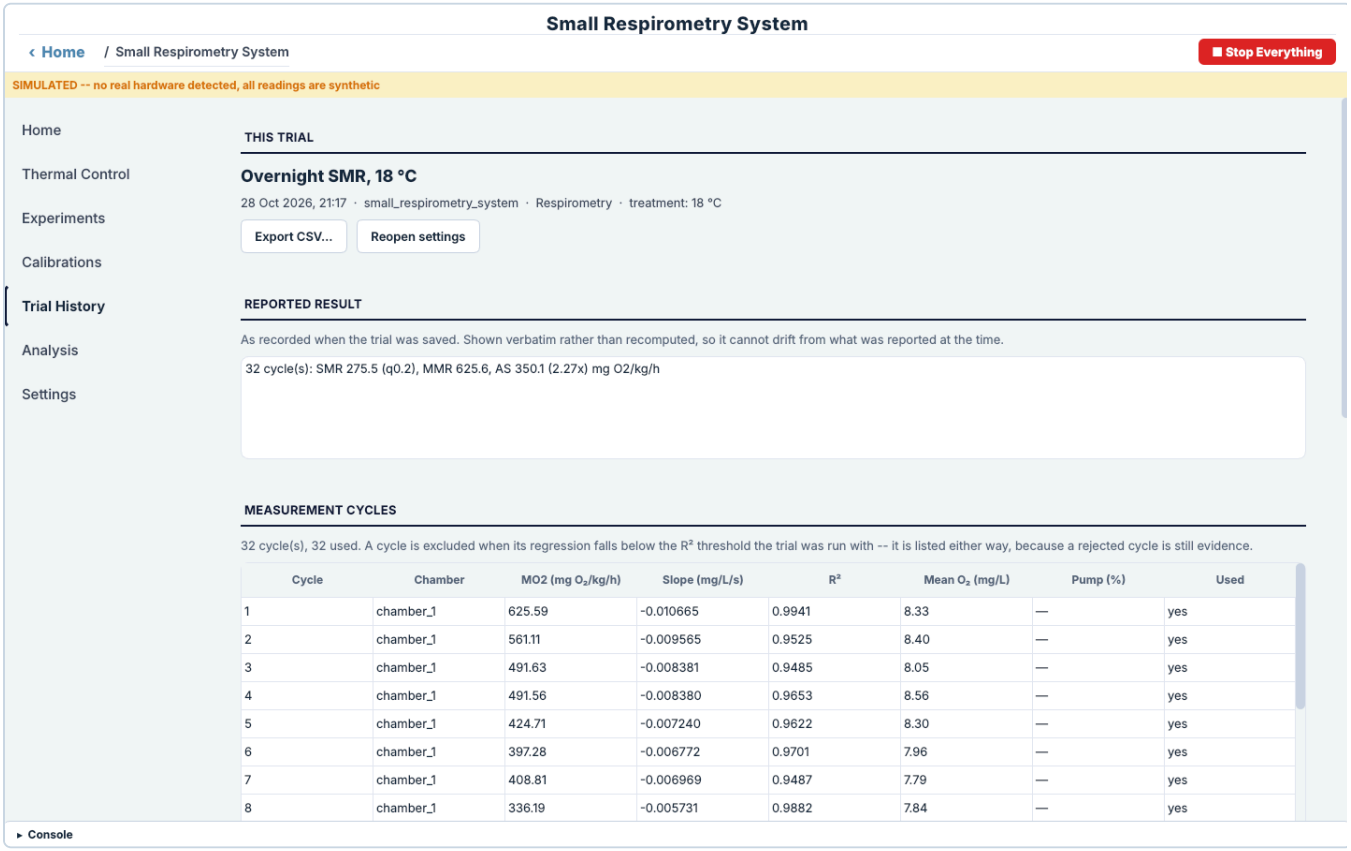
Control	What it does
Count	For example *“6 of 6 trial(s)”*.
Select all	Ticks every trial currently listed.
Delete Selected (N)	Moves the ticked trials to the trash, after asking.
Empty Trash	In the Deleted view only: permanently removes everything in the trash, after asking *“This really cannot be undone.”*

Each trial is a card showing its kind and time (in UTC), its one-line **summary** (for example *“12 cycle(s): SMR 98.4 (MLND), MMR 182.0, AS 83.6 (1.85x) mg O2/kg/h”*), and the experiment it came from. On each card:

Control	What it does
Notes	Which animal, what happened, anything odd about the run. Press **Save Notes** to keep what you typed.
Select	Ticks this trial for a bulk delete.
Inspect	Opens the trial on its own page (below).
Reload settings	Puts the trial’s settings back on the page that produced it, to run again or adjust.
Delete	Moves this trial to the trash, after asking.
Restore	In the Deleted view: brings the trial back.

> **Deleting is two steps on purpose. Delete** moves a trial to the trash; only **Empty Trash** removes it for good. Months of data are one mis-tap away on a touchscreen otherwise. The trash can also be managed from **Settings → Deleted trials** (chapter 17).

14.2 The trial page (Inspect)



One trial opened with Inspect. Modelled example data.

Inspect opens a page for one trial, with **‹ Back** to return:

Card	What it shows
This trial	The experiment name, date, machine, kind and treatment. **Export CSV...** writes the trial's cycles, one row each, for analysis elsewhere (needs a license). **Reopen settings** loads its settings into the page that produced it.
Reported result	*"As recorded when the trial was saved. Shown verbatim rather than recomputed, so it cannot drift from what was reported at the time."*
Measurement cycles	Respirometry only. One row per cycle: **Cycle**, **Chamber**, **MO_2** (mg O_2/kg/h), **Slope** (mg/L/s), **R^2**, **Mean O_2** (mg/L), **Pump (%)** and **Used**. A cycle whose fit fell below the trial's minimum R^2 is listed but marked unused: *"a rejected cycle is still evidence."*
Readings over time	For trials with recorded series, such as imported files: a chart of each series, with a chooser to show one at a time. On a machine, **Use as a temperature program...** turns the readings into a program (below).
Settings used	Every setting the trial ran with, with what each one is. *"This is what a reviewer asks for and what a repeat of the experiment has to match."*
Notes	What a person wrote about the run. **Save notes** keeps it.

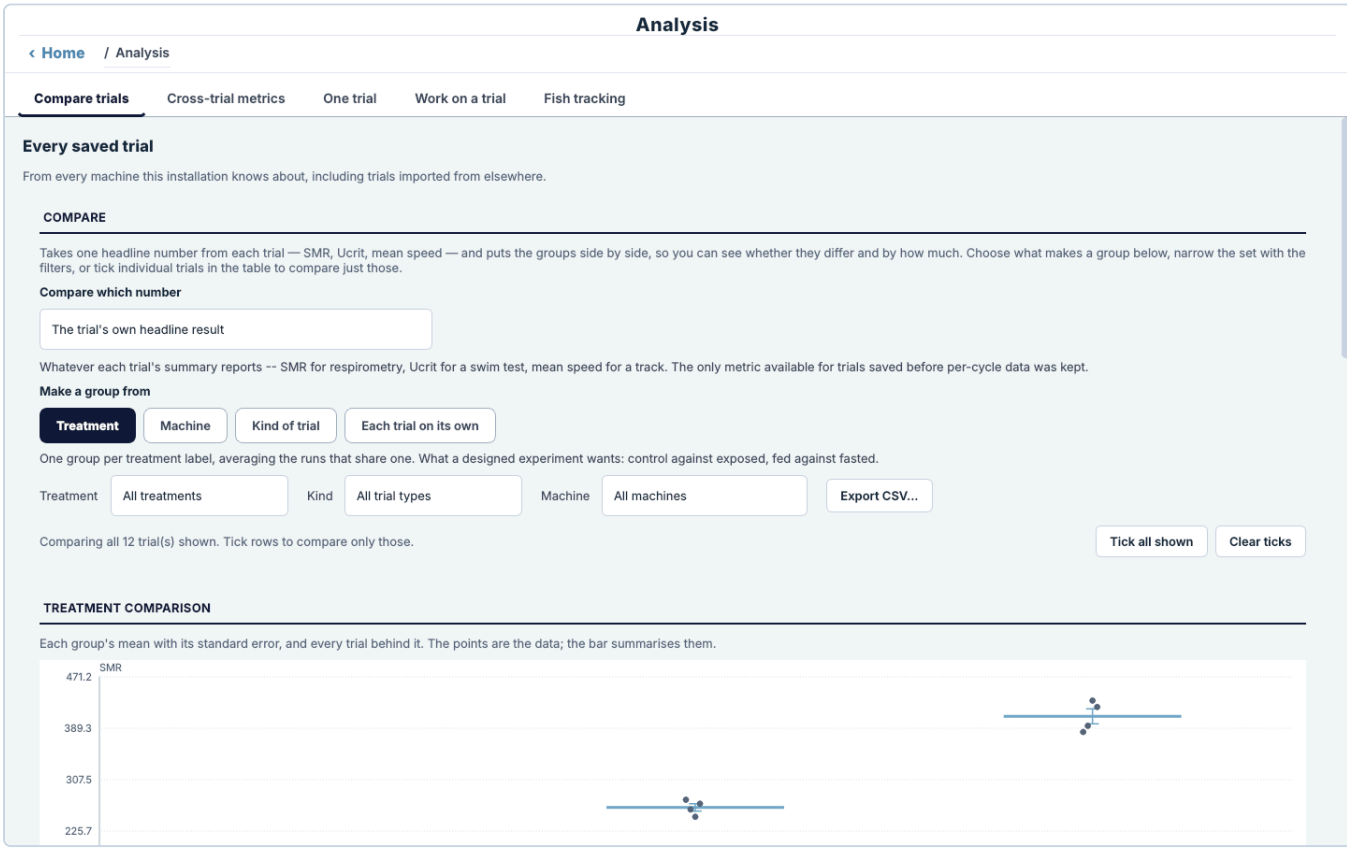
14.3 Turning a recorded trace into a temperature program

A lab with a tide-pool temperature log has the shape it wants the chamber to follow. Import the log as readings (section 20.11), inspect it, and press **Use as a temperature program…**:

Field	What it does
Temperature series	Which column holds temperature. The one named like a temperature is offered first. Choosing one that is not named that way is warned about, because a column of numbers around 8 could equally be 8 °C or 8 mg/L.
Steps	How many held steps the trace becomes. More follows it more closely and makes a longer table; the machine holds each step, so a program finer than the chamber can follow simply lags.
Repeat	How many times to run it. A single tidal cycle logged once is usually meant to run for a day.

Use as the temperature program puts the result in the Respirometry page's Temperature card (section 8.1.7), with the curve switched on, to check before running. Nothing runs until you press Start there.

15 Analysis



Analysis, Compare trials, grouping twelve trials by treatment. Modelled example data made for this guide, not measurements.

Analysis turns saved trials into the numbers and figures you publish. It is in two places, which are the same view with different trials in them:

- **A machine's Analysis section**: that machine's own trials. The question someone standing at a machine has.
- **The Launcher's Analysis card**: every saved trial from every machine on this computer, including imported ones. The place to ask whether a treatment behaved the same on two rigs, and the screen an office computer with no machine uses.

Analysis needs a license. Without one, it shows the preview cover (section 3.1). The **Trial History** list (chapter 14) is never locked.

Analysis has five tabs: **Compare trials**, **Cross-trial metrics**, **One trial**, **Work on a trial** and **Fish tracking**.

> **Nothing in Analysis changes a trial.** Every trial keeps the result it reported when it was saved. Analysis recomputes figures *beside* that result, from the cycles, and says so. A figure that silently differed from the one already published would be the worst thing this screen could show.

15.1 Example data

Before any trial has been saved, Compare shows a labelled worked example: *"Example data – not your results. … three temperature groups of four fish, plus a few swim trials."* It shows what the chart, statistics and export look like with data in them. Every row is marked as an example, nothing is stored, export is disabled, and it disappears as soon as a real trial is saved.

15.2 Compare trials

"Takes one headline number from each trial — SMR, U_{crit}, mean speed — and puts the groups side by side, so you can see whether they differ and by how much."

15.2.1 Choosing what to compare

Control	What it does
Compare which number	The metric. *The trial's own headline result* (whatever each summary reports: SMR, U_{crit} or mean speed) works for every trial. Respirometry trials saved with per-cycle data also offer: SMR by the trial's method; SMR re-estimated as a low quantile; SMR re-estimated as the lowest mode; MMR (the highest accepted cycle); aerobic scope (MMR − SMR); factorial aerobic scope (MMR ÷ SMR); mean and median MO_2 of accepted cycles; variability of the cycles (CV, %); whole-animal SMR (mg O_2/h); animal mass; mean oxygen during measurement; lowest oxygen reached; mean regression R^2; accepted cycles; rejected cycles. A line under the menu defines the chosen metric.
Make a group from	**Treatment** (one group per treatment label: what a designed experiment wants), **Machine**, **Kind of trial**, or **Each trial on its own**.
Treatment filter	Show only one treatment.
Kind	Show only one kind of trial.
Machine	Show only one machine's trials (on the Launcher's Analysis).
Export CSV…	Writes the visible trials, one row each. *"The statistics here are a summary, not a substitute for the data."*
Tick all shown / **Clear ticks**	With none ticked, every trial shown is compared. Tick rows in the Trials table to compare only those.

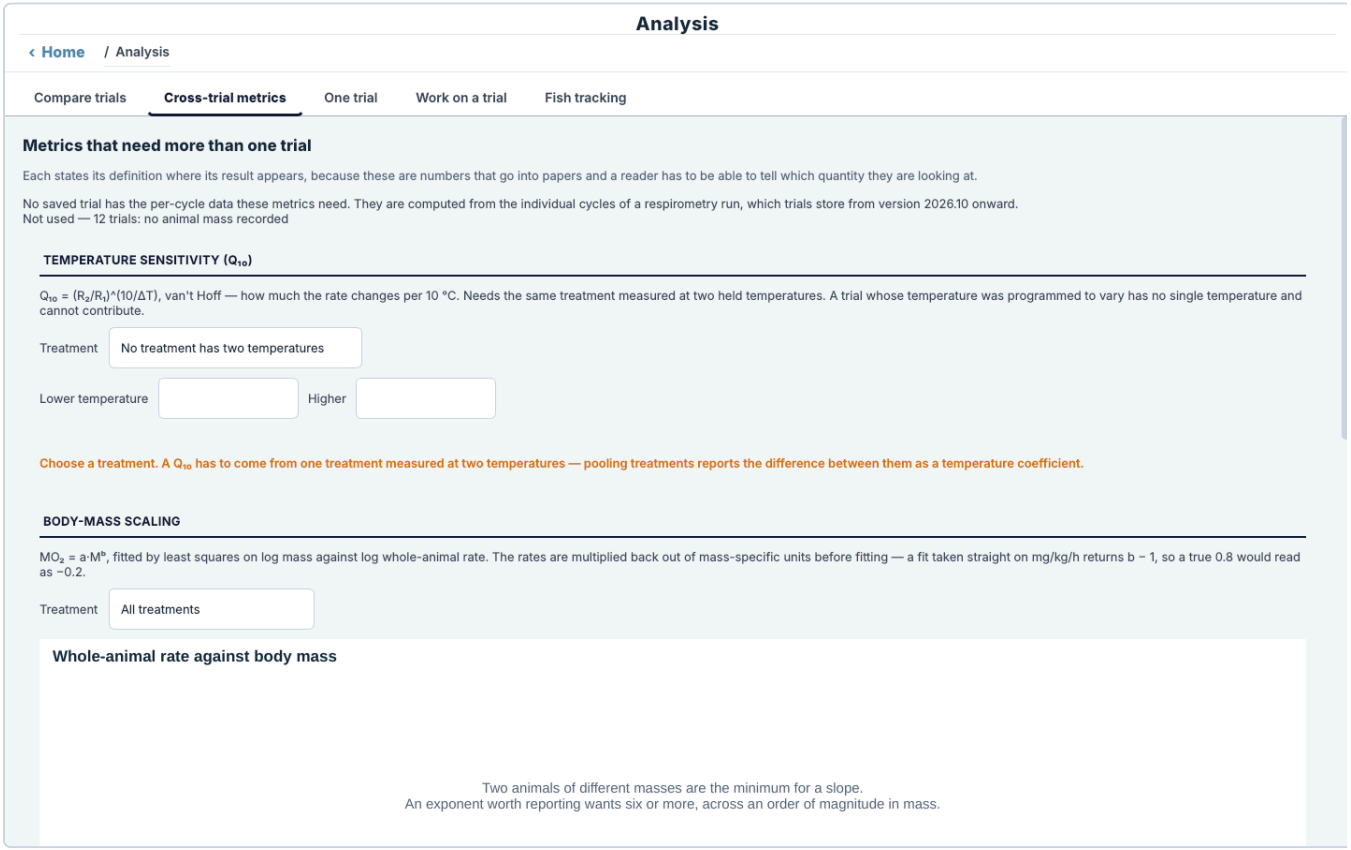
15.2.2 What it shows

Card	What it is
Treatment comparison	Each group's mean with its standard error, and every trial behind it as a point. *"The points are the data; the bar summarises them."*

Card	What it is
Traces, overlaid	One line per ticked trial, from its per-cycle data. **Plot**: *Oxygen uptake against cycle*, *Oxygen uptake against oxygen*, *Oxygen against cycle* or *Fit against cycle*. Traces are never averaged together: a mean over trials of different lengths would have to invent data.
Group statistics	Per group: **n**, **Mean**, **SD**, **SE** and **Range**. *SD describes how variable the animals were; SE describes how well the mean is pinned down. Comparing treatments wants SE.*
Trials	Every trial in the selection, with a **Compare** tick box, when, machine, kind, treatment, result and summary.
Every parameter, every trial	One row per compared trial and one column per parameter (scroll sideways). A blank cell means the parameter could not be computed for that trial, which is not the same as zero.

Trials saved before per-cycle data was kept (before version 2026.10) can be compared on their headline result but cannot be drawn as traces; the page says how many.

15.3 Cross-trial metrics

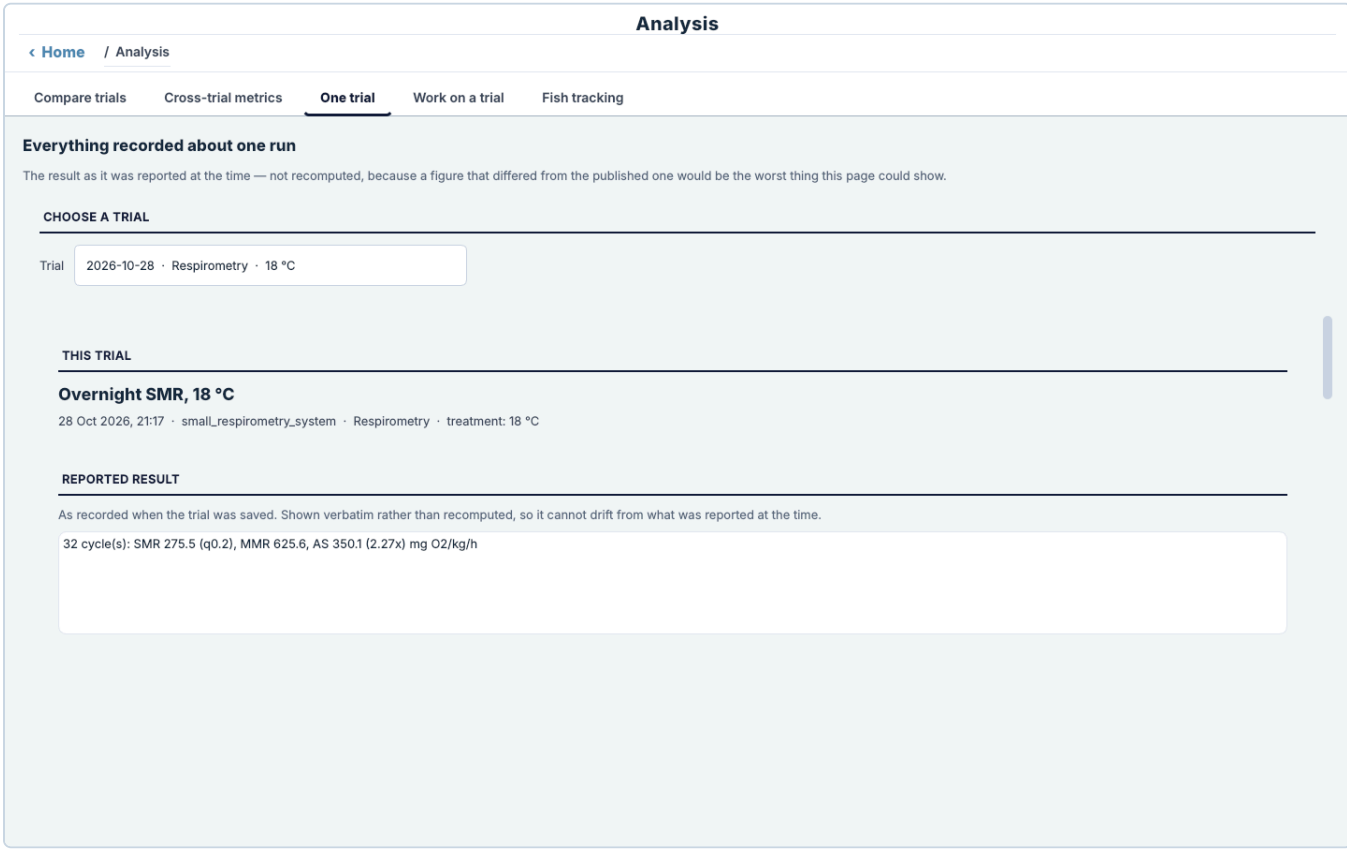


Cross-trial metrics. Modelled example data.

“Each states its definition where its result appears, because these are numbers that go into papers and a reader has to be able to tell which quantity they are looking at.”

Card	What it computes	Needs
Temperature sensitivity (Q_{10})	$Q_{10} = (R_2/R_1)^{(10/\Delta T)}$: how much the rate changes per 10 °C. Choose a **Treatment**, then a **Lower temperature** and a **Higher** one. Reports how many animals each rate is based on, and warns when the interval is short, because extrapolating a small interval to 10 °C magnifies any error.	One treatment measured at two *held* temperatures. A trial whose temperature was programmed to vary has no single temperature and cannot contribute. Pooling treatments would report the difference between them as a temperature effect, so the card will not.
Body-mass scaling	$MO_2 = a \cdot M^b$, fitted on log mass against log *whole-animal* rate. Rates are multiplied back out of mass-specific units first; a fit taken straight on mg/kg/h returns b − 1, so a true 0.8 would read as −0.2. Choose a treatment or all.	At least two animals of different masses.
What the SMR estimate came from	A histogram of every accepted cycle in the selection, with the SMR estimate marked. A clean resting run is a single peak with a tight low shoulder; a disturbed animal shows a high tail that one reported number hides. Notes when treatments or methods are pooled, and when masses span a wide range.	Respirometry trials with per-cycle data.
Oxygen at which aerobic scope has halved	The oxygen level where aerobic scope (MO_2 − SMR) has fallen to half its normoxic value, the normoxic reference being the mean of the highest-oxygen quarter of cycles. Choose an **Animal**. (Other software calls a similar figure PAAS-50; the definition here is what is computed.) This is not P_{crit}, which is where regulation fails entirely, at a lower oxygen level.	A closed-chamber run where oxygen falls through the trial. An intermittent run reflushes to the same level every cycle, so has no declining curve.

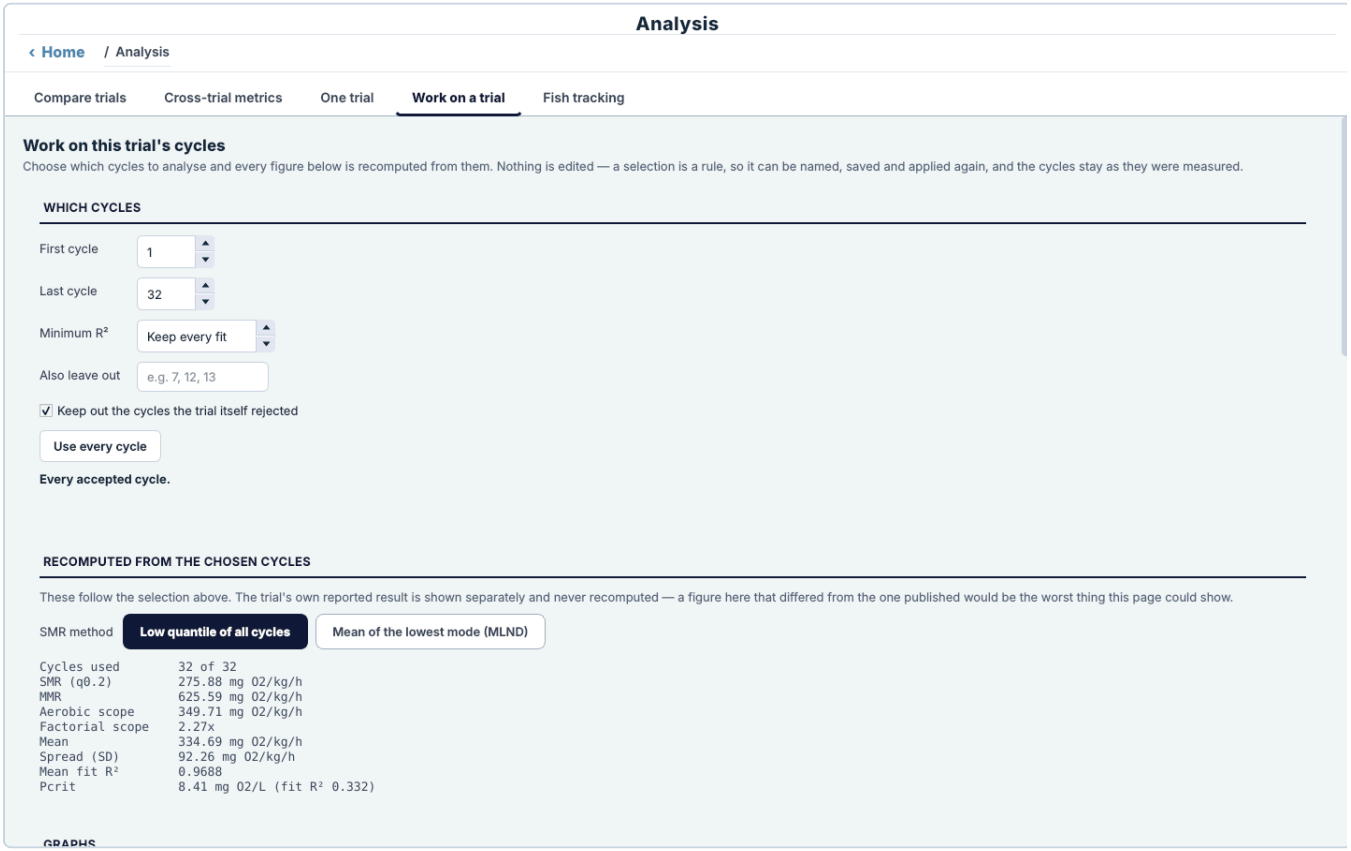
15.4 One trial



One trial. Modelled example data.

"Everything recorded about one run." Choose a **Trial** from the menu (listed by date, kind and treatment). The page is the same as **Inspect** in Trial History (section 14.2): the reported result verbatim, every measurement cycle, every setting used, notes, **Export CSV...** and **Reopen settings**.

15.5 Work on a trial



Work on a trial: choosing cycles and recomputing. Modelled example data.

"Choose which cycles to analyse and every figure below is recomputed from them. Nothing is edited — a selection is a rule, so it can be named, saved and applied again, and the cycles stay as they were measured."

This is where you deal with an animal that took three hours to settle, or one cycle where somebody bumped the bench.

15.5.1 Which cycles

Control	What it does
First cycle	The first cycle to analyse. Raise it past the cycles where the animal was still settling after handling.
Last cycle	The last cycle to analyse.
Minimum R^2	Leave out cycles whose oxygen decline fits a line worse than this. Separate from the threshold the trial ran with, which is already applied.
Also leave out	Cycle numbers to strike out individually, for example *7, 12, 13*: an outlier no threshold catches without also catching good cycles.

Control	What it does
Keep out the cycles the trial itself rejected	On by default. *"A cycle the trial rejected was rejected for a reason. Unticking this readmits it, which is a decision worth making deliberately."*
Use every cycle	Resets the selection.

15.5.2 Recomputed from the chosen cycles

Choose the **SMR method** (*Low quantile of all cycles* or *Mean of the lowest mode (MLND)*). The card shows cycles used, SMR, MMR, aerobic scope, factorial scope, mean, spread (SD), mean fit R^2, and P_{crit} (with its fit R^2) where the selection has a declining oxygen curve to find it on.

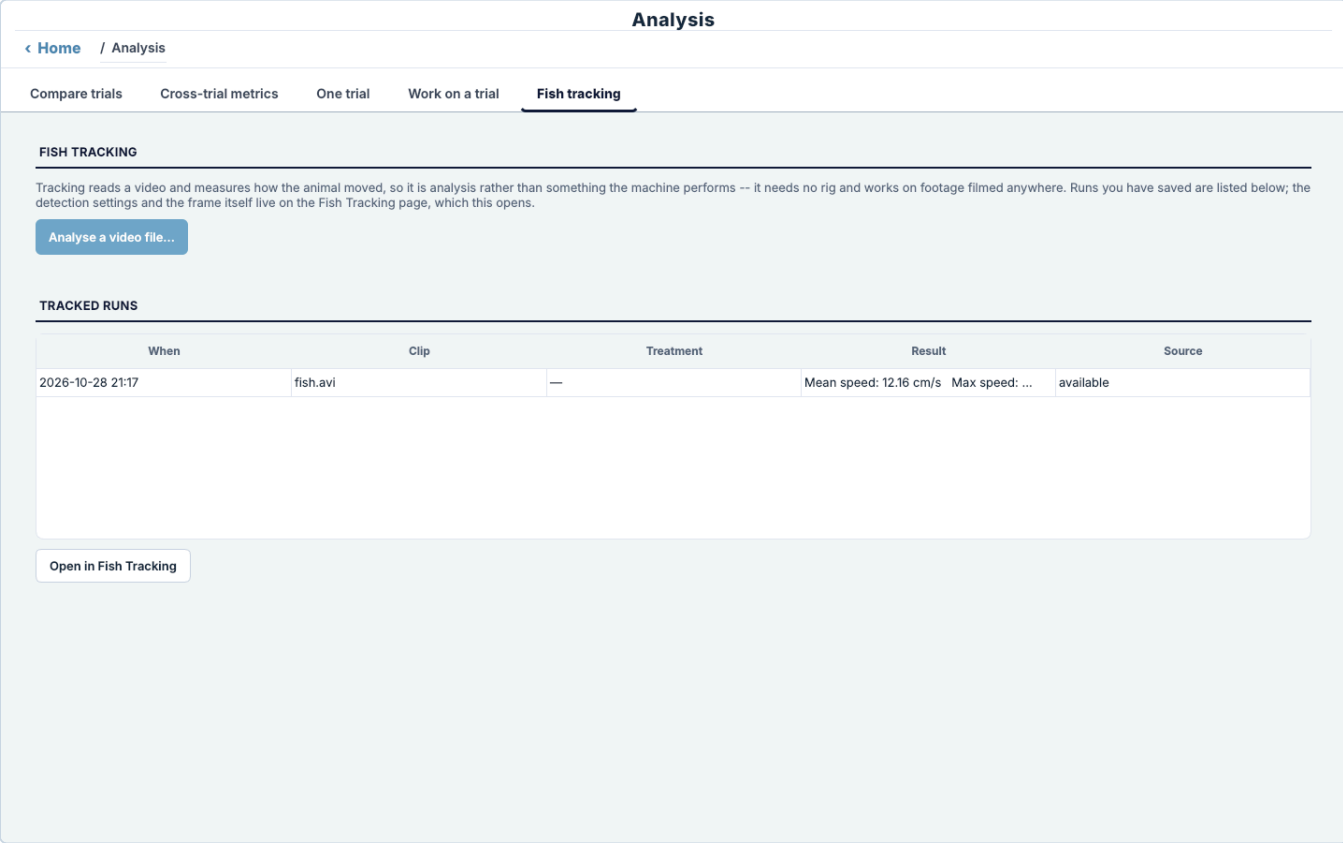
15.5.3 Graphs

MO_2 against cycle and the distribution of selected cycles. *"Filled points are in the selection; hollow ones were left out and say why on hover. The dashed verticals are the chosen bounds, and the horizontal rules are the estimates above."*

15.5.4 Saved selections

"Keep a selection under a name so the figure it produced can be reproduced. Saved with the trial, not with the software." Type a name (for example *Settled cycles only*), press **Save this selection**; **Apply selected** brings one back and **Delete selected** removes it.

15.6 Fish tracking



The Fish tracking tab of Analysis.

"Tracking reads a video and measures how the animal moved, so it is analysis rather than something the machine performs."

Control	What it does
Analyse a video file...	Opens the Fish Tracking page with the clip loaded, ready for a background frame and a scale (chapter 10).
Tracked runs	Every saved tracking run: **When**, **Clip**, **Treatment**, **Result**, **Source**.
Open in Fish Tracking	Reloads the clip and the settings the selected run used, to re-run or adjust it. Double-clicking a row does the same.

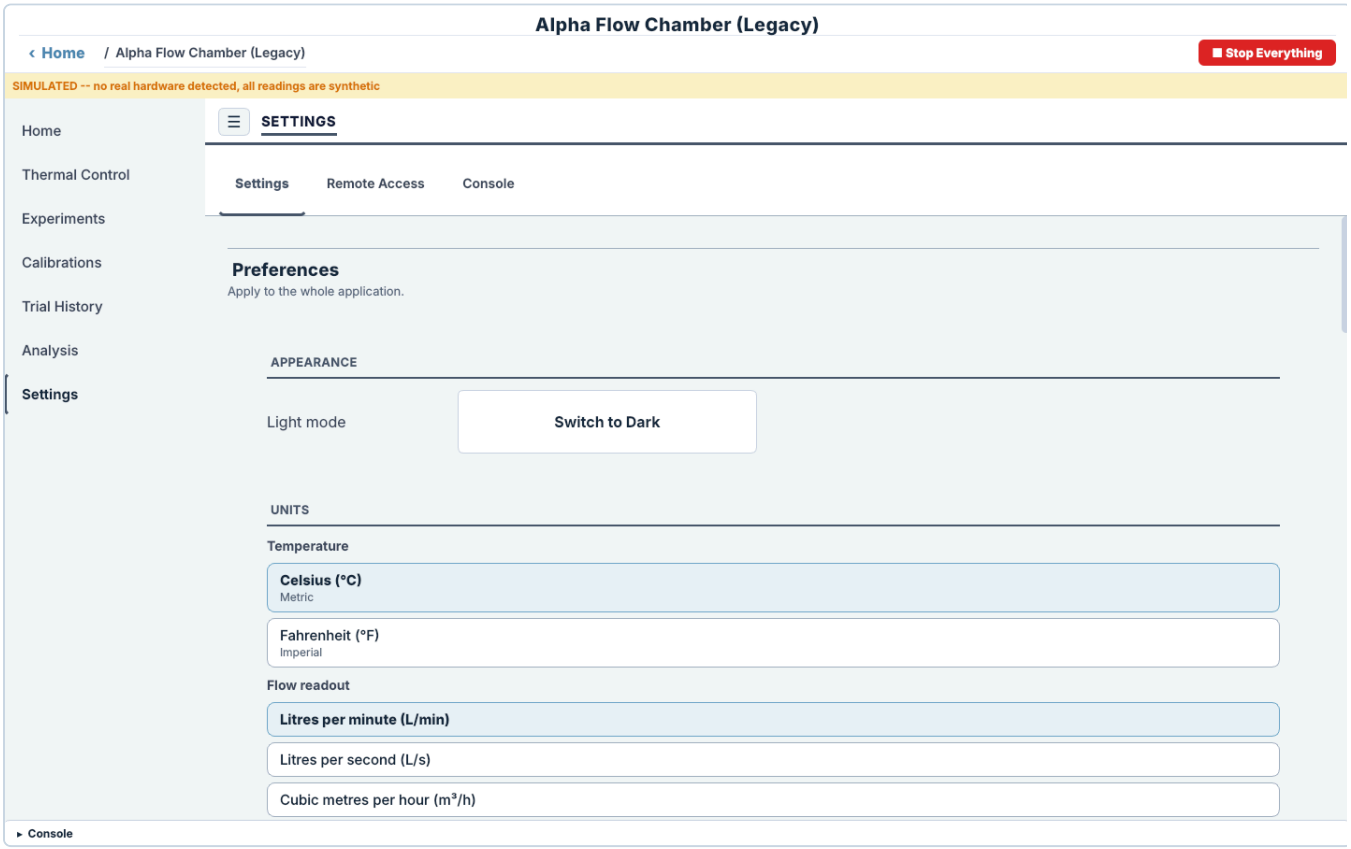
15.7 Exporting

What	Where	Gives you
One trial's cycles	**Export CSV...** on the trial page	One row per cycle.
A comparison	**Export CSV...** on Compare trials	One row per visible trial.

What	Where	Gives you
A respirometry report	**Export PDF…** on the Respirometry Live view (section 8.2)	Protocol, results, charts.
A swim report	**Export PDF…** on Swim Performance	Protocol, result, speed chart.
A chart	**Export…** on any chart’s toolbar	The visible window and series, as CSV.
A track	**Export Track…** on Fish Tracking	Positions and speeds per frame.
A vector field	**Export Vector Field…** on Flow Field (PIV)	One row per vector.

Every export needs a license.

16 Machine Settings



A machine's Settings section, on the Alpha Flow Chamber (Legacy).

A machine's **Settings** section has three tabs: **Settings**, **Remote Access** and **Console**. The Settings tab has two halves: **Preferences**, which apply to the whole application, and **This machine**, which is stored with the machine.

16.1 Preferences

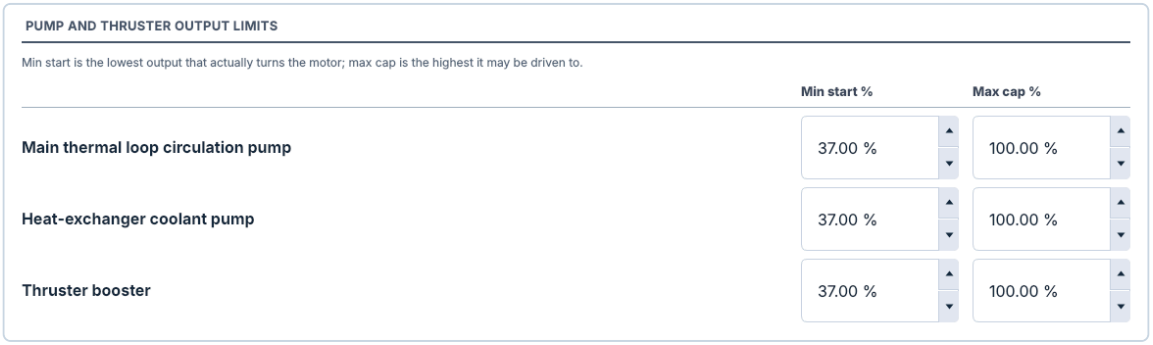
"Apply to the whole application." These are the same controls as the app-wide Settings page (chapter 17): **Appearance**, **Units**, **License**, **Software updates** and **Deleted trials**. They are repeated here so that someone at a machine does not have to close it to reach them. A change here changes the app-wide setting.

16.2 This machine's limits

"Stored with this machine, not with the software, so an update never overwrites them."

Field	What it does
Working section volume (or **Chamber N volume** on a machine with several chambers)	The water the chamber holds, in litres. **Every respirometry rate is divided by it, so a wrong value scales every result.** Measure it; do not leave the 1.0 L default.
Working section cross-section (or **Chamber N cross-section**)	Width × height of the working section, in cm². The solid-blocking correction (section 10.5) and the Chamber Velocity calibration's flow cross-check use it. Left unmeasured, the pages that need it say so rather than assuming a figure. Also needed to show flow as a chamber velocity (section 17.2).
Minimum target temperature	The coldest a thermal zone may be asked to hold. A guard against a mistyped setpoint, not a statement about what the chiller can reach.
Maximum target temperature	The warmest a thermal zone may be asked to hold.

16.3 Pump and thruster output limits



The pump and thruster output limits table.

One table, with a row for each variable-speed pump and thruster, named from the profile (for example **Main thermal loop circulation pump**), so the bands can be compared at a glance:

Column	What it does
Min start %	The lowest command that actually turns this motor. Below it the pump is commanded and does not move, which is what makes a flow curve go flat at the bottom. The Commissioning checklist (section 13.2) and the Chamber Velocity calibration's threshold search (section 13.3) both check it.
Max cap %	The highest this motor may be driven to.

A change to either column takes effect on the running machine straight away, and is kept only when you save it.

16.4 Saving, and leaving without saving

Changes to the limits and the pump and thruster table are not stored until you press **Save changes** in the bar at the bottom of the page. While there are unsaved changes the bar says so in amber: *"Unsaved changes to this machine's volumes, cross-sections, temperature limits or pump and thruster limits."* After saving it reads *"All changes saved."*

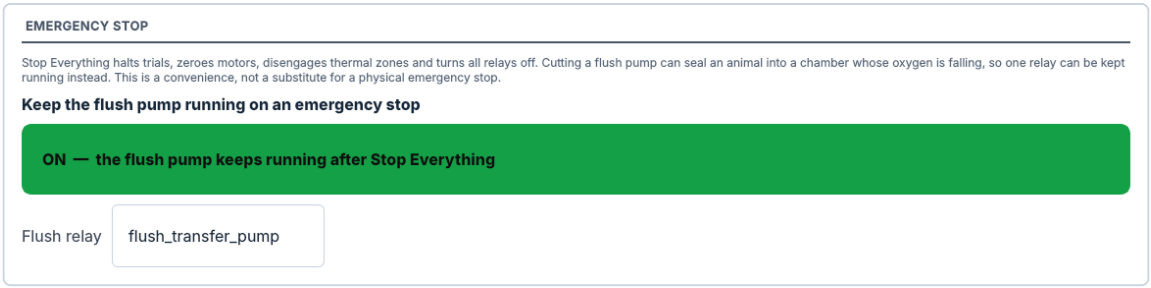
If you try to leave with unsaved changes (another section of the machine, **Back**, Edit Profile, a theme or units change, or closing the application), a window titled **Unsaved changes** asks first:

Button	What happens
Save and leave	Saves, then goes where you were going. If the save is refused (for example, a minimum temperature above the maximum), you stay on the page and it says why.
Leave without saving	Every changed value goes back to what was last saved, including the pump speeds the running machine was using.
Stay here	Nothing changes; you stay on Settings with your edits.

After a units change there is no **Stay here**: the page is redrawn in the new units either way.

The **Preferences** and the emergency-stop choice are not part of this: they save themselves as you change them.

16.5 Emergency stop



The Emergency stop card, with the flush pump kept running.

"Stop Everything halts trials, zeroes motors, disengages thermal zones and turns all relays off. Cutting a flush pump can seal an animal into a chamber whose oxygen is falling, so one relay can be kept running instead. This is a convenience, not a substitute for a physical emergency stop."

Control	What it does
Keep the flush pump running on an emergency stop	A full-width button. It reads **ON — the flush pump keeps running after Stop Everything** (filled green) or **OFF — the flush pump stops with everything else** (outlined). Press it to switch. When it is on, **Stop Everything** (section 4.5) leaves the chosen relay on.

Control	What it does
Flush relay	Which relay to keep running. Choose the pump that brings fresh water to the animal.

This choice saves itself as soon as you change it.

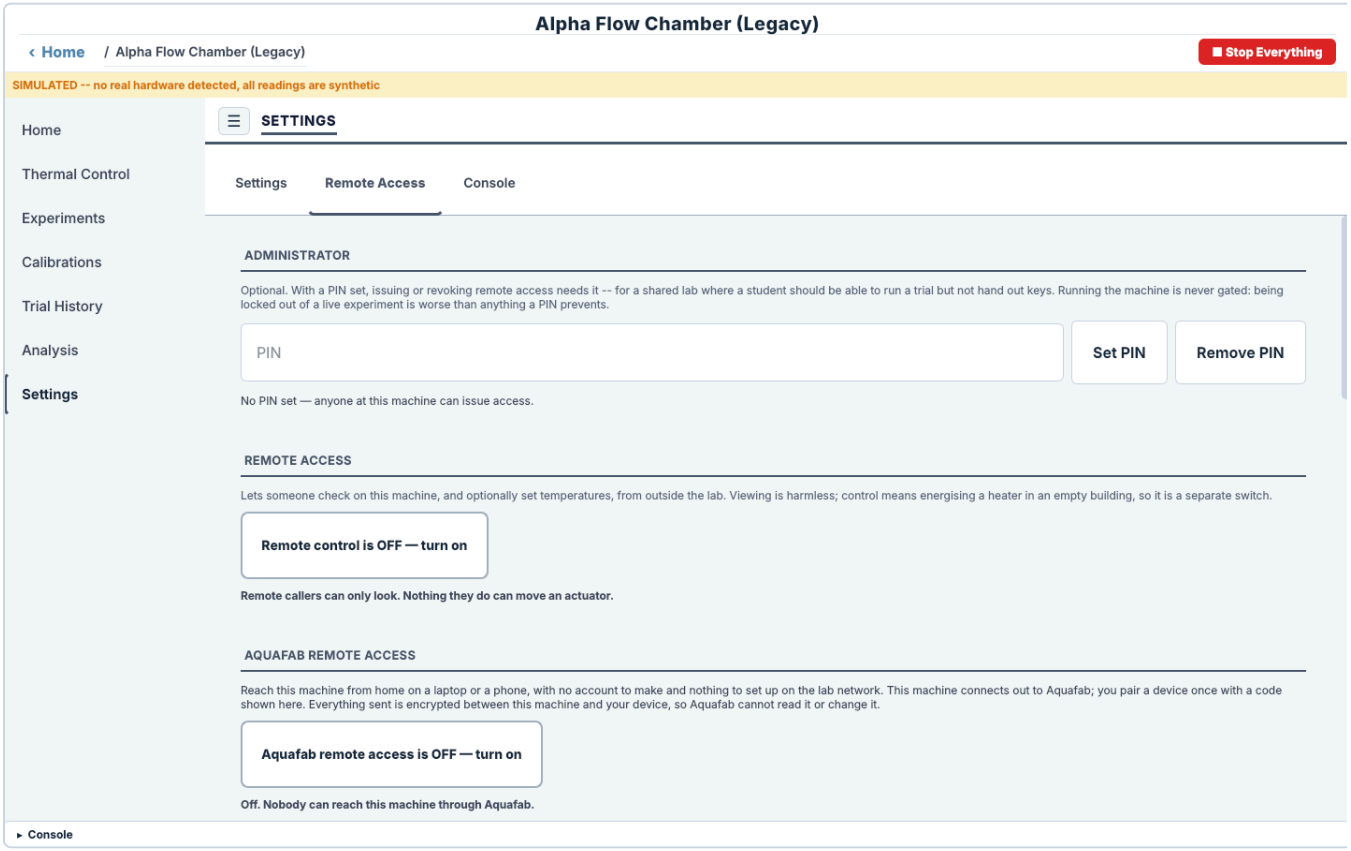
> **This is not a safety system.** Nothing in Aquafab Workbench is a rated or certified interlock. Every machine needs its own physical emergency stop and its own protection against overheating and dry-running.

16.6 Advanced

"Changes the machine's definition. Reach for these when commissioning or rewiring, not to adjust a run."

Button	What it does
Edit Profile	Opens the machine's profile in the profile editor (section 19.3): pins, relays, motors, channels, zones. Saving closes the machine's session, so the next opening picks up the change.
Calibrate O_2 Probe	On machines with an oxygen probe: opens the probe calibration (section 13.5).
Edit Dashboard Layout	Chooses which motors, relays, channels and zones this machine's pages show (section 19.4).
Delete Machine	Removes the machine from this computer, after asking. Its saved trials are kept.

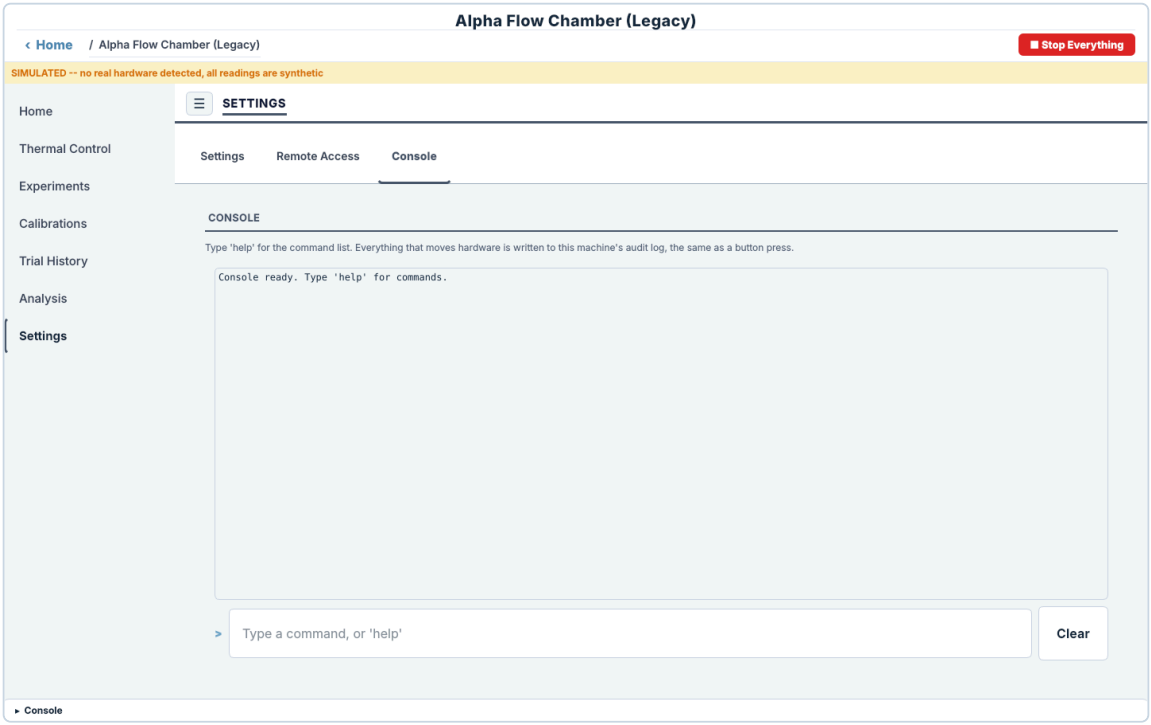
16.7 Remote Access



The Remote Access tab.

Lets someone check on the machine, and optionally set temperatures, from outside the lab, through **Aquafab remote access** with a pairing code or the older routes below it. Chapter 21 covers it in full.

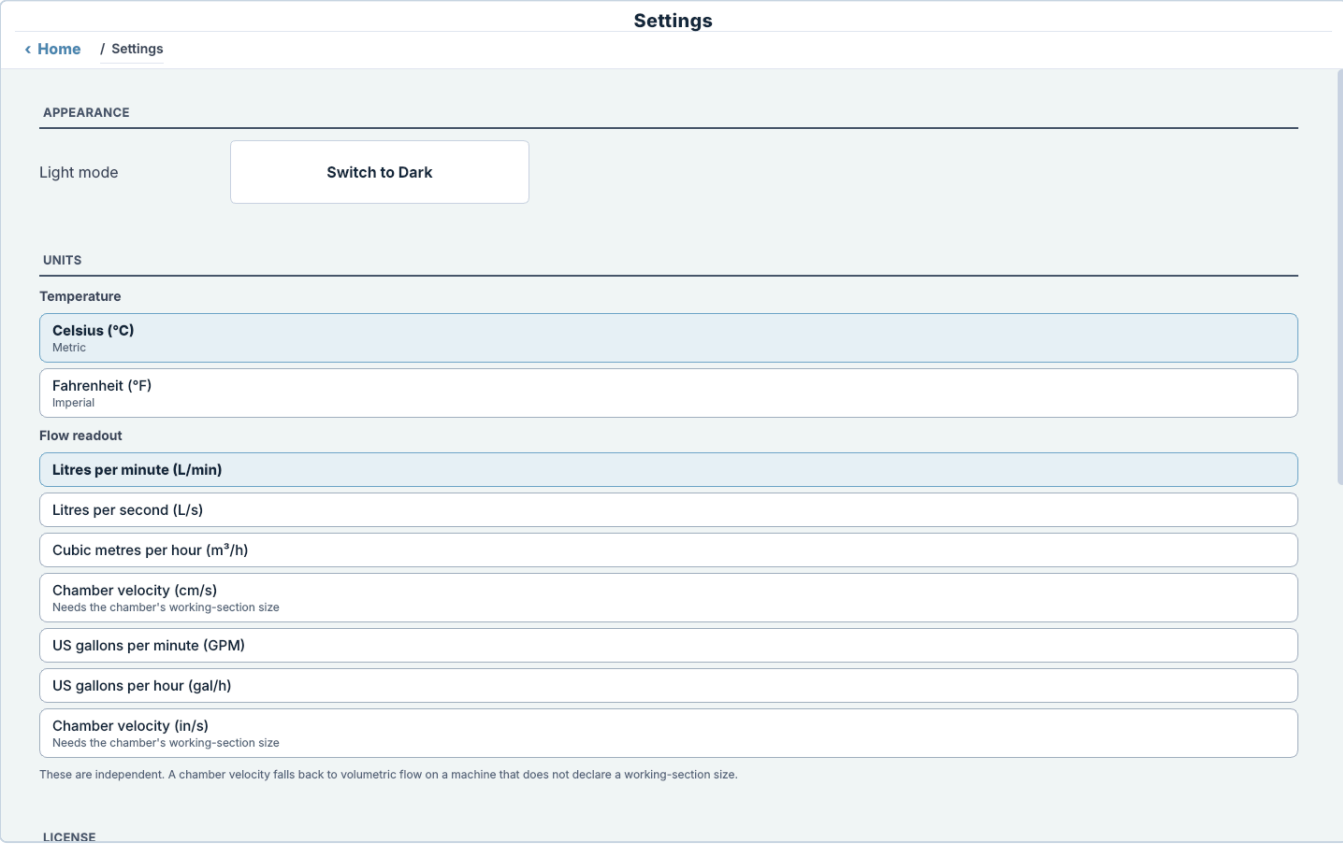
16.8 Console



The Console tab.

The machine's command console: *"Type 'help' for the command list. Everything that moves hardware is written to this machine's audit log, the same as a button press."* Type a command in the box and press **Return**; **Clear** empties the output. The same console opens over any page with **Ctrl+`**. The commands are listed in section 4.4.

17 App-wide Settings



App-wide Settings, opened from the Launcher.

Settings on the Launcher opens the settings that apply to the whole application. The same controls appear as **Preferences** at the top of every machine's Settings section (chapter 16).

17.1 Appearance

Control	What it does
Switch to Dark / **Switch to Light**	Changes the whole window between the light and dark themes. Dark is easier on the eyes in a dim aquarium room.

17.2 Units

Choice	Options
Temperature	**Celsius (°C)** or **Fahrenheit (°F)**. Every temperature on screen, every setpoint field and every chart follows it. Trials are always stored in °C, so changing it never changes data.

Choice	Options
Flow readout	Metric: **Litres per minute (L/min)**, **Litres per second (L/s)**, **Cubic metres per hour (m³/h)**, **Chamber velocity (cm/s)**. Imperial: **US gallons per minute (GPM)**, **US gallons per hour (gal/h)**, **Chamber velocity (in/s)**.

"These are independent. A chamber velocity falls back to volumetric flow on a machine that does not declare a working-section size." To show flow as a velocity, enter the working-section cross-section in the machine's Settings (section 16.2).

17.3 License

The License panel is described fully in section 3.5. In short:

Control	What it does
Status	Whether this computer has a license, and its details: seat type, when it was last renewed and when it renews by, or seats and expiry for a lab key. Without one: *"No license key has been entered, so this is a preview."*
Enter a license code or key	Type a code (`AQF-XXXX-XXXX-XXXX`) or paste a key (starting `AQFB1.`).
Activate	Activates this computer over the internet.
Activate offline…	For a computer with no internet: gives a request code to take to another computer (section 3.4).
Load activation file…	Loads the `aquafab-activation.txt` made for this computer.
Remove key from this computer	After asking: *"Anything running carries on. This computer becomes a preview…"*. Use it before moving a seat to another computer.

17.4 Software updates

Control	What it does
Installed version	The version on this computer.
Check for updates	Asks the Aquafab download site for the newest version: *"Up to date (…)"* or *"Version … is available. You have …"*.
Download and install	Downloads the new version, checks it, and installs it (chapter 22).
Open download page	Opens the website's download page, to install by hand.

17.5 Deleted trials

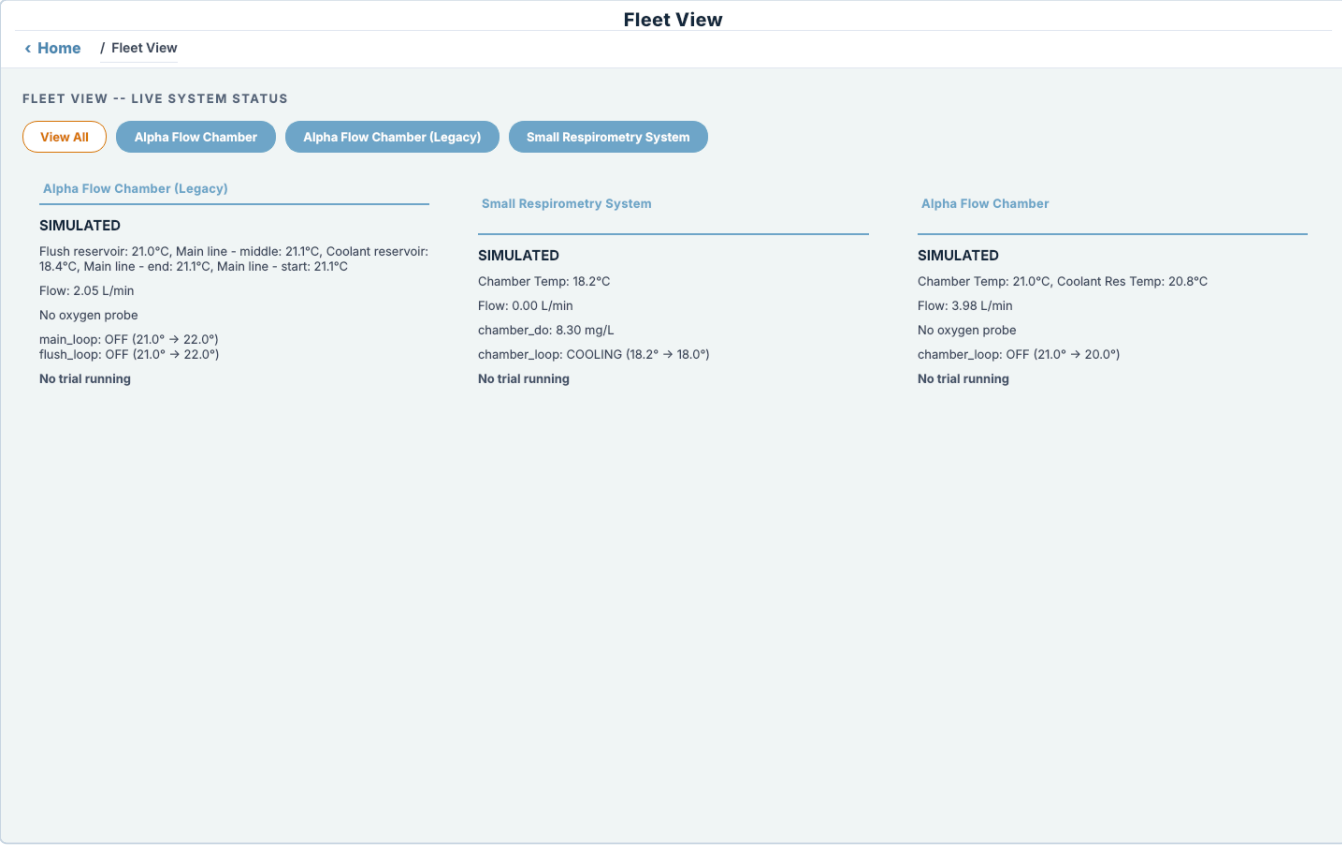
Control	What it does
Empty trash	*Never empty automatically*, or after *30*, *60*, *90* (the default) or *180 days*.

"A deleted trial goes to that machine's trash rather than being removed, and stays there until you empty it or it passes this age. Trials that have not been deleted are never touched, whatever this is set to." The trash itself is in each machine's Trial History (section 14.1).

18 Fleet View, Video Analysis and the Experiment Library

These open from the Launcher and need no machine open.

18.1 Fleet View



Fleet View with every machine shown. All four are simulated here.

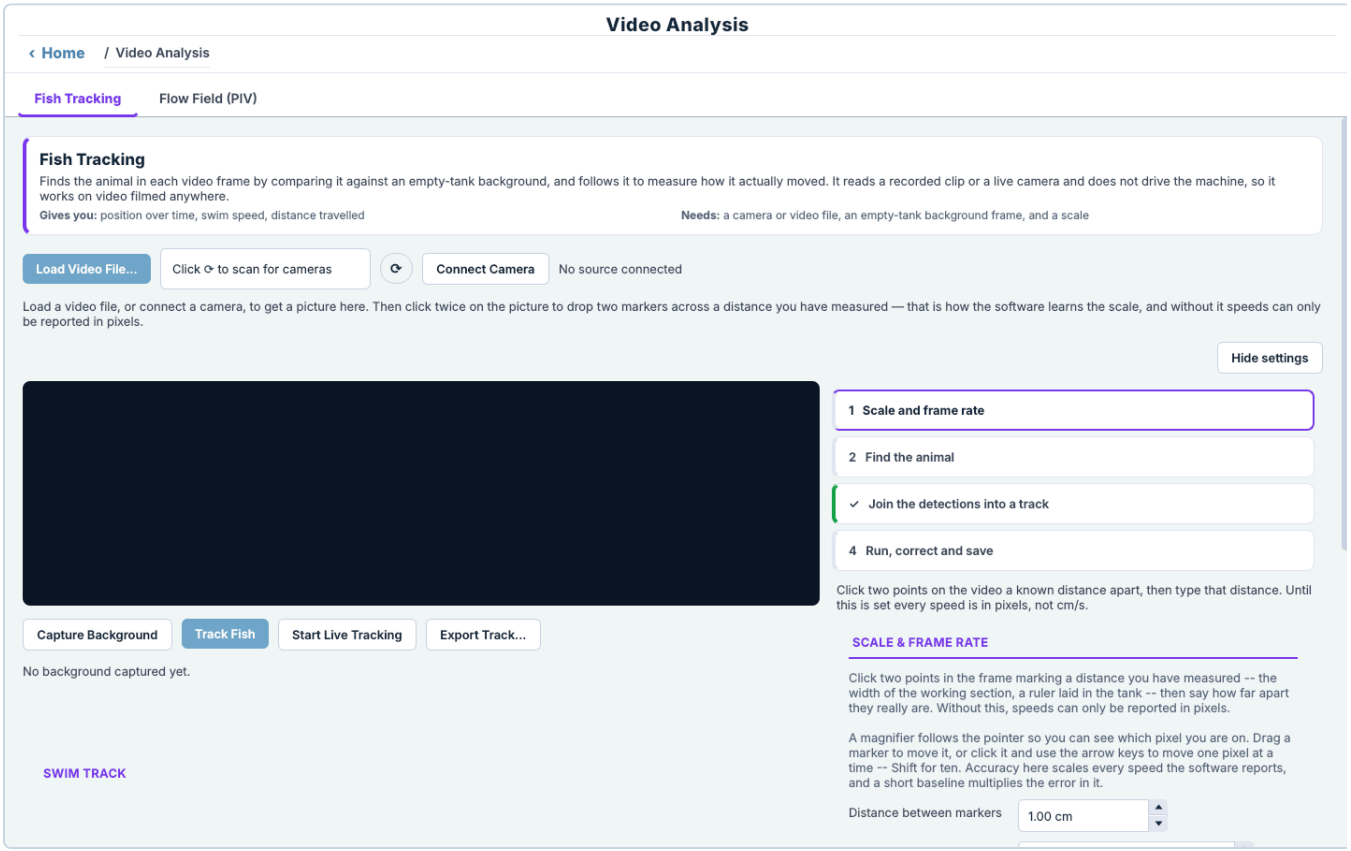
"Fleet view – live system status." Every machine this installation knows about, side by side, so a lab running several rigs can see all of them at once.

Control	What it does
View All	One card per machine.
A machine's name	Shows that machine alone, larger.

Each card shows the machine's name, **SIMULATED** if it is not connected to hardware, every temperature channel, the flow, the oxygen reading (or *"No oxygen probe"*), each thermal zone's state and its measured and target temperatures (for example *"main_loop: OFF (21.0° → 22.0°)"*), and whether a trial is running.

Fleet View only watches. To change anything, open the machine.

18.2 Video Analysis



Video Analysis, Fish Tracking tab, with no machine open.

"Track an animal, or measure a flow field by PIV, from footage on this computer." Two tabs, **Fish Tracking** and **Flow Field (PIV)**, which are the same pages as a machine's (chapters 10 and 11), with no machine attached. This is what an office computer uses to analyse footage filmed on a rig elsewhere, or anywhere else.

Runs analysed here are saved the same way as on a machine (after **Track Fish**, and after **Analyze Whole Video** completes), but under no machine: they are listed in the Launcher's **Analysis** card with the machine *video_analysis*, not in any machine's Trial History. To keep a result in a machine's own history, analyse the clip from inside that machine's Experiments section. Both tabs need a license.

18.3 Experiment Library

"Open a saved protocol to read, edit or send to a machine later." The Saved Experiments list (chapter 12) for every machine. **From** is fixed to *Every machine*, and there is no verdict on whether a machine can run each one, because no machine is open. To load an experiment, open the machine you will run it on and use its **Saved Experiments** tab.

19 Setting up machines

Aquafab systems arrive with their machine already set up. This chapter is for adding a machine, adapting a profile to a rig that has been changed, or building a profile for your own equipment.

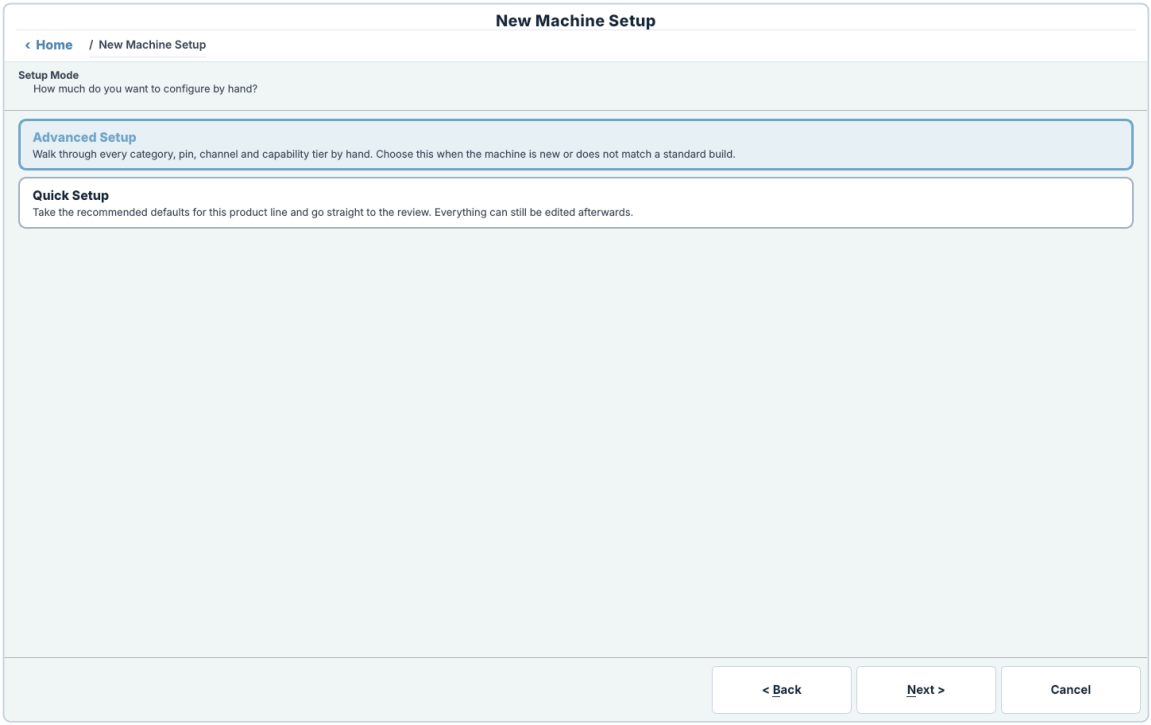
> **A profile describes hardware; it does not check it.** After any change to pins, channels or wiring, run the Commissioning checklist (section 13.2) before trusting the machine.

19.1 What a profile holds

Part	What it is
Identity	The machine ID (fixed once created; it keys the machine's trial history) and its display name.
Backend	How the machine is driven, for example *Legacy direct-GPIO (Raspberry Pi)*, and the GPIO chip (`/dev/gpiochip4` on a Raspberry Pi 5).
Temperature array	The board that reports temperatures, how it is connected (serial, baud rate), and the name of each channel.
Relays	Every switched output: pumps, valves, heaters, chillers and indicator lights. Each has a GPIO pin.
Motors	Variable-speed pumps: pin, driver, pulse range, minimum start and maximum cap.
Flow sensors	Pin, K-factor and what the K-factor means.
Oxygen sensors	Port, baud rate, and the calibration slope and offset.
Thermal zones	Each temperature-controlled loop: its probe, heater, chiller and control strategy.
Dashboard layout	Which of the above the dashboard shows.

What the machine *learns* (commissioning results, the velocity calibration, the thermal step response, probe calibrations, the limits in Settings) is stored separately, in `~/.aquafab/machines/<machine ID>/`, so an update to the software never overwrites it.

19.2 The Add Machine wizard



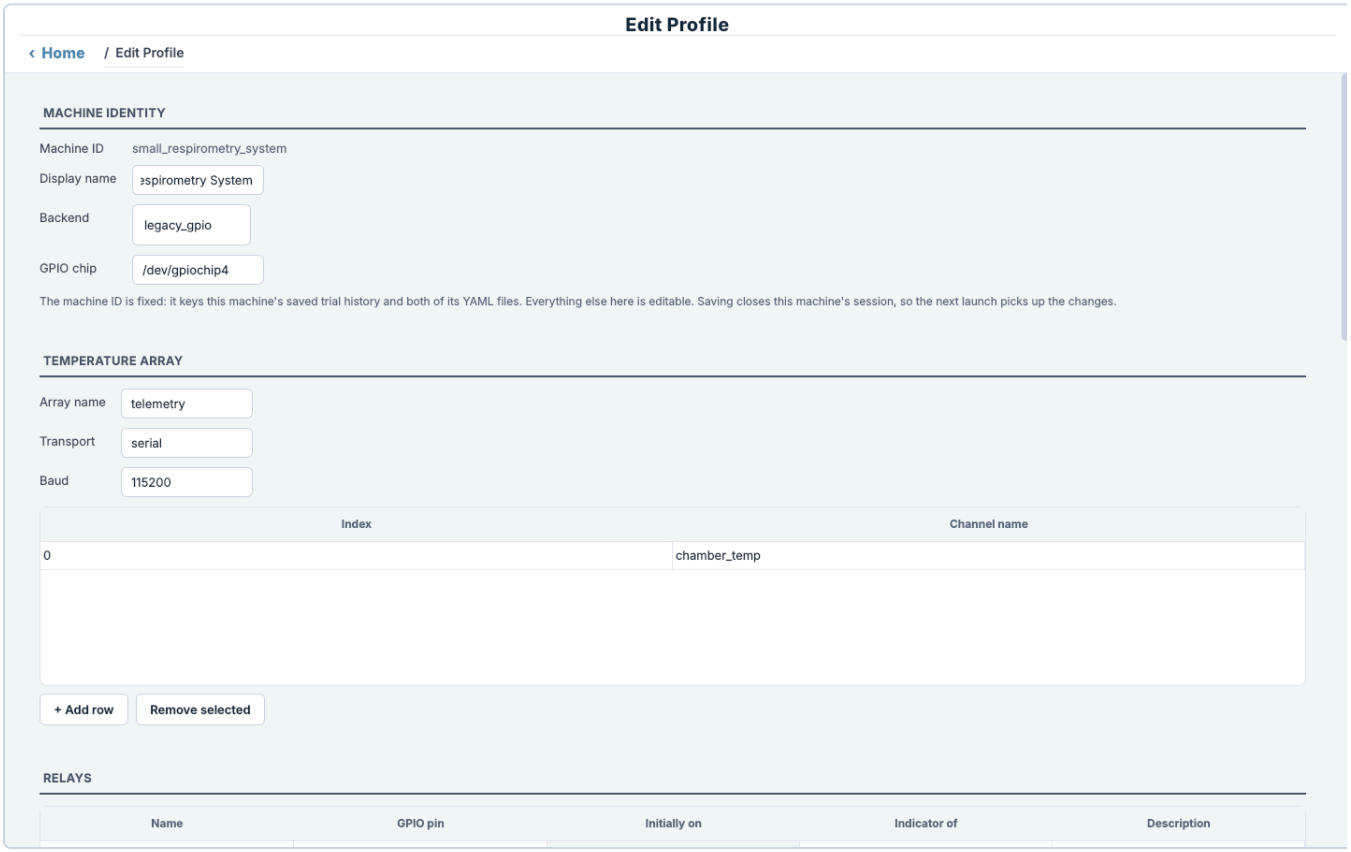
The first page of the Add Machine wizard.

+ Add Machine on the Launcher, then **Build your own…** (chapter 5), opens a wizard. For one of the bundled machines, **Add** in that dialog is quicker. **Next >** and **< Back** move between pages, **Cancel** leaves without creating anything.

1. **Setup Mode.** *"How much do you want to configure by hand?"*
    - **Quick Setup**: take the recommended defaults for a product line and go straight to the review. Everything can be edited afterwards.
    - **Advanced Setup**: walk through every category, pin, channel and capability by hand. For a new machine, or one that does not match a standard build.
2. **System Category.** *Fluid Flow Chambers*, *Respirometry Systems + Swim Tunnels*, *Life Support and Animal Care Systems*, or *Automation and Control Systems* (a bespoke build).
3. **Product Line.** For example *Alpha (Large-Scale Research-Grade)*, *Teaching Flow Chambers*, *Small (Self-Contained)*, or *Custom Build* to start from scratch.
4. **Machine Identity. Display name** (for example *Holy Cross Respirometer*) and **Machine ID** (lowercase with underscores, for example *holy_cross_respirometer*). The ID cannot be changed later.
5. **Add-on Capabilities** (Advanced). A tier for each capability: *Velocity Mapping / Logging* (turbine/paddlewheel, ultrasonic, ADV probe), *Thermal Control*, *Integrated Sensor Suite*, *Corrosion Upgrade*, *Lighting* and *Flow Rate Tier*.
6. **Backend & GPIO** (Advanced). The backend and GPIO chip.
7. **Temperature Array** (Advanced). **Array name**, **Baud rate**, and a name for each channel. Use names that say where the probe is, such as *main_line_start*; they appear on every chart.

8. **Flow Sensor** (Advanced, with velocity mapping). **Name**, **GPIO pin**, **K-factor** and **K-factor means**: *GPM per Hz* (FLOMEC/GPI QS200), *Hz per L/min* (YF-S201 quoted as 7.5), *Pulses per litre* (YF-S201 quoted as
   450. or *L/min per Hz*. *"Read the convention off the datasheet, not the number alone."* The same meter is quoted in different ways, and the wrong convention gives a reading wrong by a large factor.
9. **Thermal Zone** (Advanced, with thermal control). **Zone name** and **Default target (°C)**.
10. **Review.** The whole profile, to confirm. **Finish** creates the machine, adds it to the Launcher and returns there; press **Open ›** to start.

19.3 The profile editor



The profile editor.

Edit on a Launcher row, or **Edit Profile** in a machine's Settings, opens the profile editor: every part of the profile as editable tables.

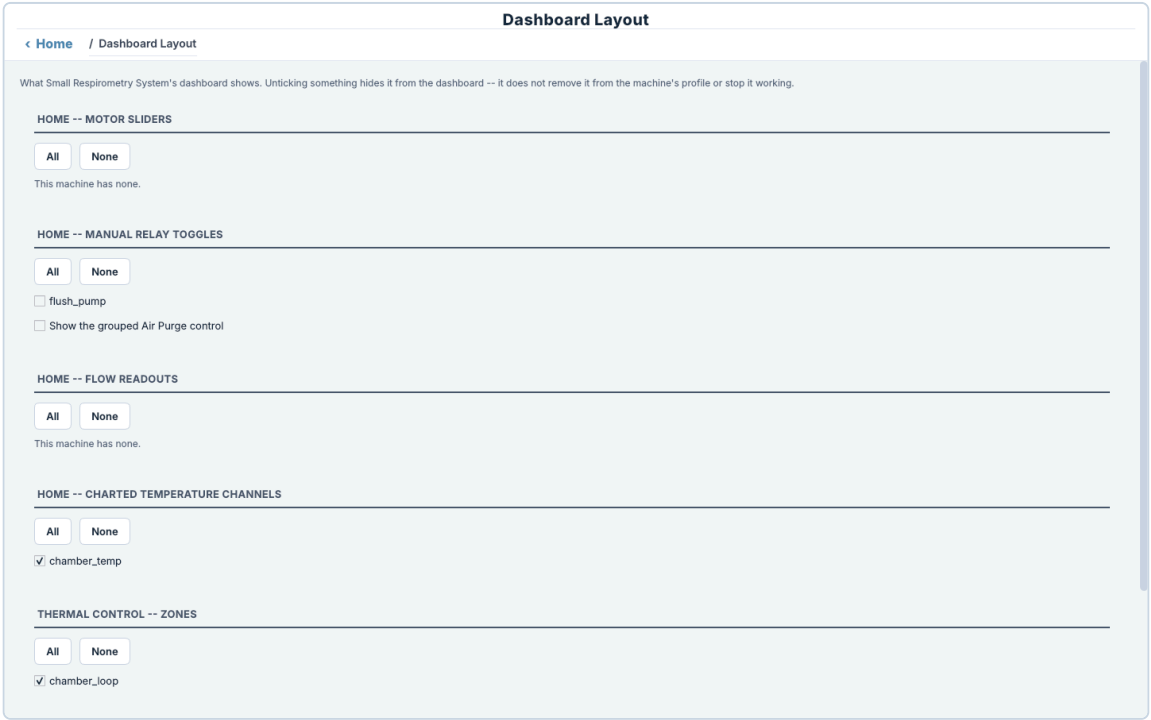
Card	Columns
Machine identity	Machine ID (fixed), **Display name**, **Backend**, **GPIO chip**.
Temperature array	**Array name**, **Transport**, **Baud**, and a table of **Index** and **Channel name**.

Card	Columns
Relays	**Name**, **GPIO pin**, **Initially on**, **Indicator of** (for a light that mirrors another output), **Description**.
Motors	**Name**, **GPIO pin**, **Driver**, **Min pulse (ms)**, **Max pulse (ms)**, **Frame (ms)**, **Min start %**, **Max cap %**, **Indicator relay**, **Description**.
Flow sensors	**Name**, **GPIO pin**, **Pull-up**, **K-factor**, **K-factor means**, **Sample interval (s)**, **Smoothing alpha**, **Hold (s)**, **Decay factor**.
Oxygen sensors	**Name**, **Transport**, **Port**, **Baud**, **Cal. slope**, **Cal. offset**, **Description**. Slope and offset are normally set by the probe calibration (section 13.5).
Thermal zones	**Name**, **Control strategy** (*simple_bangbang*, *main_loop_heat_cool* or *pid_heat_cool*; section 7.2) and **Params**, the strategy's settings as a YAML mapping.

Every table has **+ Add row** and **Remove selected**. **Save Changes** checks the profile and saves it; a profile that does not make sense (a zone naming a probe or relay that does not exist, two outputs on one pin, two relays with the same name, a chamber volume of zero) is refused with the reason. Saving closes the machine's session, so it is rebuilt from the new profile the next time it is opened. **Cancel** discards your edits.

Give each relay-controlled pump and each indicator light exactly one owner: one thermal zone, one flush system, or one indicator. Two controllers driving the same relay fight over it.

19.4 The dashboard layout editor



The dashboard layout editor.

Edit Dashboard Layout in a machine's Settings chooses what the dashboard shows. *"Unticking something hides it from the dashboard – it does not remove it from the machine's profile or stop it working."*

Group	What it controls
Home – Motor sliders	Which variable-speed pumps get a slider on Home.
Home – Manual relay toggles	Which relays get a switch on Home, and **Show the grouped Air Purge control** (on machines whose profile declares a purge group).
Home – Flow readouts	Which flow sensors are shown.
Home – Charted temperature channels	Which channels are drawn on the Live temperature chart.
Thermal Control – Zones	Which zones get a card.
Settings – Pump and thruster limits	Which motors get a row, with Min start % and Max cap %, in the table on the Settings page.

All and **None** tick or clear a whole group. **Save Changes** applies it.

Hiding a relay toggle is a good way to stop students switching something they should not, but it is not a lock: the console can still drive it.

19.5 Import, export and duplicate

On the Launcher (chapter 5):

- **Export** writes the machine's whole configuration to one `.aquafab.yaml` file.
- **↥ Import Config…** adds a machine from such a file, after checking it. It never overwrites a machine that already exists.
- **Duplicate** makes a copy under a new ID, for a second identical rig.

Machine profiles for every shipped Aquafab machine are on the website's Resource Library (*Machine profile*) as `AquafabProfile-<Name>.aquafab.yaml`.

20 How to…

Step-by-step procedures for the common jobs. Each links back to the chapter that explains the controls in full. Every procedure can be rehearsed on a simulated machine first, and it is worth doing so before the first run with an animal.

20.1 Commission a new or rewired machine

Do this once when a machine arrives, and again after any wiring change.

1. Open the machine. Check that the amber **SIMULATED** banner is **not** showing: commissioning a simulated machine checks the simulator.
2. Go to **Calibrations → Commissioning Checklist → Open** (section 13.2).
3. Type your name in **Checked by**.
4. Work down **Relays**: press **Activate**, watch and listen, then **Pass**, **Fail** or **Skip** with a short note. Press **Stop All Test Outputs** if anything keeps running.
5. **Motors**: press **Activate**; confirm the right pump turns, in the right direction, at its minimum start.
6. **Temperature probes**: hold each probe in your hand and watch **Live readings**. The named channel, and only that one, should rise.
7. **Flow**: with the loop running, **Run Check**.
8. **Configuration**: check the declared chiller, reservoir and working-section figures against the machine.
9. Press **Save Record to Profile**.
10. Fix every failed item, in the wiring or in the profile (**Settings → Edit Profile**, section 19.3), and check it again.
11. Enter the measured **Working section volume** and **cross-section** in **Settings → This machine's limits** and press **Save changes**.
12. Then run, in this order: the **Chamber Velocity** calibration (section 20.10), the **Thermal Step Test** (section 20.13) and, if the machine has one, the **dissolved-oxygen probe** calibration (section 20.12).

20.2 Hold a chamber at a temperature

1. Open the machine and go to **Thermal Control** (chapter 7).
2. On the zone's card, type the **Target °C**.
3. Press **Engage**. The status shows *HEATING* or *COOLING*, then *HOLDING* once it arrives.
4. On a machine with a flush loop, engage the flush loop at the same temperature, so a flush is not a thermal shock.
5. Press **Disengage** to stop. Disengaging turns the zone's heater, chiller and pumps off.

20.3 Run an overnight SMR trial (start a respirometry trial)

Before the day: calibrate the oxygen probe (section 20.12), and fast the animal as your protocol requires.

Set up (Experiments → Respirometry → **Set up**, chapter 8):

1. In **What are you measuring?**, choose **Respirometry**, then **Measurement: Standard metabolic rate (SMR)**. This fills in an overnight intermittent-flow schedule (about 10 hours).
2. **This experiment**: type a **Name** and a **Treatment** (for example *18C*). Type the treatment exactly the same way for every animal in that group.
3. **Chambers**: tick each chamber in use and type each **Animal mass (g)**, weighed today. Enter **Loop tubing (L)** if the chamber has a recirculation loop. Read the **respirometric volume** line and check it is sensible.
4. **Acclimation**: leave the preset (1 h, flushing) unless your species needs longer.
5. **Measurement cycle**: check the cycle count, flush, wait and measure times suit your animal and chamber. **Total protocol duration** shows the length of the run.
6. **Temperature**: set the **Constant temperature**, and leave the curve switched off.
7. **Rate analysis**: tick **Measure and subtract background respiration** for a blank cycle; set an **Oxygen floor** (for example, the lowest level your animal-care protocol permits).
8. Optional: **Save experiment**, to run the same protocol on the next animal with one click from **Saved Experiments**.

Prepare the machine (Live view):

9. Press **Prepare system**, with the chamber empty or with only water in it. Wait until every line of the Prepare card is ready: chamber at temperature, flush tank at temperature, reservoir chilled.

Run:

10. Put the animal in the chamber, close it, and check for bubbles.
11. Press **Start Trial** (or **Ctrl+Return**). To start later, for example after you have left the room, use **Start in…** and type the delay in minutes.
12. Watch the first few cycles: oxygen should fall steadily during each measurement and recover with each flush, and each cycle should add a point to the MO_2 chart.

After:

13. The trial saves itself to Trial History after every cycle, so there is nothing to press. When the status says *Complete*, the line under the buttons reads *"Trial saved to Trial History (N cycle(s))."*
14. Optionally, **Export CSV…** or **Export PDF…** (needs a license).
15. Remove the animal, and write anything unusual in the trial's **Notes** in Trial History.

20.4 Run a temperature program (tidal cycle)

1. Experiments → Respirometry → **Set up**. Choose **Temperature program only**, or **Respirometry** to measure oxygen as well.
2. In **Temperature**, switch the curve button to **ON — following the temperature curve below**.
3. Set **Low**, **High**, **Period** (minutes), **Steps** and **Repeat**. Under **Period is the length of**, choose whether that period is **Each cycle**, **Each step** or the **Whole trial**: for an hour-long trial of 12 cycles, set **Repeat** to 12, choose **Whole trial** and enter 60. Press **Generate tidal cycle**. Leave **Ramp continuously between steps** ON for a smooth curve. Edit any row of the table by hand if needed.
4. For a long cycle in a closed chamber, set **Flush for**, **Place it at the** and **Flush when**, so the animal is not driven hypoxic.
5. Read the checks under the table. A **rate warning** means the curve is steeper than the machine can follow: lengthen the period or reduce the range. A **reservoir note** says how cold to chill the reservoir before starting.
6. Set **Water speed** if the flow should be held or stepped.
7. On the Live view, press **Prepare system**, wait, then **Start Trial**. The **Programmed curve** chart shows the curve, the measured temperature and how hard the heater and chiller are working.
8. The run saves itself to Trial History as it goes.

To follow a temperature log recorded in the field instead of a generated curve, see section 20.11.

20.5 Measure P_{crit}

1. Experiments → Respirometry → **Set up** → **Respirometry**, then **Measurement: Critical oxygen level (Pcrit)**. This sets **Closed** respirometry with one long closed phase.
2. Enter the chamber, mass and temperature as for SMR.
3. **Set the Oxygen floor** in Rate analysis: the level at which the chamber is flushed to rescue the animal. Decide before the trial whether crossing it should end the run (**Abort the whole trial if the floor is crossed**).
4. Prepare, then **Start Trial**. Oxygen falls through the whole run.
5. **Derived statistics** shows P_{crit} once the oxygen range allows a breakpoint fit, with its R^2 and the number of points. The trial saves itself.
6. In **Analysis → Work on a trial** (section 15.5), choose the cycles to use and see P_{crit} recomputed with its fit.

20.6 Measure MMR and aerobic scope

MMR is usually measured straight after exhaustive exercise:

1. Set up the trial in advance with **Measurement: Maximum metabolic rate (MMR), after a chase**: no acclimation, short cycles.
2. Prepare the machine.
3. Chase the animal to exhaustion in a separate tank, following your protocol.

4. Move it into the chamber immediately and press **Start Trial**. Seconds matter: the highest rate is in the first measurement.
5. The trial saves itself after each cycle. MMR is the highest accepted cycle.

For aerobic scope in one run, choose **Measurement: Aerobic scope**; for the recovery curve after the chase, **Recovery after exercise (EPOC)**. Aerobic scope across separate SMR and MMR trials can be compared in Analysis.

20.7 Run a U_{crit} swim test

1. Calibrate chamber velocity first (section 20.10), so the result is in cm/s.
2. Experiments → **Swim Performance** (chapter 9).
3. **The staircase**: choose the **Motor**; set **Start output**, **Step size**, **Interval T_i** and **Maximum output**; enter **cm/s per % output** and **Fish length**. For cost of transport, enter **MO_2 at speed**.
4. Optional, **Tracking**: connect the camera, **Capture background** with the tunnel empty, set the scale, tick **Track the animal during the test**.
5. **Name it and run**: name, treatment, notes.
6. Put the animal in and let it settle at a low speed.
7. Press **Start Test**.
8. The moment the animal can no longer hold station, press **Fish Fatigued** (or **F**). The motor stops and the **Result** card shows U_{crit}.
9. Press **Save Result**. Optionally **Export PDF...**.

20.8 Track a fish in a video

Needs a license.

1. Experiments → **Fish Tracking** (chapter 10), or the Launcher's **Video Analysis** for footage from elsewhere.
2. **Load Video File...** (or **Connect Camera**).
3. **Scale and frame rate**: click both ends of a known distance, type **Distance between markers**, press **Apply scale from markers**.
4. **Find the animal**: step to a frame with no animal in view and press **Capture Background**. Set the **Threshold** and **Min blob area**.
5. **Join the detections into a track**: tick **Multiple fish** if there is more than one.
6. Press **Track Fish**. Check the drawn track follows the animal.
7. The run is saved to the machine's Trial History (from Video Analysis, to the Launcher's **Analysis** card, under *video_analysis*). Press **Export Track...** for the CSV.

20.9 Map the flow with PIV

Needs a license.

1. Seed the water, light it evenly and film the working section with the camera fixed and square to it, with a ruler or known width in view.
2. Experiments → **Flow Field (PIV)** (chapter 11), or **Video Analysis**.
3. **Load Video File…** or **Connect Camera**.
4. **Scale and frame rate**: two markers across the known distance; **Apply scale from markers**.
5. **Choose the region**: drag a box over the working section, away from walls and glare, and tick **Crop to region before correlating**.
6. **Correlation settings**: start with the defaults.
7. Press **Run PIV** to check one frame pair. Look at the vectors: they should be orderly, with few flagged.
8. Press **Analyze Whole Video** for the median velocity over the clip. The run is saved to Trial History. **Export Vector Field…** saves the vectors.

20.10 Calibrate chamber velocity with PIV (PIV flow calibration)

This drives the pumps through a sequence, unattended, and needs no license.

1. Fill the machine, seed the water with tracer particles, and set up the camera over the working section with even lighting. **No animal in the chamber.**
2. Make sure **Settings → This machine's limits → Working section cross-section** is entered.
3. **Calibrations → Chamber Velocity (PIV) → Open** (section 13.3).
4. **Camera & scale**: connect the camera, click two points a known distance apart (the chamber width), type **Known distance** and press **Set Scale From Two Points**.
5. Read **Before you start**. It must say **Ready**; if it says **Not ready**, fix each item it lists.
6. Leave **Find each pump's real starting threshold first** ticked.
7. Note the plan's duration, keep clear of the machine, and press **Start Calibration**. **Abort & Stop Pumps** stops everything at any time.
8. When it says *Calibration complete*, review the **Measured points**. Each should have a plausible velocity; a point at zero velocity with a pump running needs looking into.
9. Press **Save to Profile**. The Home page's flow slider now reads in cm/s.

20.11 Import readings from another logger, and follow a recorded trace

Needs a license.

1. Experiments → Respirometry → **This experiment → Import readings from a file…**, and choose the CSV.
2. The import dialog shows what it detected. Choose the **Time column** and its unit, tick the **readings to import** and say what each is measured in, and give a **Treatment**. It will not import without a time column and at least one reading.
3. Import. The trial is saved and marked as **imported** wherever it appears, so it is never mistaken for something this machine measured.
4. To make the chamber follow the trace: in **Trial History**, **Inspect** the imported trial, press **Use as a temperature program…**, choose the **Temperature series**, **Steps** and **Repeat**, and press **Use as the temperature program** (section 14.3).

5. Check the curve on the Temperature card, and run it as in section 20.4.

20.12 Calibrate the dissolved-oxygen probe

1. Prepare a **zero-oxygen solution** (for example, fresh sodium sulfite in water) and **air-saturated water** at your measuring temperature (bubble air through it for several minutes).
2. **Calibrations → Dissolved-Oxygen Probe → Open**, or **Settings → Calibrate O_2 Probe** (section 13.5).
3. Choose the **Probe**.
4. Put the probe in the zero solution. When the reading is steady, press **Capture Zero**.
5. Rinse it and put it in the saturated water. Enter **Water temperature**, **Salinity** and **Pressure**; check the **Saturated reference**. When the reading is steady, press **Capture Saturated**.
6. Press **Save Calibration**.
7. Rinse the probe thoroughly before it goes back into a chamber with an animal. Sulfite is harmful to animals.

20.13 Run a thermal step test

1. **Empty the chamber of animals.** The water is pushed beyond where any animal should be held.
2. **Calibrations → Thermal Step Test → Open** (section 13.4).
3. Choose the **Zone**, **Step size**, **Time limit per step** and **Settle between steps**. The defaults suit most machines.
4. Press **Start Step Test** and leave it; it can take most of an hour.
5. When it finishes, read **Measured response**: the heating and cooling rates, and the fitted dead time and time constant.
6. Press **Save to Profile**. This records the response; the Temperature card's rate check uses it.
7. **Only if** the zone uses PID control and you want a starting point for tuning, press **Apply Gains to Zone**, then read the warning in section 13.4. The suggested Kp is far too high for a lag-dominant chamber; start well below it.

20.14 Compare treatments and export the results

Needs a license.

1. Open **Analysis** from the Launcher (every machine) or a machine's Analysis section (chapter 15).
2. In **Compare trials**, choose **Compare which number** (for example *SMR, by the method the trial used*) and **Make a group from: Treatment**.
3. Narrow the set with the filters, or tick trials in the **Trials** table.
4. Read **Group statistics**: compare treatments by their SE.
5. Look at **Traces, overlaid** for any trial that behaved differently.
6. For a trial with unsettled early cycles, use **Work on a trial** to start from a later cycle, and **Save this selection** with a name.
7. For temperature effects, **Cross-trial metrics → Q_{10}**; for mass effects, **Body-mass scaling**.
8. **Export CSV…** on Compare for one row per trial, or **Export CSV…** on a trial for its cycles.

21 Remote access

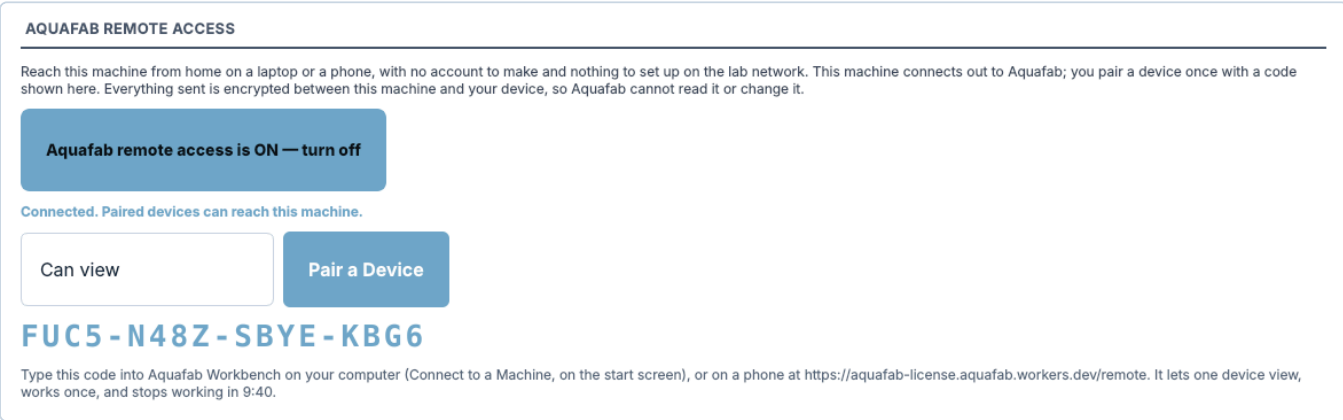
Settings → Remote Access on a machine lets someone check on it, and optionally set temperatures, from outside the lab: a phone at home, a laptop in an office. *"Viewing is harmless; control means energising a heater in an empty building, so it is a separate switch."*

Remote access sends numbers, not a picture of the screen, so it stays responsive on a slow connection, unlike a remote desktop.

> **Remote access is a convenience, not a safety system.** Someone must still be responsible for a machine running unattended, and the machine's own physical protections must not depend on it.

21.1 Aquafab remote access

The simplest way in, and the one to use unless your IT department asks for something else. There is no account to make, nothing to install, and nothing to change on the lab network: the machine connects out to Aquafab, and you pair your laptop or phone with it once, using a code shown on the machine's screen.



The Aquafab remote access card, with a pairing code showing.

It needs the machine's computer to have its **license activated online** (chapter 3). Without one the card says *"Remote access through Aquafab needs this computer's license activated online. Enter your license code under Settings, License."*

21.1.1 Turn it on, at the machine

1. Open the machine, then **Settings → Remote Access**.
2. Press **Aquafab remote access is OFF — turn on**. It now reads **Aquafab remote access is ON — turn off**, and the line under it reads *"Connected. Paired devices can reach this machine."* once the machine is online.

It stays on, and reconnects by itself, whenever the machine is open in Aquafab Workbench. If the connection drops, the line says so and it tries again on its own.

21.1.2 Pair a device, at the machine

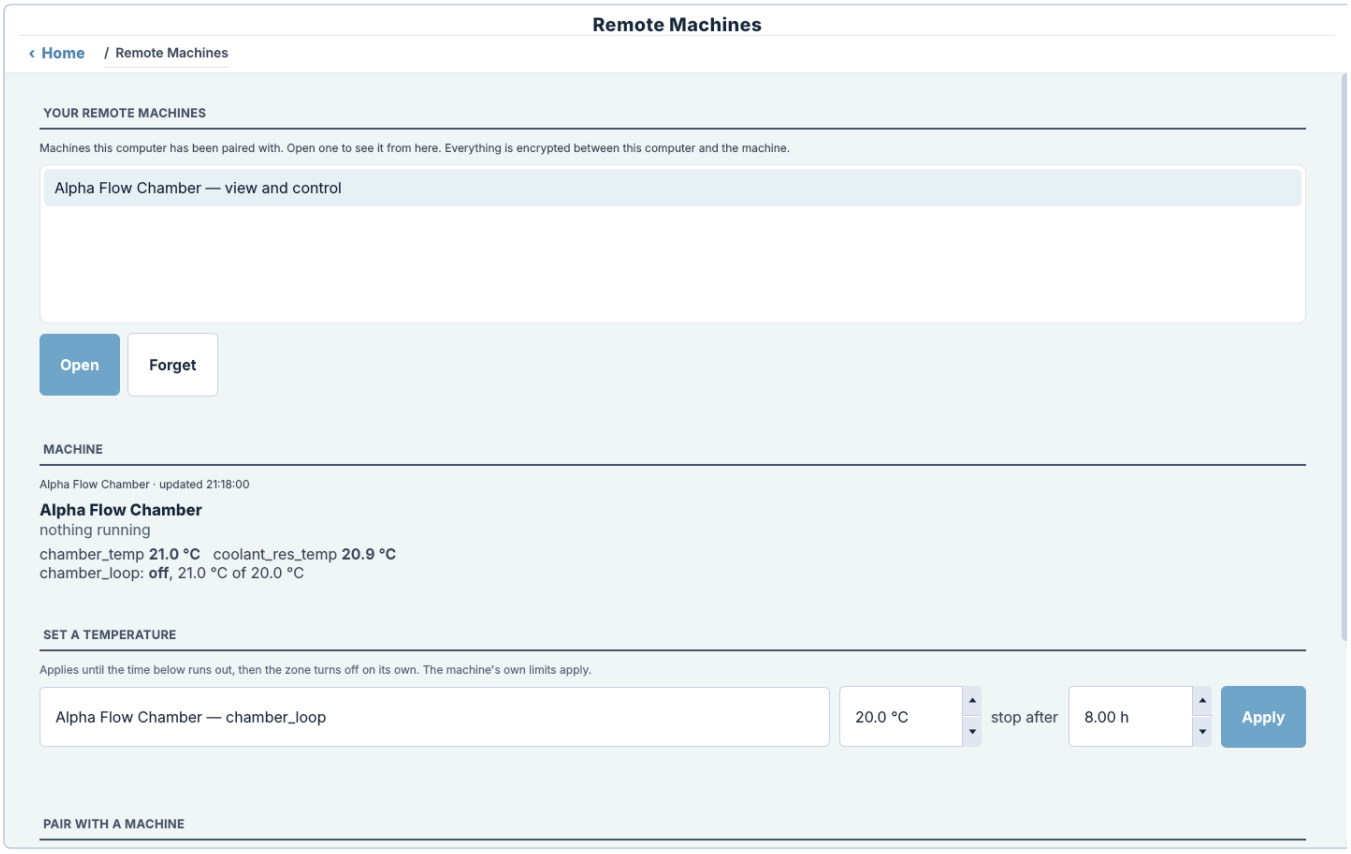
1. Choose what the device may do: **Can view** or **Can view and control**.
2. Press **Pair a Device**. A code like `DPWY-BCGQ-23VG-4KTJ` appears, with a countdown.
3. Type that code on the laptop or phone (below). The code works **once**, for **ten minutes**. Making a new code cancels the old one.

When the device has paired, the card says *"Paired. It is listed under WHO HAS ACCESS, where it can be revoked."* The device appears there as its name followed by **(paired)**.

A device paired with **Can view and control** can only set temperatures while **Remote control is ON** (below), and only within the **Limits on remote control**, exactly like any other credential.

21.1.3 Connect from a laptop

On the laptop, open Aquafab Workbench and press **Connect to a Machine** on the start screen. No license is needed on the laptop.



Connect to a Machine, with a paired machine open. The temperatures shown are simulated.

1. Under **PAIR WITH A MACHINE**, type the code into **Pairing code**.
2. **Name this computer** is how it will be listed on the machine; change it if you like.

3. Press **Pair**. The machine is added to **YOUR REMOTE MACHINES** and opens.

From then on, choose the machine in the list and press **Open**. The **MACHINE** card shows its temperatures, its zones and what trial is running, refreshed every ten seconds while the page is on screen. A device paired to control also gets **SET A TEMPERATURE**: choose the zone, the target, how many hours it should stay on (**stop after**), and press **Apply**. The zone turns itself off when that time runs out.

Forget removes the machine from this laptop. It does not stop this laptop reaching it; to do that, revoke it on the machine.

21.1.4 Connect from a phone

On the phone, open **aquafab-license.aquafab.workers.dev/remote** in the browser. Type the code into **Pairing code**, give the phone a name in **Name this device**, and press **Pair**. The machine is listed on the page from then on; tap **Open** to see it.

The phone remembers the pairing in its browser. **Forget this machine on this device** removes it.

21.1.5 What Aquafab can and cannot see

Everything between the machine and your device is encrypted with a key that only the two of them hold. Aquafab's server passes the messages along but cannot read them or change them: it never sees a temperature, a setpoint, the pairing code or the device's key. The machine applies its own limits to every request, whichever way it arrived.

On the phone, the page itself is delivered by Aquafab's server, so the phone trusts that server to deliver it unchanged. Aquafab Workbench on a laptop does not depend on that, which is one reason to prefer it where you have the choice.

21.1.6 Stop someone reaching the machine

- **One device:** select it under **WHO HAS ACCESS** and press **Revoke Selected**. It stops working immediately.
- **Everyone, through Aquafab:** press **Aquafab remote access is ON — turn off**.
- Freeing the computer's license seat also turns Aquafab remote access off.

21.2 Administrator PIN

"Optional. With a PIN set, issuing or revoking remote access needs it – for a shared lab where a student should be able to run a trial but not hand out keys."

Control	What it does
PIN box, **Set PIN**	Sets the PIN. *"PIN set. Administrative actions now require it."*
Remove PIN	Enter the current PIN, then press it.

With a PIN set, turning on Aquafab remote access and pairing a device need it too. Running the machine is never behind the PIN: *"being locked out of a live experiment is worse than anything a PIN prevents."*

21.3 Remote control on or off

Remote control is OFF — turn on / **Remote control is ON — turn off**.

- **Off** (the default): *"Remote callers can only look. Nothing they do can move an actuator."*
- **On**: anyone holding a *control* credential can change setpoints from outside the lab.

21.4 Limits on remote control

"Bounds on what a remote request may ask for, so a mistake typed on a phone cannot ask the machine for something it should never do."

Field	What it does
Setpoints between … and …	The lowest and highest temperature a remote request may set.
A remote change stops after	Hours. *"A pre-chill started from home and then forgotten disengages on its own rather than running all week."*

21.5 Who has access

Each person or device gets its own credential, so one can be revoked without affecting the others.

1. Type who it is for in **Who is this for?** (for example *lab-phone*).
2. Choose **Can view** or **Can view and control**.
3. Choose how many **days** it lasts.
4. Press **Issue Access**. The page shows the access token **once**: *"Copy it now. Only a hash is kept, so this cannot be shown again"*. Give it to the person by a private channel.

A device paired through Aquafab remote access gets its credential this way automatically; you do not issue it a token. These tokens are for the other ways in, below.

Revoke Selected removes a credential; it stops working immediately.

21.6 Other ways in

Aquafab remote access covers most labs. The two routes below are for a lab whose IT department wants the connection on its own network instead.

21.6.1 The server

"Serves a small page and a JSON API. Bound to this machine only by default – a tunnel carries it out, so nothing is opened on the lab network."

Control	What it does
Reachable from	**This machine only (use a tunnel)**, the default and the right choice; or **Anything on the local network**, for a laptop on the same network.
Port	The port the server listens on (default 8770).
Start Server / **Stop Server**	Starts or stops it. The line under it says whether it is running.

21.6.2 Connecting through Tailscale

"Do not forward a port to this machine. A tunnel dials out instead, which opens nothing on the lab network, works on a university guest network without an IT ticket, and gives this machine a name you can open from a laptop at home."

The software works with **Tailscale**, a private-network service that runs on the machine and on the device you connect from.

1. Install Tailscale on the control computer, if it is not already (the page gives the command when it is missing).
2. Ask whoever runs your Tailscale network for a one-time **setup key**. It joins the machine with no account to create and no browser login.
3. Paste it in **Setup key** and press **Join network**. The key is never stored and never shown again.
4. **Check** asks the tunnel what state it is in. Once joined, the page shows *"Open this from another machine: …"* with the address.
5. Start the server, and open that address on your phone or laptop (which must be on the same Tailscale network). Sign in with an access token.

22 Updates

Aquafab Workbench checks for a newer version a few seconds after it opens, quietly, and never interrupts you. If one is available, an **Update available** button appears in the top strip. Pressing it opens Settings, where the update is one press away. A computer with no internet simply never shows the button.

22.1 Updating from inside the software

1. Open **Settings** (from the Launcher, or a machine's Settings), **Software updates**.
2. Press **Check for updates** if the result is not already shown: *"Version … is available. You have …"*.
3. Press **Download and install**. It asks *"Download the new version (… MB)? When it has downloaded and been checked, you will be asked once more before anything is installed."*
4. The download shows its progress and can be cancelled. It is checked against its published checksum; a download that does not match is refused.
5. *"The update is downloaded and its checksum matches. Aquafab Workbench will close, install it, and open again. Your machines, calibrations and trials are kept. Install now?"* Press **Yes**.

An update is never installed while something is running. Installing closes the application, so if a trial, a calibration or a preparation is running, it says *"Finish the trial first"* and lists what is running. Finish or stop it, then update. If a trial starts while the download is running, the downloaded update waits until you press **Download and install** again.

22.2 If the check fails

"Could not check for updates: this computer could not look up …" means the computer could not find the named address on the internet. Check that it is online. If it is, the network may block that address: lab and school networks sometimes block GitHub, where the updates are kept. Ask the network's administrator to allow `github.com` and `objects.githubusercontent.com`, or update by hand (below).

22.3 Updating by hand

Open download page opens the Aquafab Workbench page on the website. Download the installer for your computer and install it over the old version, as in chapter 2. Your machines, calibrations, saved experiments, trials and license are kept, because they are stored in your home folder, not with the program (chapter 23).

On Linux, only the AppImage build can update itself. A copy run from source is updated with `git pull`.

22.4 What an update never changes

- Your **trials**, **saved experiments** and **license**.
- What each machine has **learned**: commissioning results, velocity calibration, thermal step response, probe calibration, and the limits and pump and thruster limits in Settings.
- Any **changes you made to a machine's profile**. An update can improve a shipped machine's definition, and your changes are kept on top of it.

23 Your data, and troubleshooting

23.1 Where everything is kept

Everything you make is kept in a folder called `.aquafab` in your home folder (on Windows `C:\Users\<you>\.aquafab`, on a Mac or Linux `~/.aquafab`). The folder name starts with a dot, so it is hidden by default.

Folder or file	What it holds
`trials/<machine ID>/`	Every saved trial, one file each.
`trials/<machine ID>/_deleted/`	That machine's trash.
`protocols/`	Saved experiments, one file each.
`machines/<machine ID>/`	Your changes to each machine's profile, and what each machine has learned: commissioning record, velocity calibration, thermal step response, probe calibration, limits.
`license.json`	This computer's license. It works only on this computer; copying it elsewhere does not move the license (section 3.8).
`logs/`	Each machine's activity log: everything that moved hardware, from a button or the console, and when.

The **fault log**, where the software records any error it caught, is separate: `~/Library/Logs/Aquafab Workbench/faults.log` on a Mac, and `~/.local/state/aquafab-workbench/faults.log` elsewhere.

23.1.1 Backing up

Copy the whole `.aquafab` folder. To move to a new computer, copy it into the new computer's home folder before opening Aquafab Workbench there, then activate the license on the new computer (section 3.8).

Trials are the record of your experiments. Back them up as you would any other research data: regularly, and somewhere other than the computer that recorded them.

23.2 Troubleshooting

23.2.1 Installing and opening

Problem	What to do
Windows says *"Windows protected your PC"*	Click **More info**, then **Run anyway** (section 2.3).

Problem	What to do
A Mac says it cannot check the app for malicious software	**System Settings → Privacy & Security → Open Anyway** (section 2.4).
The window is larger than the screen	It opens maximised on a small screen. Every page scrolls; press ☰ to hide the section rail. On a Pi in full screen, **Exit full screen** is in the top strip.
The window closed by itself	Look in the fault log. If the log has nothing at the time it closed, the cause was outside the software: on a Raspberry Pi, usually an under-powered supply or running out of memory. Contact Aquafab support.

23.2.2 Machines and hardware

Problem	What to do
A machine opens in simulation on its own control computer	The hardware could not be reached. On a Pi, the user may not have permission for the GPIO and serial ports yet, or may need to log out and back in. Check cables and power. Contact support if it persists.
A relay switches the wrong device	The pin in the profile is wrong. Run the Commissioning checklist (section 20.1) and correct the profile.
A pump does not turn at the bottom of the slider	Its **Min start %** is too low. Measure it with Commissioning or the Chamber Velocity calibration's threshold search, and set it in Settings (section 16.3).
The flow reading jumps between zero and a huge value	The flow meter's K-factor convention is wrong (section 19.2).
Swim Performance is missing from Experiments	The machine has no variable-speed pump.
The dissolved-oxygen calibration is greyed out	The machine's profile has no oxygen probe.

23.2.3 Trials

Problem	What to do
Start Trial says the machine is not prepared	It is waiting for the chamber, flush tank or reservoir to reach temperature. Wait, or press **Start anyway** (section 8.2.1).
Start Trial says a license is needed	Trials on connected hardware need a license. Enter one in Settings (chapter 3). Simulated machines run without one.
A finished trial is not in Trial History	Swim tests are saved only when you press **Save Result**; respirometry saves itself after every cycle. Tracking and PIV runs from Video Analysis are under the Launcher's **Analysis** card (machine *video_analysis*), not in a machine's Trial History.

Problem	What to do
A trial says *Interrupted*	The software stopped before the run finished (a power cut, a crash, the computer slept). The cycles measured up to then are kept and can be analysed; the run cannot be resumed.
MO_2 values are implausibly high or low	Check, in order: the **Working section volume** in Settings, the **Animal mass**, the **respirometric volume** line on the Chambers card, and the probe calibration. Every rate scales directly with the first two.
SMR looks high	The animal may not have settled. In **Analysis → Work on a trial**, start from a later cycle, and look at the distribution of cycles (section 15.3).
Many cycles are rejected	Their oxygen decline does not fit a straight line: the animal moved, the chamber had not mixed (lengthen **Wait**), or a bubble is in the chamber.
A temperature curve is not followed	The curve is steeper than the machine can manage. The rate warning under the program table says so; run the Thermal Step Test to measure what the machine can do.

23.2.4 Video

Problem	What to do
No camera in the list	Plug it in and press ⟳. If another page is using it, press **Disconnect** there first.
Speeds are in pixels	Set the scale: two markers across a known distance, then **Apply scale from markers**.
The track jumps around the tank	Raise the **Threshold** or **Min blob area**; capture a fresh background; check the camera did not move.
Many PIV vectors flagged	Too few particles per window, particles moving too far between frames, or glare. Try a larger window, more passes or a smaller region (section 11.7).

23.2.5 License

Problem	What to do
"...needs a license" over a page	The software is a preview on this computer. Enter a license code in **Settings → License**.
Activation fails with no internet	Use **Activate offline...** (section 3.4).
"This code is already in use on another computer"	Each code activates one computer. Free the seat in the license dashboard (section 3.6), or use another code.

23.3 Getting help

Email **sales@aquafabindustries.com** or call **+1 (513) 781-8010**. It helps to include the version (**Settings → Software updates**), the machine, what you did, what happened, and the fault log if there is one.

24 Reference

24.1 How the numbers are calculated

Quantity	How it is computed
MO_2 (mg O_2 kg^{-1} h^{-1})	|slope| × respirometric volume × 3600 ÷ animal mass (kg). The slope is the least-squares line through the oxygen decline (mg L^{-1} s^{-1}) in one measurement phase. The respirometric volume is the chamber volume plus loop tubing, minus the water the animal displaces.
Background correction	The rate measured in a blank cycle with no animal is subtracted from every cycle, assuming background respiration is constant through the trial (Svendsen et al. 2016).
SMR, low quantile	The chosen quantile (0.2 recommended) of the accepted cycles' MO_2 (Chabot et al. 2016).
SMR, MLND	The mean of the lowest normal distribution fitted to the accepted cycles, falling back to the quantile when there is no distinct low mode (Chabot et al. 2016).
MMR	The highest accepted cycle.
Aerobic scope	MMR − SMR (absolute) and MMR ÷ SMR (factorial).
P_{crit}	Two straight lines joined at a breakpoint, fitted to MO_2 against oxygen (segmented regression); the breakpoint is P_{crit}. Reported with its R^2. Treat a P_{crit} from a handful of cycles as indicative, not publishable.
U_{crit}	$U + (T \div T_i) \times U_i$ (Brett 1964).
Cost of transport	MO_2 ÷ distance per hour, in mg O_2 kg^{-1} m^{-1}. U_{opt} is the speed with the lowest cost.
Q_{10}	$(R_2 \div R_1)^{(10 \div \Delta T)}$, from one treatment at two held temperatures.
Mass scaling	$MO_2 = a \cdot M^b$, fitted on log mass against log whole-animal rate.
Oxygen solubility	García & Gordon (1992), from temperature, salinity and pressure.
Solid blocking	Bell & Terhune (1970). Verify the coefficients before publication use.
Suggested PID gains	Ziegler–Nichols open-loop rules (1942) from a first-order-plus-dead-time fit. **Not valid on lag-dominant chambers** (section 13.4).
PIV	Cross-correlation of interrogation windows between frame pairs, with signal-to-noise validation and outlier replacement, using OpenPIV.
Fish tracking	Background subtraction and blob detection per frame, linked into tracks; multi-animal linking uses trackpy.

24.2 Published methods referenced

- Bell, W. H. & Terhune, L. D. B. (1970). *Water tunnel design for fisheries research.* Fisheries Research Board of Canada Technical Report 195.
- Brett, J. R. (1964). The respiratory metabolism and swimming performance of young sockeye salmon. *Journal of the Fisheries Research Board of Canada* 21, 1183–1226.
- Chabot, D., Steffensen, J. F. & Farrell, A. P. (2016). The determination of standard metabolic rate in fishes. *Journal of Fish Biology* 88, 81–121.
- García, H. E. & Gordon, L. I. (1992). Oxygen solubility in seawater: better fitting equations. *Limnology and Oceanography* 37, 1307–1312.
- Svendsen, M. B. S., Bushnell, P. G. & Steffensen, J. F. (2016). Design and setup of intermittent-flow respirometry system for aquatic organisms. *Journal of Fish Biology* 88, 26–50.
- Ziegler, J. G. & Nichols, N. B. (1942). Optimum settings for automatic controllers. *Transactions of the ASME* 64, 759–768.

24.3 Keyboard shortcuts

The shortcuts are listed in section 4.6.

24.4 Limitations and safety

- **Aquafab Workbench is not a safety system.** No interlock, limit, floor or emergency stop in it is rated or certified. Every machine needs its own physical emergency stop, and its own protection against overheating, dry running and flooding.
- **Simulated machines are not your machine.** Readings, rates and calibration results on a simulated machine describe the built-in simulator. Never report a simulated figure as a measurement.
- **A calibration is only as good as its reference.** A wrong scale distance, probe reference or chamber volume scales every result built on it, and the result still looks plausible.
- **Suggested PID gains are a starting point, not an answer**, and on these chambers they are too aggressive as given (section 13.4).
- **The oxygen floor is a rescue, not a protocol.** Set it for every closed and P_{crit} run, and still watch the animal.
- **Remote control energises heaters in an empty building.** It is off by default; turn it on only with someone responsible for the machine.

24.5 Version of this guide

This guide describes Aquafab Workbench as built from the version named on its cover. It is updated with the software. The newest copy is always on the Aquafab Workbench page of the Aquafab website.